

Mathematical Analysis I

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Feel free to open issues or pull requests for corrections.

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Preface

Throughout this textbook, **logical formulas** are interpreted according to the following conventions:

- **Parentheses** have the highest precedence. For example,

$$(P \vee Q) \wedge R$$

is evaluated by first computing $P \vee Q$.

- The **logical connectives** are interpreted in the order

$$\neg > \wedge > \oplus > \vee > \implies, \iff > \iff$$

Hence

$$P \oplus Q \implies \neg R \vee S \wedge T \iff U \iff V$$

is understood as

$$((P \oplus Q) \implies ((\neg R) \vee (S \wedge T))) \iff (U \iff V)$$

- A **quantifier** binds the entire formula that follows it. Thus

$$\forall x, P(x) \wedge Q(x) \implies R(x)$$

means

$$\forall x ((P(x) \wedge Q(x)) \implies R(x))$$

- The symbol $\Downarrow$ is **not** a logical connective; it is used only to indicate the flow of a proof or an explanation. For example,

$$\begin{array}{c} P \\ \Downarrow \\ Q \end{array}$$

merely expresses that P , and therefore Q in the current discussion; it is not itself a proposition.

Chapter 1

The Number Systems

1.1 The Natural Number System

Axiom 1.1.1. The set of **natural numbers** $\mathbb{N}$ satisfies the following axioms:

- $1 \in \mathbb{N}$
- $\forall n \in \mathbb{N}, s(n) \in \mathbb{N}$
- $\nexists n \in \mathbb{N}, s(n) = 1$
- $\forall m, n \in \mathbb{N}, s(m) = s(n) \implies m = n$
- $\forall S \subseteq \mathbb{N}, (1 \in S \wedge (\forall n \in S, s(n) \in S)) \implies S = \mathbb{N}$

$s(n)$ is called the **successor** of n . These axioms are called the **Peano axioms**.

Remark. The **Peano axioms** do not guarantee the *existence* of $\mathbb{N}$, which is guaranteed by the **axiom of infinity** in the **ZFC axioms**. In this textbook, the **ZFC axioms** are not discussed, so we will just assume that $\mathbb{N}$ exists.

Theorem 1.1.2. Suppose P is a predicate with domain $\mathbb{N}$. Then

$$P(1) \wedge (\forall n \in \mathbb{N}, P(n) \implies P(s(n))) \implies \forall n \in \mathbb{N}, P(n)$$

This method of proof is called the **principle of mathematical induction**.

Proof. Suppose

- $P(1)$

- $\forall n \in \mathbb{N}, P(n) \implies P(s(n))$

Define

$$S := \{n \in \mathbb{N} \mid P(n)\} \subseteq \mathbb{N}$$

Then

- $1 \in S$
- $\forall n \in S, s(n) \in S$

By [axiom 1.1.1](#),

$$\begin{aligned} S &= \mathbb{N} \\ \Downarrow \\ \forall n \in \mathbb{N}, P(n) \end{aligned}$$

□

Definition 1.1.3. Suppose $n, m \in \mathbb{N}$.

- **Addition** $+$ on $\mathbb{N}$ is defined by
 - $n + 1 := s(n)$
 - $n + s(m) := s(n + m)$
- **Multiplication** $\cdot$ on $\mathbb{N}$ is defined by
 - $n \cdot 1 := n$
 - $n \cdot s(m) := n \cdot m + n$

Remark. Strictly speaking, [definition 1.1.3](#) requires the **recursion theorem** to be strictly defined. In this textbook, the **recursion theorem** is not discussed, so we will just assume that the operations on $\mathbb{N}$ are **well-defined**.

Theorem 1.1.4. $\mathbb{N}$ has the **well-ordering property**, which states

$$\forall S \subseteq \mathbb{N}, S \neq \emptyset \implies \exists m \in S, m = \min S$$

Proof. Assume

$$\exists S \subseteq \mathbb{N}, S \neq \emptyset \wedge (\nexists m \in S, m = \min S)$$

Define

- $T := \mathbb{N} \setminus S \subseteq \mathbb{N}$

$$\bullet L := \{n \in \mathbb{N} \mid \{1, 2, \dots, n\} \subseteq T\} \subseteq \mathbb{N}$$

We will show

$$\begin{aligned} L &= \mathbb{N} \\ \Downarrow \\ T &= \mathbb{N} \\ \Downarrow \\ S &= \emptyset \\ \Downarrow \\ &\perp \end{aligned}$$

• **Base case:** Assume $1 \in S$. Then

$$\begin{aligned} 1 &= \min S \\ \Downarrow \\ &\perp \end{aligned}$$

Therefore,

$$\begin{aligned} 1 &\notin S \\ \Downarrow \\ 1 &\in T \\ \Downarrow \\ 1 &\in L \end{aligned}$$

• **Inductive step:** Suppose $n \in L$. Then

$$\begin{aligned} n &\in L \\ \Downarrow \\ \{1, 2, \dots, n\} &\subseteq T \\ \Downarrow \\ \{1, 2, \dots, n\} \cap S &= \emptyset \end{aligned}$$

Assume $n + 1 \in S$. Then

$$\begin{aligned} n + 1 &= \min S \\ \Downarrow \\ &\perp \end{aligned}$$

Therefore,

$$\begin{aligned} n + 1 &\notin S \\ \Downarrow \\ n + 1 &\in T \\ \Downarrow \\ \{1, 2, \dots, n + 1\} &\subseteq T \\ \Downarrow \\ n + 1 &\in L \end{aligned}$$

By [theorem 1.1.2](#),

$$L = \mathbb{N}$$

□

1.2 Relations

Definition 1.2.1. A **relation** R from A to B is defined by

$$R \subseteq A \times B$$

In particular, a relation R on A is defined by

$$R \subseteq A \times A$$

The notation xRy denotes $(x, y) \in R$.

Definition 1.2.2. A relation $\sim$ on S is called an **equivalence relation** if it satisfies

- **Reflexivity:** $\forall x \in S, x \sim x$
- **Symmetry:** $\forall x, y \in S, x \sim y \implies y \sim x$
- **Transitivity:** $\forall x, y, z \in S, x \sim y \wedge y \sim z \implies x \sim z$

Definition 1.2.3. A collection $\{A_\alpha\}_{\alpha \in I}$ is called a **partition** of S if

- **Nonemptiness:** $\forall \alpha \in I, \emptyset \neq A_\alpha \subseteq S$
- **Disjointness:** $\forall \alpha, \beta \in I, \alpha \neq \beta \implies A_\alpha \cap A_\beta = \emptyset$
- **Exhaustivity:** $\bigcup_{\alpha \in I} A_\alpha = S$

Definition 1.2.4. Suppose

- $\sim$ is an equivalence relation on S
- $x \in S$

The **equivalence class** of x is defined by

$$[x] := \{y \in S \mid y \sim x\}$$

The set of all equivalence classes of S is denoted by

$$S/\sim := \{[x] \mid x \in S\}$$

Lemma 1.2.5. *Suppose $\sim$ is an equivalence relation on S . Then*

$$\forall x, y \in S, [x] \cap [y] \neq \emptyset \implies [x] = [y]$$

Proof. Suppose

- $x, y \in S$
- $[x] \cap [y] \neq \emptyset$

Then

$$\begin{aligned} \exists z \in S, z \in [x] \cap [y] \\ \Downarrow \\ z \sim x \wedge z \sim y \\ \Downarrow \\ x \sim z \wedge z \sim y \\ \Downarrow \\ x \sim y \end{aligned}$$

Therefore,

$$\begin{aligned} \forall w \in S, w \sim x \implies w \sim y \\ \Downarrow \\ \forall w \in [x], w \in [y] \\ \Downarrow \\ [x] \subseteq [y] \end{aligned}$$

Similarly,

$$[y] \subseteq [x]$$

□

Theorem 1.2.6. *Suppose $\sim$ is an equivalence relation on S . Then*

$$\{[\alpha]\}_{[\alpha] \in S/\sim} = S/\sim$$

is a partition of S .

Proof. By [definition 1.2.3](#),

- **Nonemptiness:**

$$\forall [\alpha] \in S / \sim, \alpha \in [\alpha] \subseteq S$$

- **Disjointness:** By [lemma 1.2.5](#),

$$\forall [\alpha], [\beta] \in S / \sim, [\alpha] \neq [\beta] \implies [\alpha] \cap [\beta] = \emptyset$$

- **Exhaustivity:**

– ($\subseteq$)

$$\begin{aligned} & \forall [\alpha] \in S / \sim, [\alpha] \subseteq S \\ & \quad \downarrow \\ & \bigcup_{[\alpha] \in S / \sim} [\alpha] \subseteq S \end{aligned}$$

– ($\supseteq$)

$$\begin{aligned} & \forall \beta \in S, [\beta] \in S / \sim \\ & \quad \downarrow \\ & \forall \beta \in S, \beta \in [\beta] \subseteq \bigcup_{[\alpha] \in S / \sim} [\alpha] \end{aligned}$$

□

Theorem 1.2.7. Suppose $\{A_\alpha\}_{\alpha \in I}$ is a partition of S . Define $\sim$ on S by

$$\forall x, y \in S, x \sim y \iff \exists \alpha \in I, x \in A_\alpha \wedge y \in A_\alpha$$

Then

(A) $\sim$ is an equivalence relation on S

- (B) • $\forall x \in S, \exists \beta \in I, [x] = A_\beta$
 • $\forall \beta \in I, \exists x \in S, [x] = A_\beta$

(C) $\{A_\alpha\}_{\alpha \in I} = S / \sim$

Proof.

- (A) By [definition 1.2.2](#),

- **Reflexivity:** Suppose $x \in S$. Then

$$\begin{aligned}
 x &\in \bigcup_{\alpha \in I} A_\alpha \\
 &\Downarrow \\
 \exists \beta \in I, x &\in A_\beta \\
 &\Downarrow \\
 x &\sim x
 \end{aligned}$$

- **Symmetry:** Suppose

- $x, y \in S$
- $x \sim y$

Then

$$\begin{aligned}
 \exists \alpha \in I, x &\in A_\alpha \wedge y \in A_\alpha \\
 &\Downarrow \\
 y &\in A_\alpha \wedge x \in A_\alpha \\
 &\Downarrow \\
 y &\sim x
 \end{aligned}$$

- **Transitivity:** Suppose

- $x, y, z \in S$
- $x \sim y$
- $y \sim z$

Then

- $\exists \alpha \in I, x \in A_\alpha \wedge y \in A_\alpha$
- $\exists \beta \in I, y \in A_\beta \wedge z \in A_\beta$

Therefore,

$$\begin{aligned}
 y &\in A_\alpha \wedge y \in A_\beta \\
 &\Downarrow \\
 A_\alpha \cap A_\beta &\neq \emptyset \\
 &\Downarrow \\
 \alpha &= \beta \\
 &\Downarrow \\
 x &\in A_\alpha \wedge z \in A_\alpha \\
 &\Downarrow \\
 x &\sim z
 \end{aligned}$$

- (B) • Suppose $x \in S$. Then

$$\begin{aligned}
 x &\in \bigcup_{\alpha \in I} A_\alpha \\
 &\Downarrow \\
 \exists \beta \in I, x &\in A_\beta
 \end{aligned}$$

- Suppose $\beta \in I$. Then

$$\exists x \in A_\beta \subseteq S$$

We will show $[x] = A_\beta$.

- $(\subseteq)$ Suppose $y \in [x]$. Then

$$\begin{aligned} \exists \gamma \in I, y \in A_\gamma \wedge x \in A_\gamma \\ \Downarrow \\ x \in A_\gamma \cap A_\beta \\ \Downarrow \\ \gamma = \beta \\ \Downarrow \\ y \in A_\beta \end{aligned}$$

- $(\supseteq)$ Suppose $y \in A_\beta$. Then

$$\begin{aligned} y \sim x \\ \Downarrow \\ y \in [x] \end{aligned}$$

(C) By [theorem 1.2.7](#),

- $(\subseteq)$

$$\forall A_\beta \in \{A_\alpha\}_{\alpha \in I}, \exists x \in S, A_\beta = [x] \in S / \sim$$

- $(\supseteq)$

$$\forall [x] \in S / \sim, \exists \beta \in I, [x] = A_\beta \in \{A_\alpha\}_{\alpha \in I}$$

□

1.3 The Integer and Rational Number Systems

Theorem 1.3.1. Define $\sim$ on $\mathbb{N} \times \mathbb{N}$ by

$$\forall (a, b), (c, d) \in \mathbb{N} \times \mathbb{N}, (a, b) \sim (c, d) \iff a + d = b + c$$

Then $\sim$ is an equivalence relation on $\mathbb{N} \times \mathbb{N}$.

Proof. By [definition 1.2.2](#),

- **Reflexivity:** Suppose $(a, b) \in \mathbb{N} \times \mathbb{N}$. Then

$$\begin{aligned} a + b &= b + a \\ \Downarrow \\ (a, b) &\sim (a, b) \end{aligned}$$

- **Symmetry:** Suppose

- $(a, b), (c, d) \in \mathbb{N} \times \mathbb{N}$
- $(a, b) \sim (c, d)$

Then

$$\begin{aligned} a + d &= b + c \\ \Downarrow \\ c + b &= d + a \\ \Downarrow \\ (c, d) &\sim (a, b) \end{aligned}$$

- **Transitivity:** Suppose

- $(a, b), (c, d), (e, f) \in \mathbb{N} \times \mathbb{N}$
- $(a, b) \sim (c, d)$
- $(c, d) \sim (e, f)$

Then

$$\begin{aligned} a + d &= b + c \wedge c + f = d + e \\ \Downarrow \\ a + d + f &= b + c + f = b + d + e \\ \Downarrow \\ a + f &= b + e \\ \Downarrow \\ (a, b) &\sim (e, f) \end{aligned}$$

□

Remark. Strictly speaking, [theorem 1.3.1](#) requires several **properties** of operations on $\mathbb{N}$. In this textbook, the properties are not discussed, so just **accepted** as true. The same issue arises for [theorem 1.3.4](#).

Definition 1.3.2. Recall the relation $\sim$ defined in [theorem 1.3.1](#). The set of **integers** $\mathbb{Z}$ is defined by

$$\mathbb{Z} := (\mathbb{N} \times \mathbb{N}) / \sim$$

Definition 1.3.3. Suppose $[(a, b)], [(c, d)] \in \mathbb{Z}$.

- **Addition** $+$ on $\mathbb{Z}$ is defined by

$$[(a, b)] + [(c, d)] := [(a + c, b + d)]$$

- **Multiplication** $\cdot$ on $\mathbb{Z}$ is defined by

$$[(a, b)] \cdot [(c, d)] := [(a \cdot c + b \cdot d, a \cdot d + b \cdot c)]$$

Remark. The operations on $\mathbb{Z}$ are **well-defined**; that is, they do not depend on the **choice of representatives** of the equivalence classes. Indeed, if

- $(a, b) \sim (a', b')$
- $(c, d) \sim (c', d')$

then one can verify that

- $(a + c, b + d) \sim (a' + c', b' + d')$
- $(a \cdot c + b \cdot d, a \cdot d + b \cdot c) \sim (a' \cdot c' + b' \cdot d', a' \cdot d' + b' \cdot c')$

The same issue arises for operations on $\mathbb{Q}$, whose **well-definedness** could be verified in exactly the same manner.

Theorem 1.3.4. Define

$$\mathbb{Z}^+ := \{[(n + 1, 1)] \in \mathbb{Z} \mid n \in \mathbb{N}\} \subseteq \mathbb{Z}$$

Define $\sim$ on $\mathbb{Z} \times \mathbb{Z}^+$ by

$$\forall (a, b), (c, d) \in \mathbb{Z} \times \mathbb{Z}^+, (a, b) \sim (c, d) \iff a \cdot d = b \cdot c$$

Then $\sim$ is an equivalence relation on $\mathbb{Z} \times \mathbb{Z}^+$.

Proof. By [definition 1.2.2](#),

- **Reflexivity:** Suppose $(a, b) \in \mathbb{Z} \times \mathbb{Z}^+$. Then

$$\begin{aligned} a \cdot b &= b \cdot a \\ \Downarrow \\ (a, b) &\sim (a, b) \end{aligned}$$

- **Symmetry:** Suppose

- $(a, b), (c, d) \in \mathbb{Z} \times \mathbb{Z}^+$
- $(a, b) \sim (c, d)$

Then

$$\begin{aligned} a \cdot d &= b \cdot c \\ \Downarrow \\ c \cdot b &= d \cdot a \\ \Downarrow \\ (c, d) &\sim (a, b) \end{aligned}$$

- **Transitivity:** Suppose

- $(a, b), (c, d), (e, f) \in \mathbb{Z} \times \mathbb{Z}^+$
- $(a, b) \sim (c, d)$
- $(c, d) \sim (e, f)$

Then

$$\begin{aligned} a \cdot d &= b \cdot c \wedge c \cdot f = d \cdot e \\ \Downarrow \\ a \cdot d \cdot f &= b \cdot c \cdot f = b \cdot d \cdot e \\ \Downarrow \\ a \cdot f &= b \cdot e \\ \Downarrow \\ (a, b) &\sim (e, f) \end{aligned}$$

□

Definition 1.3.5. Recall the relation $\sim$ defined in [theorem 1.3.4](#). The set of **rational numbers** $\mathbb{Q}$ is defined by

$$\mathbb{Q} := (\mathbb{Z} \times \mathbb{Z}^+) / \sim$$

Definition 1.3.6. Suppose $[(a, b)], [(c, d)] \in \mathbb{Q}$.

- **Addition** $+$ on $\mathbb{Q}$ is defined by

$$[(a, b)] + [(c, d)] := [(ad + bc, bd)]$$

- **Multiplication** $\cdot$ on $\mathbb{Q}$ is defined by

$$[(a, b)] \cdot [(c, d)] := [(ac, bd)]$$

Remark. Strictly speaking, the constructions of $\mathbb{N}$, $\mathbb{Z}$, and $\mathbb{Q}$ produce **different kind** of sets. For example,

- $n \in \mathbb{N}$
- $[(n + 1, 1)] \in \mathbb{Z}$
- $[(n, [(2, 1)])] \in \mathbb{Q}$

are distinct objects in the underlying set-theoretic construction. Nevertheless, there are **canonical embeddings**

- $\mathbb{N} \rightarrow \mathbb{Z}$ such that $n \mapsto [(n + 1, 1)]$
- $\mathbb{Z} \rightarrow \mathbb{Q}$ such that $a \mapsto [(a, [(2, 1)])]$

which preserve the **algebraic operations**. Therefore, we identify each number system with its image in the next larger one and write

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q}$$

This identification is justified by the fact that the embeddings are canonical, requiring **no arbitrary choices**. Throughout this textbook, inclusions such as

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$$

should be understood as canonical identifications rather than literal inclusions arising directly from the set-theoretic constructions.

Remark. Define

$$0 := [(1, 1)] \in \mathbb{Z}$$

The notations below denote subsets of $\mathbb{Z}$:

- $[0, \infty)_{\mathbb{N}} := \mathbb{N} \cup \{0\} \subseteq \mathbb{Z}$
- $\forall n \in [0, \infty)_{\mathbb{N}}, [n, \infty)_{\mathbb{N}} := \{k \in [0, \infty)_{\mathbb{N}} \mid k \geq n\}$
- $\forall n, m \in [0, \infty)_{\mathbb{N}}, [n, m]_{\mathbb{N}} := \{k \in [n, \infty)_{\mathbb{N}} \mid k \leq m\}$

1.4 Ordered Sets

Definition 1.4.1. Suppose a relation $<$ on S satisfies the following axioms:

- $\forall x, y, z \in S, x < y \wedge y < z \implies x < z$
- $\forall x, y \in S, x \neq y \implies x < y \oplus y < x$

Then

- The relation $<$ is called an **(total) order** on S
- The structure $(S, <)$ is called an **(totally) ordered set**

The notation

- $x > y$ denotes $y < x$
- $x \geq y$ denotes $x > y \vee x = y$
- $x \leq y$ denotes $y \geq x$

Remark. An ordered set $(S, <)$ is a **structure**, distinct from its **underlying set** S . For any structure $\mathcal{S}$, the notation $|\mathcal{S}|$ denotes its underlying set, not its **cardinality**. That is,

$$S = |(S, <)| \neq (S, <)$$

Example 1.4.2. $(\mathbb{N}, <)$, $(\mathbb{Z}, <)$, $(\mathbb{Q}, <)$ are ordered sets with

- $\forall n, m \in \mathbb{N}, m < n \iff \exists k \in \mathbb{N}, n = m + k$
- $\forall n, m \in \mathbb{Z}, m < n \iff \exists k \in \mathbb{Z}^+, n = m + k$
- $\forall p, q \in \mathbb{Q}, q < p \iff \exists r \in \mathbb{Q}^+, p = q + r$

where

$$\mathbb{Q}^+ := \{[(p, q)] \in \mathbb{Q} \mid p, q \in \mathbb{Z}^+\} \subseteq \mathbb{Q}$$

Throughout this textbook, the **number systems** use the **standard order relations** above, unless otherwise specified.

Definition 1.4.3. Suppose $E \subseteq |(S, <)|$. Then

- An element $m \in S$ is called a **lower bound** of E if

$$\forall x \in E, m \leq x$$

The set of all lower bounds of E is denoted by

$$E^\ell := \{m \in S \mid \forall x \in E, m \leq x\}$$

E is said to be **bounded below** if

$$E^\ell \neq \emptyset$$

- An element $M \in S$ is called an **upper bound** of E if

$$\forall x \in E, x \leq M$$

The set of all upper bounds of E is denoted by

$$E^u := \{M \in S \mid \forall x \in E, x \leq M\}$$

E is said to be **bounded above** if

$$E^u \neq \emptyset$$

Definition 1.4.4. Suppose $E \subseteq |(S, <)|$. Then

- An element $m \in E$ is called a **minimum** of E if

$$m \in E^\ell$$

Such an m is denoted by

$$m = \min E$$

- An element $M \in E$ is called a **maximum** of E if

$$M \in E^u$$

Such an M is denoted by

$$M = \max E$$

Theorem 1.4.5.

$$(A) \quad \forall E \subseteq \mathbb{Z}, E \neq \emptyset \wedge E^\ell \neq \emptyset \implies \exists m \in E, m = \min E$$

$$(B) \quad \forall E \subseteq \mathbb{Z}, E \neq \emptyset \wedge E^u \neq \emptyset \implies \exists M \in E, M = \max E$$

Proof. Suppose

- $E \subseteq \mathbb{Z}$
- $E \neq \emptyset$

Then

(A) Suppose $E^\ell \neq \emptyset$. Then

$$\exists k \in \mathbb{Z}, \forall \alpha \in E, k \leq \alpha$$

Define

$$F := \{\alpha - k + 1 \mid \alpha \in E\} \subseteq \mathbb{N}$$

By [theorem 1.1.4](#),

$$\begin{aligned} \exists n \in F, n &= \min F \\ &\Downarrow \\ \forall \alpha \in E, n &\leq \alpha - k + 1 \\ &\Downarrow \\ \forall \alpha \in E, n + k - 1 &\leq \alpha \end{aligned}$$

Therefore,

$$\begin{aligned} \exists m \in E, n &= m - k + 1 \\ &\Downarrow \\ m &= n + k - 1 = \min E \end{aligned}$$

(B) Suppose $E^u \neq \emptyset$. Then

$$\exists k \in \mathbb{Z}, \forall \alpha \in E, \alpha \leq k$$

Define

$$F := \{-\alpha + k + 1 \mid \alpha \in E\} \subseteq \mathbb{N}$$

By [theorem 1.1.4](#),

$$\begin{aligned} \exists n \in F, n &= \min F \\ &\Downarrow \\ \forall \alpha \in E, n &\leq -\alpha + k + 1 \\ &\Downarrow \\ \forall \alpha \in E, \alpha &\leq k - n + 1 \end{aligned}$$

Therefore,

$$\begin{aligned} \exists M \in E, n &= -M + k + 1 \\ &\Downarrow \\ M &= k - n + 1 = \max E \end{aligned}$$

□

Definition 1.4.6. Suppose $E \subseteq |(S, <)|$. Then

- Suppose $E^u \neq \emptyset$. An element $\alpha \in S$ is called a **supremum** of E if

$$\alpha = \min E^u$$

Such an α is denoted by

$$\alpha = \sup E$$

- Suppose $E^\ell \neq \emptyset$. An element $\alpha \in S$ is called an **infimum** of E if

$$\alpha = \max E^\ell$$

Such an α is denoted by

$$\alpha = \inf E$$

Definition 1.4.7. An ordered set $(S, <)$ is said to have the **least upper bound property** if

$$\forall E \subseteq S, E \neq \emptyset \wedge E^u \neq \emptyset \implies \exists \alpha \in S, \alpha = \sup E$$

Example 1.4.8. Define

- $S := \{p \in \mathbb{Q}^+ \mid p^2 < 2\} \subseteq \mathbb{Q}$
- $\forall p \in \mathbb{Q}^+, q := p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2}$

Then

- If $p \in S$, then

$$q^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2} < 0$$

$$\Downarrow$$

$$q \in S$$

$$\Downarrow$$

$$p \notin S^u$$

$$q > p$$

- Otherwise,

$$p^2 \geq 2$$

$$\Downarrow$$

$$p^2 > 2$$

$$\nexists p \in \mathbb{Q}^+, p^2 = 2$$

Then

$$q^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2} > 0$$

$$\Downarrow$$

$$q \in S^u$$

$$\Downarrow$$

$$p \neq \min S^u$$

$$q < p$$

Therefore, $(\mathbb{Q}, <)$ does not have the **least upper bound property**.

Theorem 1.4.9. Suppose $(S, <)$ have the **least upper bound property**. Then

$$\forall E \subseteq S, E \neq \emptyset \wedge E^\ell \neq \emptyset \implies \exists \alpha \in S, \alpha = \inf E$$

Proof. Suppose

- $E \subseteq S$
- $E \neq \emptyset$
- $E^\ell \neq \emptyset$

Then

$$\begin{aligned} & \exists x \in E \\ & \Downarrow \\ & \forall y \in E^\ell, y \leq x \\ & \Downarrow \\ & x \in (E^\ell)^u \\ & \Downarrow \\ & \exists \alpha \in S, \alpha = \sup E^\ell \end{aligned}$$

Assume $\alpha \notin E^\ell$. Then

$$\begin{aligned} & \exists \beta \in E, \beta < \alpha \\ & \Downarrow \\ & \beta \notin (E^\ell)^u \\ & \Downarrow \\ & \exists \gamma \in E^\ell, \beta < \gamma \\ & \Downarrow \\ & \gamma \leq \beta \\ & \Downarrow \\ & \perp \end{aligned} \quad \begin{aligned} & \alpha = \sup E^\ell \\ & \\ & \beta \in E \end{aligned}$$

Therefore,

$$\begin{aligned} & \alpha \in E^\ell \\ & \Downarrow \\ & \alpha = \max E^\ell \end{aligned} \quad \alpha = \sup E^\ell$$

□

1.5 Fields and Ordered Fields

Definition 1.5.1. Suppose two operations

- **Addition** $+$
- **Multiplication** $\cdot$

on F satisfy the following axioms:

(A) Axioms for **addition**

- $\forall x, y \in F, x + y \in F$
- $\forall x, y \in F, x + y = y + x$
- $\forall x, y, z \in F, (x + y) + z = x + (y + z)$
- $\exists \alpha \in F, \forall x \in F, x + \alpha = x$
- $\forall x \in F, \exists \beta \in F, x + \beta = \alpha$

(B) Axioms for **multiplication**

- $\forall x, y \in F, x \cdot y \in F$
- $\forall x, y \in F, x \cdot y = y \cdot x$
- $\forall x, y, z \in F, (x \cdot y) \cdot z = x \cdot (y \cdot z)$
- $\exists \gamma \in F, \gamma \neq \alpha \wedge (\forall x \in F, x \cdot \gamma = x)$
- $\forall x \in F, x \neq \alpha \implies \exists \omega \in F, x \cdot \omega = \gamma$

(C) The **distributive law**

- $\forall x, y, z \in F, x \cdot (y + z) = x \cdot y + x \cdot z$

Then the structure $(F, +, \cdot)$ is called a **field**.

Definition 1.5.2. Suppose

- $(F, +, \cdot)$
- $(F, <)$

satisfy the following axioms:

- $\forall x, y, z \in F, x < y \implies x + z < y + z$
- $\forall x, y \in F, x > 0 \wedge y > 0 \implies x \cdot y > 0$

Then the structure $(F, +, \cdot, <)$ is called an **ordered field**.

Remark. $(\mathbb{Q}, +, \cdot, <)$ is an ordered field. Also, there exists an ordered field $(\mathbb{R}, +, \cdot, <)$ which has the *least upper bound property*. In this textbook, the construction of $\mathbb{R}$ is not discussed, so we will just assume that $(\mathbb{R}, +, \cdot, <)$ exists and satisfies [axiom 1.6.1](#).

1.6 The Real Number System

Axiom 1.6.1. $(\mathbb{R}, +, \cdot, <)$ has the *least upper bound property*.

Theorem 1.6.2. $(\mathbb{R}, +, \cdot, <)$ has the **Archimedean property**, which states

$$\forall x \in \mathbb{R}^+, \forall y \in \mathbb{R}, \exists n \in \mathbb{N}, nx > y$$

where

$$\mathbb{R}^+ := \{x \in \mathbb{R} \mid x > 0\}$$

Proof. Suppose

- $x \in \mathbb{R}^+$
- $y \in \mathbb{R}$

Define

$$E := \{nx \mid nx \leq y, n \in \mathbb{N}\} \subseteq \mathbb{R}$$

Then

- If $E = \emptyset$, then

$$\begin{aligned} 1 \cdot x &\notin E \\ \Downarrow \\ 1 \cdot x &> y \end{aligned}$$

- Otherwise,

$$\begin{aligned} E &\neq \emptyset \wedge y \in E^u \\ \Downarrow \\ \exists \alpha \in \mathbb{R}, \alpha &= \sup E \\ \Downarrow \\ \alpha - x &\notin E^u \\ \Downarrow \\ \exists m \in \mathbb{N}, mx &> \alpha - x \\ \Downarrow \\ (m+1)x &> \alpha \\ \Downarrow \\ (m+1)x &\notin E \\ \Downarrow \\ (m+1)x &> y \end{aligned}$$

Axiom 1.6.1

$\alpha = \sup E$

□

Lemma 1.6.3.

$$\forall x \in [-1, +\infty), \forall n \in \mathbb{N}, (1+x)^n \geq 1+nx$$

This inequality is called **Bernoulli's inequality**.

Proof. Suppose $x \in [-1, +\infty)$. Define

$$\forall n \in \mathbb{N}, P(n) \iff (1+x)^n \geq 1+nx$$

Then

- **Base Case:**

$$\begin{aligned} (1+x)^1 &= 1+1 \cdot x \\ &\Downarrow \\ P(1) \end{aligned}$$

- **Inductive step:** Suppose $P(n)$. Then

$$\begin{aligned} (1+x)^n &\geq 1+nx \\ &\Downarrow \\ (1+x)^{n+1} &\geq (1+nx)(1+x) = 1+(n+1)x+nx^2 & 1+x \geq 0 \\ &\Downarrow & x^2 \geq 0 \\ (1+x)^{n+1} &\geq 1+(n+1)x \\ &\Downarrow \\ P(n+1) \end{aligned}$$

By [theorem 1.1.2](#),

$$\forall n \in \mathbb{N}, P(n)$$

□

Theorem 1.6.4.

$$\forall \varepsilon \in \mathbb{R}^+, \forall b \in (1, +\infty), \exists n \in \mathbb{N}, b^{-n} < \varepsilon$$

Proof. Suppose

- $\varepsilon \in \mathbb{R}^+$
- $b \in (1, +\infty)$

Then

$$\begin{aligned}
 & b - 1 \in \mathbb{R}^+ \\
 & \Downarrow \text{Theorem 1.6.2} \\
 & \exists n \in \mathbb{N}, n(b - 1) > \frac{1}{\varepsilon} \\
 & \Downarrow \text{Lemma 1.6.3} \\
 & (1 + (b - 1))^n \geq 1 + n(b - 1) > \frac{1}{\varepsilon} \\
 & \Downarrow \\
 & (1 + (b - 1))^{-n} < \varepsilon \\
 & \Downarrow \\
 & b^{-n} < \varepsilon
 \end{aligned}$$

□

Theorem 1.6.5.

$$(A) \quad \forall E \subseteq \mathbb{Z}, E \neq \emptyset \wedge (\exists \alpha \in \mathbb{R}, \forall x \in E, \alpha \leq x) \implies \exists m \in E, m = \min E$$

$$(B) \quad \forall E \subseteq \mathbb{Z}, E \neq \emptyset \wedge (\exists \beta \in \mathbb{R}, \forall x \in E, x \leq \beta) \implies \exists M \in E, M = \max E$$

Proof. Suppose

- $E \subseteq \mathbb{Z}$
- $E \neq \emptyset$

Then

(A) Suppose

$$\exists \alpha \in \mathbb{R}, \forall x \in E, \alpha \leq x$$

Then

- If $\alpha \geq 0$, then

$$\begin{aligned}
 & \forall x \in E, 0 \leq \alpha \leq x \\
 & \Downarrow \\
 & 0 \in E^\ell
 \end{aligned}$$

- Otherwise, by [theorem 1.6.2](#),

$$\begin{aligned}
 & \exists n \in \mathbb{N}, n > -\alpha \\
 & \Downarrow \\
 & \forall x \in E, -n < \alpha \leq x \\
 & \Downarrow \\
 & -n \in E^\ell
 \end{aligned}$$

Therefore,

$$\begin{aligned} E^\ell &\neq \emptyset \\ &\Downarrow \\ \exists m \in E, m &= \min E \end{aligned}$$

Theorem 1.4.5

(B) Suppose

$$\exists \beta \in \mathbb{R}, \forall x \in E, x \leq \beta$$

By theorem 1.6.2,

$$\begin{aligned} \exists n \in \mathbb{N}, n &> \beta \\ &\Downarrow \\ \forall x \in E, x &\leq \beta < n \\ &\Downarrow \\ n &\in E^u \\ &\Downarrow \\ \exists M \in E, M &= \max E \end{aligned}$$

Theorem 1.4.5

□

Definition 1.6.6. Suppose $x \in \mathbb{R}$.

- The **floor function** $\lfloor x \rfloor$ is defined by

$$\lfloor x \rfloor := \max\{k \in \mathbb{Z} \mid k \leq x\}$$

- The **ceiling function** $\lceil x \rceil$ is defined by

$$\lceil x \rceil := \min\{k \in \mathbb{Z} \mid k \geq x\}$$

By theorem 1.6.5,

$$\forall x \in \mathbb{R}, \exists k, m \in \mathbb{Z}, k = \lfloor x \rfloor \wedge m = \lceil x \rceil$$

Theorem 1.6.7. $\mathbb{Q}$ is *dense* in $\mathbb{R}$, which states

$$\forall x, y \in \mathbb{R}, x < y \implies \exists p \in \mathbb{Q}, x < p < y$$

Proof. Suppose

- $x, y \in \mathbb{R}$
- $x < y$

Define

$$\varepsilon := y - x > 0$$

By [theorem 1.6.2](#),

$$\exists n \in \mathbb{N}, \frac{1}{n} < \varepsilon$$

Define

$$S := \left\{ k \in \mathbb{Z} \mid \frac{k}{n} \leq x \right\} \subseteq \mathbb{Z}$$

Then

$$\begin{aligned} \lfloor x \rfloor &\leq x < \lfloor x \rfloor + 1 \\ &\Downarrow \\ n\lfloor x \rfloor &\leq nx < n\lfloor x \rfloor + n \\ &\Downarrow \\ n\lfloor x \rfloor &\in S \end{aligned}$$

Also,

$$\forall k \in S, k \leq nx$$

By [theorem 1.6.5](#),

$$\exists M \in \mathbb{Z}, M = \max S$$

Then

$$\begin{aligned} M &\in S \wedge M + 1 \notin S \\ &\Downarrow \\ \frac{M}{n} &\leq x < \frac{M + 1}{n} \\ &\Downarrow \\ x &< \frac{M + 1}{n} \leq x + \frac{1}{n} < x + \varepsilon = y \end{aligned}$$

□

Lemma 1.6.8.

$$\forall n \in \mathbb{N}, \forall a, b \in \mathbb{R}^+, a < b \implies b^n - a^n \leq (b - a)nb^{n-1}$$

Proof. Suppose

- $n \in \mathbb{N}$
- $a, b \in \mathbb{R}^+$
- $a < b$

Then

$$b^n - a^n = (b - a) \sum_{k=0}^{n-1} b^{n-1-k} a^k \leq (b - a) n b^{n-1}$$

□

Lemma 1.6.9.

$$\forall n \in \mathbb{N}, \forall a, b \in \mathbb{R}^+, a < b \iff a^n < b^n$$

Proof. Suppose

- $n \in \mathbb{N}$
- $a, b \in \mathbb{R}^+$

Then

$$\begin{aligned} b^n - a^n &= (b - a) \sum_{k=0}^{n-1} b^{n-1-k} a^k \\ &\Downarrow \\ a < b &\iff a^n < b^n \end{aligned}$$

□

Theorem 1.6.10.

$$\forall x \in \mathbb{R}^+, \forall n \in \mathbb{N}, \exists! y \in \mathbb{R}^+, y^n = x$$

Proof. Suppose

- $x \in \mathbb{R}^+$
- $n \in \mathbb{N}$

Define

$$E := \{t \in \mathbb{R}^+ \mid t^n < x\} \subseteq \mathbb{R}^+$$

Then

- If $x \geq 1$, then

$$\begin{aligned} 2^{-n} &< 1 \leq x \\ &\Downarrow \\ 2^{-1} &\in E \end{aligned}$$

Also,

$$\begin{aligned} \forall t \in E, t^n &< x \leq x^n \\ &\Downarrow \\ \forall t \in E, t &< x \\ &\Downarrow \\ x &\in E^u \end{aligned}$$

Lemma 1.6.9

- Otherwise,

$$\begin{aligned} x &< 1 \\ &\Downarrow \\ x^{2n} &< x \\ &\Downarrow \\ x^2 &\in E \end{aligned}$$

Also,

$$\begin{aligned} \forall t \in E, t^n &< x < 1 = 1^n \\ &\Downarrow \\ \forall t \in E, t &< 1 \\ &\Downarrow \\ 1 &\in E^u \end{aligned}$$

Lemma 1.6.9

Therefore,

$$\begin{aligned} E &\neq \emptyset \wedge E^u \neq \emptyset \\ &\Downarrow \\ \exists \alpha \in \mathbb{R}^+, \alpha &= \sup E \end{aligned}$$

Axiom 1.6.1

We will show

$$\begin{aligned} \alpha^n &\not< x \wedge \alpha^n \not> x \\ &\Downarrow \\ \alpha^n &= x \end{aligned}$$

- Assume $\alpha^n < x$. By [theorem 1.6.7](#),

$$\begin{aligned} \exists h \in \mathbb{Q}, 0 < h &< \min \left\{ 1, \frac{x - \alpha^n}{n(\alpha + 1)^{n-1}} \right\} \\ &\Downarrow \\ (\alpha + h)^n - \alpha^n &\leq hn(\alpha + h)^{n-1} < hn(\alpha + 1)^{n-1} \\ &\Downarrow \\ (\alpha + h)^n &< \alpha^n + hn(\alpha + 1)^{n-1} < \alpha^n + (x - \alpha^n) = x \\ &\Downarrow \\ \alpha + h &\in E \\ &\Downarrow \\ &\perp \end{aligned}$$

Lemma 1.6.8

$\alpha = \sup E$

- Assume $\alpha^n > x$. By [theorem 1.6.7](#),

$$\begin{aligned}
 & \exists h \in \mathbb{Q}, 0 < h < \min \left\{ \alpha, \frac{\alpha^n - x}{n\alpha^{n-1}} \right\} \\
 & \quad \downarrow \\
 & \alpha^n - (\alpha - h)^n \leq hn\alpha^{n-1} && \text{Lemma 1.6.8} \\
 & \quad \downarrow \\
 & (\alpha - h)^n \geq \alpha^n - hn\alpha^{n-1} > \alpha^n - (\alpha^n - x) = x \\
 & \quad \downarrow \\
 & \forall t \in E, t^n < x < (\alpha - h)^n \\
 & \quad \downarrow \\
 & \forall t \in E, t < \alpha - h && \text{Lemma 1.6.9} \\
 & \quad \downarrow \\
 & \alpha - h \in E^u \\
 & \quad \downarrow \\
 & \perp && \alpha = \sup E
 \end{aligned}$$

This implies

$$\exists y \in \mathbb{R}^+, y^n = x$$

Suppose

- $y_1, y_2 \in \mathbb{R}^+$
- $(y_1)^n = (y_2)^n = x$

Assume $y_1 \neq y_2$ (WLOG $y_1 < y_2$). Then

$$\begin{aligned}
 & y_1 < y_2 \\
 & \quad \downarrow \\
 & (y_1)^n < (y_2)^n && \text{Lemma 1.6.9} \\
 & \quad \downarrow \\
 & \perp
 \end{aligned}$$

Therefore,

$$y_1 = y_2$$

This implies

$$\exists! y \in \mathbb{R}^+, y^n = x$$

□

Corollary 1.6.11.

$$\forall x \in [0, \infty), \forall n \in \mathbb{N}, \exists! y \in [0, \infty), y^n = x$$

Such an y is denoted by

$$y = \sqrt[n]{x} = x^{\frac{1}{n}}$$

Proof. Suppose

- $x \in [0, \infty)$
- $n \in \mathbb{N}$

Then

- If $x = 0$, then
 - $0^n = 0$
 - $\forall y \in [0, \infty), y \neq 0 \implies y^n \neq 0$
- Otherwise,

$$\begin{array}{c} x \in \mathbb{R}^+ \\ \Downarrow \\ \exists! y \in \mathbb{R}^+, y^n = x \end{array}$$

Theorem 1.6.10

□

Definition 1.6.12. Suppose

- $y \in [0, 1)$
- $b \in [2, \infty)_{\mathbb{N}}$

A sequence $(a_n)_{n=1}^{\infty}$ in $[0, b-1]_{\mathbb{N}}$ is called a **greedy base b expansion** of y if

$$\forall n \in \mathbb{N}, 0 \leq y - \sum_{k=1}^n \frac{a_k}{b^k} < \frac{1}{b^n}$$

Such an $(a_n)_{n=1}^{\infty}$ is denoted by

$$(a_n)_{n=1}^{\infty} = \mathcal{G}_b(y)$$

Theorem 1.6.13.

$$\forall b \in [2, \infty)_{\mathbb{N}}, \forall y \in [0, 1), \exists! (a_n)_{n=1}^{\infty} \in \prod_{n=1}^{\infty} [0, b-1]_{\mathbb{N}}, (a_n)_{n=1}^{\infty} = \mathcal{G}_b(y)$$

Proof. Suppose

- $b \in [2, \infty)_{\mathbb{N}}$

- $y \in [0, 1)$

Define

$$\begin{array}{lll}
 t_1 := by & T_1 := \{t \in [0, b-1]_{\mathbb{N}} \mid t \leq t_1\} & a_1 := \max T_1 \\
 t_2 := b^2y - ba_1 & T_2 := \{t \in [0, b-1]_{\mathbb{N}} \mid t \leq t_2\} & a_2 := \max T_2 \\
 \vdots & \vdots & \vdots \\
 t_n := b^ny - \sum_{k=1}^{n-1} a_k b^{n-k} & T_n := \{t \in [0, b-1]_{\mathbb{N}} \mid t \leq t_n\} & a_n := \max T_n \\
 \vdots & \vdots & \vdots
 \end{array}$$

Define

$$\forall n \in \mathbb{N}, P(n) \iff \forall k \in [1, n]_{\mathbb{N}}, 0 \leq t_k < b$$

Then

- **Base case:**

$$\begin{array}{c}
 0 \leq y < 1 \\
 \Downarrow \\
 0 \leq t_1 = by < b \\
 \Downarrow \\
 P(1)
 \end{array}$$

- **Inductive step:** Suppose $P(n)$. Then

$$\begin{array}{c}
 \forall k \in [1, n]_{\mathbb{N}}, 0 \in T_k \wedge b-1 \in (T_k)^u \\
 \Downarrow \\
 \forall k \in [1, n]_{\mathbb{N}}, \exists M \in [0, b-1]_{\mathbb{N}}, M = \max T_k
 \end{array}$$

Theorem 1.4.5

Therefore, $(a_k)_{k=1}^n$ is well-defined. Then

$$\begin{aligned}
 t_{n+1} &= b^{n+1}y - \sum_{k=1}^n a_k b^{n+1-k} \\
 &= b \left(b^ny - \sum_{k=1}^n a_k b^{n-k} \right) \\
 &= b \left(b^ny - \sum_{k=1}^{n-1} a_k b^{n-k} - a_n \right) \\
 &= b(t_n - a_n)
 \end{aligned}$$

Since $P(n)$,

$$\begin{aligned}
 a_n &\leq t_n < a_n + 1 \\
 &\Downarrow \\
 0 &\leq t_n - a_n < 1 \\
 &\Downarrow \\
 0 &\leq t_{n+1} = b(t_n - a_n) < b \\
 &\Downarrow \\
 &P(n+1)
 \end{aligned}$$

By [theorem 1.1.2](#),

$$\forall n \in \mathbb{N}, P(n)$$

Therefore, $(t_n)_{n=1}^\infty$ is well-defined. Then

$$\begin{aligned}
 \forall n \in \mathbb{N}, 0 &\in T_n \wedge b-1 \in (T_n)^u \\
 &\Downarrow \\
 \forall n \in \mathbb{N}, \exists M &\in [0, b-1]_{\mathbb{N}}, M = \max T_n
 \end{aligned}$$

[Theorem 1.4.5](#)

Therefore, $(a_n)_{n=1}^\infty$ is well-defined. Then

$$\begin{aligned}
 \forall n \in \mathbb{N}, a_n &\leq b^n y - \sum_{k=1}^{n-1} a_k b^{n-k} < a_n + 1 \\
 &\Downarrow \\
 \forall n \in \mathbb{N}, 0 &\leq b^n y - \sum_{k=1}^n a_k b^{n-k} < 1 \\
 &\Downarrow \\
 \forall n \in \mathbb{N}, 0 &\leq y - \sum_{k=1}^n \frac{a_k}{b^k} < \frac{1}{b^n}
 \end{aligned}$$

This implies

$$\exists (a_n)_{n=1}^\infty \in \prod_{n=1}^\infty [0, b-1]_{\mathbb{N}}, (a_n)_{n=1}^\infty = \mathcal{G}_b(y)$$

Suppose $(a_n)_{n=1}^\infty = (c_n)_{n=1}^\infty = \mathcal{G}_b(y)$. Define

$$\forall n \in \mathbb{N}, Q(n) \iff \forall k \in [1, n]_{\mathbb{N}}, a_k = c_k$$

Then

• **Base case:** By [definition 1.6.12](#),

$$\begin{array}{ccc}
 0 \leq y - \frac{a_1}{b} < \frac{1}{b} & & 0 \leq y - \frac{c_1}{b} < \frac{1}{b} \\
 \Downarrow & & \Downarrow \\
 \frac{a_1}{b} \leq y < \frac{a_1+1}{b} & & \frac{c_1}{b} \leq y < \frac{c_1+1}{b}
 \end{array}$$

Then

$$\begin{aligned} \frac{a_1}{b} \leq y < \frac{c_1 + 1}{b} \\ \Downarrow \\ a_1 < c_1 + 1 \\ \Downarrow \\ a_1 \leq c_1 \end{aligned}$$

$$\begin{aligned} \frac{c_1}{b} \leq y < \frac{a_1 + 1}{b} \\ \Downarrow \\ c_1 < a_1 + 1 \\ \Downarrow \\ c_1 \leq a_1 \end{aligned}$$

Therefore,

$$\begin{aligned} a_1 &= c_1 \\ \Downarrow \\ Q(1) \end{aligned}$$

• **Inductive step:** Suppose $Q(n)$. By [definition 1.6.12](#),

$$\begin{aligned} 0 \leq y - \sum_{k=1}^{n+1} \frac{a_k}{b^k} < \frac{1}{b^{n+1}} \\ \Downarrow \\ \sum_{k=1}^{n+1} \frac{a_k}{b^k} \leq y < \sum_{k=1}^{n+1} \frac{a_k}{b^k} + \frac{1}{b^{n+1}} \end{aligned}$$

$$\begin{aligned} 0 \leq y - \sum_{k=1}^{n+1} \frac{c_k}{b^k} < \frac{1}{b^{n+1}} \\ \Downarrow \\ \sum_{k=1}^{n+1} \frac{c_k}{b^k} \leq y < \sum_{k=1}^{n+1} \frac{c_k}{b^k} + \frac{1}{b^{n+1}} \end{aligned}$$

Then

$$\begin{aligned} \sum_{k=1}^{n+1} \frac{a_k}{b^k} \leq y < \sum_{k=1}^{n+1} \frac{c_k}{b^k} + \frac{1}{b^{n+1}} \\ \Downarrow \\ \sum_{k=1}^{n+1} \frac{a_k - c_k}{b^k} < \frac{1}{b^{n+1}} \end{aligned}$$

$$\begin{aligned} \sum_{k=1}^{n+1} \frac{c_k}{b^k} \leq y < \sum_{k=1}^{n+1} \frac{a_k}{b^k} + \frac{1}{b^{n+1}} \\ \Downarrow \\ \sum_{k=1}^{n+1} \frac{c_k - a_k}{b^k} < \frac{1}{b^{n+1}} \end{aligned}$$

Since $Q(n)$,

$$\begin{aligned} \frac{a_{n+1} - c_{n+1}}{b^{n+1}} < \frac{1}{b^{n+1}} \\ \Downarrow \\ a_{n+1} - c_{n+1} < 1 \\ \Downarrow \\ a_{n+1} \leq c_{n+1} \end{aligned}$$

$$\begin{aligned} \frac{c_{n+1} - a_{n+1}}{b^{n+1}} < \frac{1}{b^{n+1}} \\ \Downarrow \\ c_{n+1} - a_{n+1} < 1 \\ \Downarrow \\ c_{n+1} \leq a_{n+1} \end{aligned}$$

Therefore,

$$\begin{aligned} a_{n+1} &= c_{n+1} \\ \Downarrow \\ Q(n+1) \end{aligned}$$

By [theorem 1.1.2](#),

$$\forall n \in \mathbb{N}, Q(n)$$

This implies

$$\exists! (a_n)_{n=1}^{\infty} \in \prod_{n=1}^{\infty} [0, b-1]_{\mathbb{N}}, (a_n)_{n=1}^{\infty} = \mathcal{G}_b(y)$$

□

Theorem 1.6.14.

$$\forall b \in [2, \infty)_{\mathbb{N}}, \forall a, c \in [0, 1), \mathcal{G}_b(a) = \mathcal{G}_b(c) \implies a = c$$

Proof. Suppose

- $b \in [2, \infty)_{\mathbb{N}}$
- $a, c \in [0, 1)$
- $a \neq c$ (*WLOG* $a < c$)

By [theorem 1.6.4](#),

$$\exists n \in \mathbb{N}, \frac{1}{b^n} < c - a$$

By [theorem 1.6.13](#), let

- $\{a_n\}_{n=1}^{\infty} = \mathcal{G}_b(a)$
- $\{c_n\}_{n=1}^{\infty} = \mathcal{G}_b(c)$

By [definition 1.6.12](#),

$$\begin{array}{ccc} 0 \leq a - \sum_{k=1}^n \frac{a_k}{b^k} < \frac{1}{b^n} & & 0 \leq c - \sum_{k=1}^n \frac{c_k}{b^k} < \frac{1}{b^n} \\ \Downarrow & & \Downarrow \\ \sum_{k=1}^n \frac{a_k}{b^k} + \frac{1}{b^n} \leq a + \frac{1}{b^n} & & c < \sum_{k=1}^n \frac{c_k}{b^k} + \frac{1}{b^n} \end{array}$$

Therefore,

$$\begin{array}{ccc} \sum_{k=1}^n \frac{a_k}{b^k} + \frac{1}{b^n} \leq a + \frac{1}{b^n} < c < \sum_{k=1}^n \frac{c_k}{b^k} + \frac{1}{b^n} \\ \Downarrow & & \\ \sum_{k=1}^n \frac{a_k}{b^k} < \sum_{k=1}^n \frac{c_k}{b^k} \\ \Downarrow & & \\ \mathcal{G}_b(a) \neq \mathcal{G}_b(c) \end{array}$$

□

Theorem 1.6.15.

$$\begin{aligned} & \forall b \in [2, \infty)_{\mathbb{N}}, \forall (a_n)_{n=1}^{\infty} \in \prod_{n=1}^{\infty} [0, b-1]_{\mathbb{N}}, \\ & (\nexists y \in [0, 1), (a_n)_{n=1}^{\infty} = \mathcal{G}_b(y)) \implies \exists N \in \mathbb{N}, \forall n \in [N, \infty)_{\mathbb{N}}, a_n = b-1 \end{aligned}$$

Proof. Suppose

- $b \in [2, \infty)_{\mathbb{N}}$
- $(a_n)_{n=1}^{\infty} \in \prod_{n=1}^{\infty} [0, b-1]_{\mathbb{N}}$
- $\forall N \in \mathbb{N}, \exists n \in [N, \infty)_{\mathbb{N}}, a_n \neq b-1$

Define

$$\forall n \in \mathbb{N}, x_n := \sum_{k=1}^n \frac{a_k}{b^k}$$

Then

- Suppose
 - $n, m \in \mathbb{N}$
 - $n \leq m$

Then

$$\begin{aligned} & \forall k \in \mathbb{N}, a_k \geq 0 \\ & \Downarrow \\ & x_n \leq x_m \end{aligned}$$

- Suppose
 - $n, m \in \mathbb{N}$
 - $n > m$

Then

$$\begin{aligned} x_n &= x_m + \sum_{k=m+1}^n \frac{a_k}{b^k} \leq x_m + \sum_{k=m+1}^n \frac{b-1}{b^k} \\ & \Downarrow \\ x_n &\leq x_m + \frac{1}{b^m} - \frac{1}{b^n} \end{aligned}$$

Let $N = 1$. Then

$$\begin{aligned} \exists m \in \mathbb{N}, a_m \neq b - 1 \\ \Downarrow \\ a_m + 1 \leq b - 1 \end{aligned}$$

This implies

$$\begin{aligned} x_m + \frac{1}{b^m} &= \sum_{k=1}^{m-1} \frac{a_k}{b^k} + \frac{a_m + 1}{b^m} \\ &\leq \sum_{k=1}^{m-1} \frac{a_k}{b^k} + \frac{b - 1}{b^m} \\ &\leq \sum_{k=1}^m \frac{b - 1}{b^k} = 1 - \frac{1}{b^m} \end{aligned}$$

Then

- $\forall n \in [1, m]_{\mathbb{N}}, x_n \leq x_m$
- $\forall n \in [m + 1, \infty)_{\mathbb{N}}, x_n < x_m + \frac{1}{b^m} \leq 1 - \frac{1}{b^m}$

Therefore,

$$\begin{aligned} 1 - \frac{1}{b^m} &\in (\{x_n\}_{n=1}^{\infty})^u \\ &\Downarrow \\ \exists y \in \mathbb{R}, y &= \sup\{x_n\}_{n=1}^{\infty} \\ &\Downarrow \\ y &\leq 1 - \frac{1}{b^m} < 1 \\ &\Downarrow \\ y &\in [0, 1) \end{aligned}$$

Axiom 1.6.1

$$0 \leq x_1 \leq y$$

Suppose $k \in \mathbb{N}$. Then

$$\begin{aligned} \exists m \in [k + 1, \infty)_{\mathbb{N}}, a_m \neq b - 1 \\ \Downarrow \\ x_m = x_k + \sum_{n=k+1}^m \frac{a_n}{b^n} < x_k + \sum_{n=k+1}^m \frac{b - 1}{b^n} \\ \Downarrow \\ x_m < x_k + \frac{1}{b^k} - \frac{1}{b^m} \\ \Downarrow \\ x_m + \frac{1}{b^m} < x_k + \frac{1}{b^k} \end{aligned}$$

Therefore,

$$\begin{aligned}
 x_m + \frac{1}{b^m} &\in (\{x_n\}_{n=1}^\infty)^u \\
 &\Downarrow \\
 x_k &\leq y \leq x_m + \frac{1}{b^m} \\
 &\Downarrow \\
 0 &\leq y - x_k < \frac{1}{b^k}
 \end{aligned}
 \qquad y = \sup\{x_n\}_{n=1}^\infty$$

Since $k \in \mathbb{N}$ was arbitrary,

$$(a_n)_{n=1}^\infty = \mathcal{G}_b(y)$$

□

Theorem 1.6.16.

$$\begin{aligned}
 &\forall b \in [2, \infty)_{\mathbb{N}}, \forall (a_n)_{n=1}^\infty \in \prod_{n=1}^\infty [0, b-1]_{\mathbb{N}}, \\
 &(\exists N \in \mathbb{N}, \forall n \in [N, \infty)_{\mathbb{N}}, a_n = b-1) \implies \nexists y \in [0, 1), (a_n)_{n=1}^\infty = \mathcal{G}_b(y)
 \end{aligned}$$

Proof. Suppose

- $b \in [2, \infty)_{\mathbb{N}}$
- $(a_n)_{n=1}^\infty \in \prod_{n=1}^\infty [0, b-1]_{\mathbb{N}}$
- $\exists N \in \mathbb{N}, \forall n \in [N, \infty)_{\mathbb{N}}, a_n = b-1$

Assume

$$\exists y \in [0, 1), (a_n)_{n=1}^\infty = \mathcal{G}_b(y)$$

Define

$$\forall n \in \mathbb{N}, x_n := \sum_{k=1}^n \frac{a_k}{b^k}$$

Then

$$\begin{aligned}
 \forall n \in [N+1, \infty)_{\mathbb{N}}, x_n &= x_N + \sum_{k=N+1}^n \frac{b-1}{b^k} = x_N + \frac{1}{b^N} - \frac{1}{b^n} \\
 &\Downarrow \\
 \forall n \in [N+1, \infty)_{\mathbb{N}}, y - x_n &= y - x_N - \frac{1}{b^N} + \frac{1}{b^n} \geq 0 \\
 &\Downarrow \\
 \forall n \in [N+1, \infty)_{\mathbb{N}}, \frac{1}{b^n} &\geq x_N + \frac{1}{b^N} - y
 \end{aligned}$$

By [definition 1.6.12](#),

$$\begin{aligned}
 0 &\leq y - x_N < \frac{1}{b^N} \\
 &\Downarrow \\
 x_N + \frac{1}{b^N} - y &> 0 \\
 &\Downarrow \\
 \exists m \in \mathbb{N}, \frac{1}{b^m} &< x_N + \frac{1}{b^N} - y && \text{Theorem 1.6.4} \\
 &\Downarrow \\
 \forall n \in [\max\{m, N\} + 1, \infty)_{\mathbb{N}}, \frac{1}{b^n} &< \frac{1}{b^m} < x_N + \frac{1}{b^N} - y \\
 &\Downarrow \\
 &\perp
 \end{aligned}$$

Therefore,

$$\nexists y \in [0, 1), (a_n)_{n=1}^{\infty} = \mathcal{G}_b(y)$$

□

1.7 The Extended Real Number System

Definition 1.7.1. The extended real number system $\overline{\mathbb{R}}$ is defined by

$$\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, +\infty\}$$

$(\overline{\mathbb{R}}, <)$ is an ordered set with

$$\forall x \in \mathbb{R}, -\infty < x < +\infty$$

Definition 1.7.2. Operations on $\overline{\mathbb{R}}$ are defined by

- $\forall x \in (-\infty, +\infty], x + (+\infty) = (+\infty) + x := +\infty$
- $\forall x \in [-\infty, +\infty), x + (-\infty) = (-\infty) + x := -\infty$
- $\forall x \in (0, +\infty], x \cdot (\pm\infty) = (\pm\infty) \cdot x := \pm\infty$
- $\forall x \in [-\infty, 0), x \cdot (\pm\infty) = (\pm\infty) \cdot x := \mp\infty$
- $\forall x \in \mathbb{R}, \frac{x}{\pm\infty} := 0$

Undefined operations in $\overline{\mathbb{R}}$ are

- $(\pm\infty) + (\mp\infty)$
- $0 \cdot (\pm\infty)$
- $\frac{(\pm\infty)}{(\pm\infty)}, \frac{(\pm\infty)}{(\mp\infty)}$
- $\forall x \in \overline{\mathbb{R}}, \frac{x}{0}$

Theorem 1.7.3.

(A) $\forall E \subseteq \overline{\mathbb{R}}, \exists x \in \overline{\mathbb{R}}, x = \sup E$

(B) $\forall E \subseteq \overline{\mathbb{R}}, \exists y \in \overline{\mathbb{R}}, y = \inf E$

Proof. Suppose $E \subseteq \overline{\mathbb{R}}$. Then

- If $E = \emptyset$, then
 - (A)** $\sup E = -\infty$
 - (B)** $\inf E = +\infty$
- If $E = \{-\infty\}$, then
 - (A)** $\sup E = -\infty$
 - (B)** $\inf E = -\infty$
- If $E = \{+\infty\}$, then
 - (A)** $\sup E = +\infty$
 - (B)** $\inf E = +\infty$
- Otherwise,

(A) – If

$$\exists \beta \in \mathbb{R}, \forall x \in E, x \leq \beta$$

then

$$\begin{aligned}
 E \setminus \{-\infty\} &\subseteq \mathbb{R} \\
 \Downarrow \\
 \exists x \in \mathbb{R}, x &= \sup E \setminus \{-\infty\} \\
 \Downarrow \\
 x &= \sup E
 \end{aligned}$$

Axiom 1.6.1

– Otherwise,

$$\sup E = +\infty$$

(B) – If

$$\exists \alpha \in \mathbb{R}, \forall x \in E, \alpha \leq x$$

then

$$E \setminus \{+\infty\} \subseteq \mathbb{R}$$

$$\Downarrow$$

$$\exists x \in \mathbb{R}, x = \inf E \setminus \{+\infty\}$$

$$\Downarrow$$

$$x = \inf E$$

– Otherwise,

$$\inf E = -\infty$$

Theorem 1.4.9

□

1.8 The Complex Number System

Definition 1.8.1. The complex number system $\mathbb{C}$ is defined by

$$\mathbb{C} := \{(x, y) \mid x, y \in \mathbb{R}\}$$

Definition 1.8.2. The operations on $\mathbb{C}$ are defined by

- $\forall (x, y), (u, v) \in \mathbb{C}, (x, y) + (u, v) := (x + u, y + v)$
- $\forall (x, y), (u, v) \in \mathbb{C}, (x, y) \cdot (u, v) := (xu - yv, xv + yu)$

Remark. One can verify that $(\mathbb{C}, +, \cdot)$ is a field with

- The **additive identity** is $(0, 0)$
- The **multiplicative identity** is $(1, 0)$
- The **additive inverse** of (x, y) is $(-x, -y)$
- The **multiplicative inverse** of $(x, y) \neq (0, 0)$ is

$$\left(\frac{x}{x^2 + y^2}, -\frac{y}{x^2 + y^2} \right)$$

Definition 1.8.3. The **imaginary unit** i is defined by

$$i := (0, 1) \in \mathbb{C}$$

Remark. There is a **canonical embedding**

$$\mathbb{R} \rightarrow \mathbb{C} \text{ such that } x \mapsto (x, 0)$$

Therefore, we identify $\mathbb{R}$ with its image in $\mathbb{C}$ and write

$$\mathbb{R} \subseteq \mathbb{C}$$

Then

$$\begin{aligned} \forall (x, y) \in \mathbb{C}, (x, y) &= (x, 0) + (0, y) \\ &= (x, 0) + (y, 0) \cdot (0, 1) \\ &= x + yi \end{aligned}$$

Definition 1.8.4. Suppose $z = (x, y) \in \mathbb{C}$.

- The **real part** of z is defined by

$$\begin{aligned} - \Re(z) &:= x \\ - \operatorname{Re}(z) &:= x \end{aligned}$$

- The **imaginary part** of z is defined by

$$\begin{aligned} - \Im(z) &:= y \\ - \operatorname{Im}(z) &:= y \end{aligned}$$

- The **complex conjugate** of z is defined by

$$\bar{z} := (x, -y) = x - yi$$

Definition 1.8.5. Suppose $z = (x, y) \in \mathbb{C}$. The **modulus** of z is defined by

$$|z| := \sqrt{z\bar{z}} = \sqrt{x^2 + y^2} \in [0, \infty)$$

By [corollary 1.6.11](#),

$$\forall z \in \mathbb{C}, \exists! r \in [0, \infty), r = |z|$$

Theorem 1.8.6.

$$\forall n \in \mathbb{N}, \forall \{a_k\}_{k=1}^n, \{b_k\}_{k=1}^n \subseteq \mathcal{P}(\mathbb{C}), \left| \sum_{k=1}^n a_k \overline{b_k} \right|^2 \leq \sum_{k=1}^n |a_k|^2 \cdot \sum_{k=1}^n |b_k|^2$$

This theorem is called the **Cauchy-Schwarz inequality**.

Proof. Suppose

- $n \in \mathbb{N}$
- $\{a_k\}_{k=1}^n, \{b_k\}_{k=1}^n \subseteq \mathcal{P}(\mathbb{C})$

Then

$$\begin{aligned} \forall i, j \in [1, n]_{\mathbb{N}}, |a_i b_j - a_j b_i|^2 &= (a_i b_j - a_j b_i) \overline{(a_i b_j - a_j b_i)} \\ &= |a_i|^2 |b_j|^2 + |a_j|^2 |b_i|^2 - (a_i \overline{b_i} \cdot \overline{a_j} b_j + \overline{a_i} b_i \cdot a_j \overline{b_j}) \end{aligned}$$

Consider

$$\begin{aligned} \sum_{i < j} (|a_i|^2 |b_j|^2 + |a_j|^2 |b_i|^2) &= \sum_{i < j} (|a_i|^2 |b_j|^2) + \sum_{j < i} (|a_i|^2 |b_j|^2) \\ &= \sum_{i \neq j} (|a_i|^2 |b_j|^2) \\ &= \sum_{k=1}^n |a_k|^2 \cdot \sum_{k=1}^n |b_k|^2 - \sum_{k=1}^n |a_k|^2 |b_k|^2 \end{aligned}$$

Also,

$$\begin{aligned} \sum_{i < j} (a_i \overline{b_i} \cdot \overline{a_j} b_j + \overline{a_i} b_i \cdot a_j \overline{b_j}) &= \sum_{i \neq j} (a_i \overline{b_i} \cdot \overline{a_j} b_j) \\ &= \sum_{i=1}^n a_i \overline{b_i} \cdot \sum_{j=1}^n \overline{a_j} b_j - \sum_{k=1}^n a_k \overline{b_k} \cdot \overline{a_k} b_k \\ &= \left| \sum_{k=1}^n a_k \overline{b_k} \right|^2 - \sum_{k=1}^n |a_k|^2 |b_k|^2 \end{aligned}$$

Therefore,

$$\sum_{i < j} |a_i b_j - a_j b_i|^2 = \sum_{k=1}^n |a_k|^2 \cdot \sum_{k=1}^n |b_k|^2 - \left| \sum_{k=1}^n a_k \overline{b_k} \right|^2 \geq 0$$

□

Corollary 1.8.7.

$$\forall n \in \mathbb{N}, \forall \{a_k\}_{k=1}^n, \{b_k\}_{k=1}^n \subseteq \mathcal{P}(\mathbb{R}), \left(\sum_{k=1}^n a_k b_k \right)^2 \leq \sum_{k=1}^n a_k^2 \cdot \sum_{k=1}^n b_k^2$$

Theorem 1.8.8.

$$(A) \quad \forall z, w \in \mathbb{C}, |zw| = |z| \cdot |w|$$

$$(B) \quad \forall z, w \in \mathbb{C}, |z + w| \leq |z| + |w|$$

$$(C) \quad \forall z, w \in \mathbb{C}, ||z| - |w|| \leq |z - w|$$

Proof. Suppose

$$\bullet \quad z = (x, y) \in \mathbb{C}$$

$$\bullet \quad w = (u, v) \in \mathbb{C}$$

Then

$$(A) \quad |zw|^2 = (xu - yv)^2 + (xv + yu)^2 = (x^2 + y^2)(u^2 + v^2) = |z|^2 \cdot |w|^2$$

(B) By [corollary 1.8.7](#),

$$\begin{aligned} |z + w|^2 &= (x + u)^2 + (y + v)^2 \\ &= x^2 + y^2 + u^2 + v^2 + 2(xu + yv) \\ &\leq x^2 + y^2 + u^2 + v^2 + 2\sqrt{x^2 + y^2} \cdot \sqrt{u^2 + v^2} \\ &= (|z| + |w|)^2 \end{aligned}$$

(C) By [corollary 1.8.7](#),

$$\begin{aligned} |z - w|^2 &= (x - u)^2 + (y - v)^2 \\ &= x^2 + y^2 + u^2 + v^2 - 2(xu + yv) \\ &\geq x^2 + y^2 + u^2 + v^2 - 2\sqrt{x^2 + y^2} \cdot \sqrt{u^2 + v^2} \\ &= (|z| - |w|)^2 \end{aligned}$$

□

1.9 Euclidean Spaces

Definition 1.9.1. Suppose $n \in \mathbb{N}$. The **Euclidean n -space** $\mathbb{R}^n$ is defined by

$$\mathbb{R}^n := \{\mathbf{x} = (x_1, x_2, \dots, x_n) \mid \forall k \in [1, n]_{\mathbb{N}}, x_k \in \mathbb{R}\}$$

$x_1, x_2, \dots, x_n$ are called the **coordinates** of $\mathbf{x}$. The elements of $\mathbb{R}^n$ for $n > 1$ are called **points** and are represented by bold symbols.

Definition 1.9.2. The operations on $\mathbb{R}^n$ are defined by

- $\forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n, \mathbf{x} + \mathbf{y} := (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n)$
- $\forall \alpha \in \mathbb{R}, \forall \mathbf{x} \in \mathbb{R}^n, \alpha \mathbf{x} := (\alpha x_1, \alpha x_2, \dots, \alpha x_n)$

Definition 1.9.3. Suppose $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.

- The **inner product** of $\mathbf{x}$ and $\mathbf{y}$ is defined by

$$\mathbf{x} \cdot \mathbf{y} := \sum_{k=1}^n x_k y_k$$

- The **norm** of $\mathbf{x}$ is defined by

$$\|\mathbf{x}\| := \sqrt{\mathbf{x} \cdot \mathbf{x}} = \sqrt{\sum_{k=1}^n x_k^2}$$

Theorem 1.9.4.

- (A) $\forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n, |\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\| \cdot \|\mathbf{y}\|$
- (B) $\forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n, \|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$
- (C) $\forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n, \|\mathbf{x} - \mathbf{y}\| \geq \|\mathbf{x}\| - \|\mathbf{y}\|$

Proof. Suppose $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. Then

(A) By [corollary 1.8.7](#),

$$|\mathbf{x} \cdot \mathbf{y}|^2 = \left| \sum_{k=1}^n x_k y_k \right|^2 \leq \sum_{k=1}^n x_k^2 \cdot \sum_{k=1}^n y_k^2 = \|\mathbf{x}\|^2 \cdot \|\mathbf{y}\|^2$$

(B) By part (A),

$$\begin{aligned}
 \|x + y\|^2 &= (x + y) \cdot (x + y) \\
 &= \|x\|^2 + 2x \cdot y + \|y\|^2 \\
 &\leq \|x\|^2 + 2|x \cdot y| + \|y\|^2 \\
 &\leq \|x\|^2 + 2\|x\| \cdot \|y\| + \|y\|^2 \\
 &= (\|x\| + \|y\|)^2
 \end{aligned}$$

(C) By part (B),

- If $\|x\| \geq \|y\|$, then

$$\begin{aligned}
 \|x\| &= \|(x - y) + y\| \leq \|x - y\| + \|y\| \\
 &\quad \downarrow \\
 \|x\| - \|y\| &\leq \|x - y\|
 \end{aligned}$$

- Otherwise,

$$\begin{aligned}
 \|y\| &= \|(y - x) + x\| \leq \|y - x\| + \|x\| \\
 &\quad \downarrow \\
 \|y\| - \|x\| &\leq \|x - y\|
 \end{aligned}$$

□

Chapter 2

Basic Topology

2.1 Functions

Definition 2.1.1. Suppose a relation G from X to Y satisfies

$$\forall x \in X, \exists! y \in Y, (x, y) \in G$$

Then the ordered triple

$$f := (X, Y, G)$$

is called a **function**, where

- X is called the **domain** of f
- Y is called the **codomain** of f
- G is called the **graph** of f

The notation

- $f : X \rightarrow Y$ denotes that f is a function from X to Y
- $f(x) = y$ denotes $(x, y) \in G$

Definition 2.1.2. Suppose $f : X \rightarrow Y$.

- The **image** of $E \subseteq X$ under f is defined by

$$f(E) := \{f(x) \mid x \in E\} \subseteq Y$$

- The **inverse image** of $E \subseteq Y$ under f is defined by

$$f^{-1}(E) := \{x \in X \mid f(x) \in E\} \subseteq X$$

Definition 2.1.3. Suppose $f : X \rightarrow Y$.

- f is said to be **injective** if

$$\forall x_1, x_2 \in X, f(x_1) = f(x_2) \implies x_1 = x_2$$

The notation $f : X \hookrightarrow Y$ denotes that f is injective.

- f is said to be **surjective** if

$$\forall y \in Y, \exists x \in X, f(x) = y$$

or equivalently,

$$f(X) = Y$$

The notation $f : X \twoheadrightarrow Y$ denotes that f is surjective.

- f is said to be **bijective** if

- f is injective
- f is surjective

or equivalently,

$$\forall y \in Y, \exists! x \in X, f(x) = y$$

The notation $f : X \xrightarrow{\sim} Y$ denotes that f is bijective.

Definition 2.1.4. Suppose $f : X \xrightarrow{\sim} Y$. The **inverse function** $f^{-1} : Y \rightarrow X$ of f is defined by

$$\forall (y, x) \in Y \times X, f^{-1}(y) = x \iff f(x) = y$$

By [definition 2.1.3](#),

$$\begin{aligned} \forall y \in Y, \exists! x \in X, f(x) = y \\ \Downarrow \\ \forall y \in Y, \exists! x \in X, f^{-1}(y) = x \end{aligned}$$

Therefore, f^{-1} is well-defined.

Theorem 2.1.5.

$$\forall f : X \xrightarrow{\sim} Y, f^{-1} : Y \xrightarrow{\sim} X$$

Proof. By [definition 2.1.1](#),

$$\begin{aligned} \forall x \in X, \exists! y \in Y, f(x) = y \\ \Downarrow \\ \forall x \in X, \exists! y \in Y, f^{-1}(y) = x \end{aligned}$$

□

Definition 2.1.6. Suppose $f : X \rightarrow Y$. Then

- Suppose $E \subseteq X$. The **restriction** $f|_E : E \rightarrow Y$ of f to E is defined by

$$\forall x \in E, f|_E(x) = f(x)$$

- Suppose $f(X) \subseteq F \subseteq Y$. The **corestriction** $f|^F : X \rightarrow F$ of f to F is defined by

$$\forall x \in X, f|^F(x) = f(x)$$

- Suppose

- $E \subseteq X$
- $f(E) \subseteq F \subseteq Y$

The **bi-restriction** $f|_E^F : E \rightarrow F$ of f to E and F is defined by

$$\forall x \in E, f|_E^F(x) = f(x)$$

Theorem 2.1.7. Suppose

- $f : X \rightarrow Y$
- $E \subseteq X$
- $f(E) \subseteq F \subseteq Y$

Then

$$(A) \quad f : X \hookrightarrow Y \implies f|_E^F : E \hookrightarrow F$$

$$(B) \quad f(E) = F \implies f|_E^F : E \twoheadrightarrow F$$

$$(C) \quad f : X \hookrightarrow Y \wedge f(E) = F \implies f|_E^F : E \xrightarrow{\sim} F$$

Proof.

(A) Suppose

- $f : X \hookrightarrow Y$
- $x_1, x_2 \in E$
- $f|_E^F(x_1) = f|_E^F(x_2)$

Then

$$\begin{aligned} f(x_1) &= f(x_2) \\ \Downarrow \\ x_1 &= x_2 \end{aligned}$$

(B) Suppose $f(E) = F$. Then

$$\begin{aligned} \forall y \in F, \exists x \in E, f(x) &= y \\ \Downarrow \\ \forall y \in F, \exists x \in E, f|_E^F(x) &= y \end{aligned}$$

(C) follows from (A) and (B).

□

Definition 2.1.8. Suppose

- $f : X \rightarrow Y$
- $g : Y \rightarrow Z$

The **composition** $g \circ f : X \rightarrow Z$ of g with f is defined by

$$\forall x \in X, (g \circ f)(x) = g(f(x))$$

Theorem 2.1.9. Suppose

- $f : X \rightarrow Y$
- $g : Y \rightarrow Z$

Then

$$(A) \quad f : X \hookrightarrow Y \wedge g : Y \hookrightarrow Z \implies g \circ f : X \hookrightarrow Z$$

$$(B) \quad f : X \twoheadrightarrow Y \wedge g : Y \twoheadrightarrow Z \implies g \circ f : X \twoheadrightarrow Z$$

$$(C) \quad f : X \xrightarrow{\sim} Y \wedge g : Y \xrightarrow{\sim} Z \implies g \circ f : X \xrightarrow{\sim} Z$$

Proof.

(A) Suppose

- $f : X \hookrightarrow Y$
- $g : Y \hookrightarrow Z$
- $x_1, x_2 \in X$
- $(g \circ f)(x_1) = (g \circ f)(x_2)$

Then

$$\begin{aligned} f(x_1) &= f(x_2) \\ \Downarrow \\ x_1 &= x_2 \end{aligned}$$

(B) Suppose

- $f : X \twoheadrightarrow Y$
- $g : Y \twoheadrightarrow Z$

Then

- $\forall z \in Z, \exists y \in Y, g(y) = z$
- $\forall y \in Y, \exists x \in X, f(x) = y$

Therefore,

$$\forall z \in Z, \exists x \in X, (g \circ f)(x) = z$$

(C) follows from (A) and (B).

□

2.2 Finite Sets

Theorem 2.2.1. Define $\sim$ by

$$\forall X \forall Y, X \sim Y \iff \exists f : X \xrightarrow{\sim} Y$$

Then $\sim$ is an equivalence relation.

Proof. By [definition 1.2.2](#),

- **Reflexivity:** Define $f : X \rightarrow X$ by

$$\forall x \in X, f(x) := x$$

Then

- **Injectivity:** Suppose

$$* \ x_1, x_2 \in X$$

$$* \ f(x_1) = f(x_2)$$

Then

$$x_1 = x_2$$

- **Surjectivity:** Suppose $y \in X$. Then

$$f(y) = y$$

Therefore,

$$\begin{array}{c} f : X \xrightarrow{\sim} X \\ \Downarrow \\ X \sim X \end{array}$$

- **Symmetry:** Suppose $X \sim Y$. Then

$$\begin{array}{c} \exists f : X \xrightarrow{\sim} Y \\ \Downarrow \\ f^{-1} : Y \xrightarrow{\sim} X \\ \Downarrow \\ Y \sim X \end{array}$$

Theorem 2.1.5

- **Transitivity:** Suppose

$$- \ X \sim Y$$

$$- \ Y \sim Z$$

Then

$$- \ \exists f : X \xrightarrow{\sim} Y$$

$$- \ \exists g : Y \xrightarrow{\sim} Z$$

By [theorem 2.1.9](#),

$$\begin{array}{c} g \circ f : X \xrightarrow{\sim} Z \\ \Downarrow \\ X \sim Z \end{array}$$

□

Remark. Since there is no **set of all sets** in the underlying set theory, the relation defined in [theorem 2.2.1](#) is a **proper-class** relation rather than a set. Its restriction to any set S becomes an equivalence relation on S .

Definition 2.2.2. Recall the relation $\sim$ defined in [theorem 2.2.1](#). A set X is said to be **finite** if

$$\exists n \in [0, \infty)_{\mathbb{N}}, [1, n]_{\mathbb{N}} \sim X$$

Such an n is called a **cardinality** of X , and is denoted by

$$|X| = n$$

The notation $|X| < \aleph_0$ denotes that X is finite.

Definition 2.2.3. A set X is said to be

- **infinite** if $\neg(|X| < \aleph_0)$
- **countable** if $\mathbb{N} \sim X$
- **at most countable** if $|X| < \aleph_0 \vee \mathbb{N} \sim X$
- **uncountable** if $\neg(|X| < \aleph_0 \vee \mathbb{N} \sim X)$

The notation

- $|X| \geq \aleph_0$ denotes that X is infinite
- $|X| = \aleph_0$ denotes that X is countable
- $|X| \leq \aleph_0$ denotes that X is at most countable
- $|X| > \aleph_0$ denotes that X is uncountable

The order symbols above are not **total orders**; they are just convenient notations.

Lemma 2.2.4.

$$\forall n \in [0, \infty)_{\mathbb{N}}, \forall A \subseteq [1, n]_{\mathbb{N}}, |A| < \aleph_0$$

Proof. Define

$$\forall n \in [0, \infty)_{\mathbb{N}}, P(n) \iff \forall A \subseteq [1, n]_{\mathbb{N}}, |A| < \aleph_0$$

Then

- **Base case:** Define $f : [1, 0]_{\mathbb{N}} \rightarrow [1, 0]_{\mathbb{N}}$ by

$$\forall k \in [1, 0]_{\mathbb{N}}, f(k) := k$$

Then

$$\begin{aligned} [1, 0]_{\mathbb{N}} &= \emptyset \\ \Downarrow \\ f : [1, 0]_{\mathbb{N}} &\xrightarrow{\sim} [1, 0]_{\mathbb{N}} \\ \Downarrow \\ |[1, 0]_{\mathbb{N}}| &< \aleph_0 \end{aligned}$$

Therefore,

$$\begin{aligned} \forall A \subseteq [1, 0]_{\mathbb{N}}, A &= [1, 0]_{\mathbb{N}} \\ \Downarrow \\ \forall A \subseteq [1, 0]_{\mathbb{N}}, |A| &< \aleph_0 \\ \Downarrow \\ P(0) \end{aligned}$$

- **Inductive step:** Suppose

- $P(n)$
- $A \subseteq [1, n+1]_{\mathbb{N}}$

Then

- If $n+1 \notin A$, then

$$\begin{aligned} A &\subseteq [1, n]_{\mathbb{N}} \\ \Downarrow \\ |A| &< \aleph_0 \end{aligned}$$

$P(n)$

- Otherwise, define

$$B := A \setminus \{n+1\} \subseteq [1, n]_{\mathbb{N}}$$

Since $P(n)$,

$$\begin{aligned} |B| &< \aleph_0 \\ \Downarrow \\ \exists m \in [0, \infty)_{\mathbb{N}}, \exists f : [1, m]_{\mathbb{N}} &\xrightarrow{\sim} B \end{aligned}$$

Define $g : [1, m+1]_{\mathbb{N}} \rightarrow A$ by

$$\forall k \in [1, m+1]_{\mathbb{N}}, g(k) := \begin{cases} n+1, & \text{if } k = m+1 \\ f(k), & \text{otherwise} \end{cases}$$

Then

- * **Injectivity:** Suppose

- $k_1, k_2 \in [1, m+1]_{\mathbb{N}}$
- $g(k_1) = g(k_2)$

Then

- If $k_1 = m+1$, then

$$g(k_1) = n+1$$

$$\Downarrow$$

$$g(k_2) = n+1$$

$$\Downarrow$$

$$k_2 = m+1$$

$$n+1 \notin B$$

- Otherwise,

$$k_1 \in [1, m]_{\mathbb{N}}$$

$$\Downarrow$$

$$g(k_1) = f(k_1)$$

$$\Downarrow$$

$$g(k_2) = f(k_1)$$

$$\Downarrow$$

$$g(k_2) = f(k_2) = f(k_1)$$

$$\Downarrow$$

$$k_1 = k_2$$

$$f(k_1) \neq n+1$$

$$f : [1, m]_{\mathbb{N}} \xrightarrow{\sim} B$$

* **Surjectivity:** Suppose $a \in A$.

- If $a = n+1$, then

$$g(m+1) = a$$

- Otherwise,

$$a \in B$$

$$\Downarrow$$

$$\exists k \in [1, m]_{\mathbb{N}}, f(k) = a$$

$$\Downarrow$$

$$g(k) = f(k) = a$$

$$f : [1, m]_{\mathbb{N}} \xrightarrow{\sim} B$$

Therefore,

$$g : [1, m+1]_{\mathbb{N}} \xrightarrow{\sim} A$$

$$\Downarrow$$

$$|A| < \aleph_0$$

Therefore, $P(n+1)$.

By [theorem 1.1.2](#),

$$\forall n \in [0, \infty)_{\mathbb{N}}, P(n)$$

□

Theorem 2.2.5.

$$\forall X \forall Y, |X| < \aleph_0 \wedge Y \subseteq X \implies |Y| < \aleph_0$$

Proof. Suppose

- $|X| < \aleph_0$
- $Y \subseteq X$

Then

$$\exists n \in [0, \infty)_{\mathbb{N}}, \exists f : [1, n]_{\mathbb{N}} \xrightarrow{\sim} X$$

Define

$$Z := f^{-1}(Y) \subseteq [1, n]_{\mathbb{N}}$$

By [lemma 2.2.4](#),

$$|Z| < \aleph_0$$

By [theorem 2.1.7](#),

$$\begin{array}{c} f|_Z^Y : Z \xrightarrow{\sim} Y \\ \downarrow \\ |Y| < \aleph_0 \end{array}$$

[Theorem 2.2.1](#)

□

Corollary 2.2.6.

$$\forall X \forall Y, |Y| \geq \aleph_0 \wedge Y \subseteq X \implies |X| \geq \aleph_0$$

Proof. Suppose

- $|Y| \geq \aleph_0$
- $Y \subseteq X$

Assume $|X| < \aleph_0$. By [theorem 2.2.5](#),

$$\begin{array}{c} |Y| < \aleph_0 \\ \downarrow \\ \perp \end{array}$$

Therefore,

$$|X| \geq \aleph_0$$

□

Lemma 2.2.7.

$$\forall Y, (\exists n \in [0, \infty)_{\mathbb{N}}, \exists f : [1, n]_{\mathbb{N}} \twoheadrightarrow Y) \implies |Y| < \aleph_0$$

Proof. Suppose

$$\exists n \in [0, \infty)_{\mathbb{N}}, \exists f : [1, n]_{\mathbb{N}} \twoheadrightarrow Y$$

Define

$$\forall y \in Y, N_y := f^{-1}(\{y\}) \neq \emptyset$$

By [theorem 1.1.4](#),

$$\forall y \in Y, \exists n_y \in N_y, n_y = \min N_y$$

Define

- $M := \{n_y \mid y \in Y\} \subseteq [1, n]_{\mathbb{N}}$
- $g : M \rightarrow Y$ by $g \equiv f|_M^Y$

Then

- **Injectivity:** Suppose

- $n_{y_1}, n_{y_2} \in M$
- $g(n_{y_1}) = g(n_{y_2})$

Then

$$\begin{aligned} f(n_{y_1}) &= f(n_{y_2}) \\ \Downarrow \\ y_1 &= y_2 \\ \Downarrow \\ N_{y_1} &= N_{y_2} \\ \Downarrow \\ n_{y_1} &= n_{y_2} \end{aligned}$$

- **Surjectivity:** Suppose $y \in Y$. Then

$$\begin{aligned} \exists n_y \in N_y, n_y &= \min N_y \\ \Downarrow \\ g(n_y) &= y \end{aligned}$$

Therefore,

$$g : M \xrightarrow{\sim} Y$$

By [lemma 2.2.4](#),

$$\begin{aligned} |M| &< \aleph_0 \\ \Downarrow \\ |Y| &< \aleph_0 \end{aligned}$$

[Theorem 2.2.1](#)

□

Theorem 2.2.8.

$$\forall X \forall Y, |X| < \aleph_0 \wedge \exists f : X \rightarrow Y \implies |Y| < \aleph_0$$

Proof. Suppose

- $|X| < \aleph_0$
- $\exists f : X \rightarrow Y$

Then

$$\begin{aligned} \exists n \in [0, \infty)_{\mathbb{N}}, \exists g : [1, n]_{\mathbb{N}} &\xrightarrow{\sim} X \\ \downarrow & \\ f \circ g : [1, n]_{\mathbb{N}} &\rightarrow Y \\ \downarrow & \\ |Y| &< \aleph_0 \end{aligned}$$

Theorem 2.1.9

Lemma 2.2.7

□

Remark. Suppose

- $|X| < \aleph_0$
- $X \neq \emptyset$

Then

$$\exists n \in \mathbb{N}, \exists f : [1, n]_{\mathbb{N}} \xrightarrow{\sim} X$$

This implies

$$X = \{f(1), f(2), \dots, f(n)\}$$

where

$$\forall i, j \in [1, n]_{\mathbb{N}}, i \neq j \implies f(i) \neq f(j)$$

Theorem 2.2.9.

$$\forall (X, <), |X| < \aleph_0 \wedge X \neq \emptyset \implies \exists m, M \in X, m = \min X \wedge M = \max X$$

Proof. Define

$$\forall n \in \mathbb{N}, P(n) \iff \forall (X, <), |X| = n \implies \exists m, M \in X, m = \min X \wedge M = \max X$$

Then

- **Base case:** Suppose $X = \{a_1\}$. Then

- $\forall x \in X, a_1 \leq x$
- $\forall x \in X, x \leq a_1$

Therefore, $P(1)$.

- **Inductive step:** Suppose

- $P(n)$
- $X = \{a_1, a_2, \dots, a_{n+1}\}$
- $Y = \{a_1, a_2, \dots, a_n\} \subseteq X$

Since $P(n)$,

- $\exists k \in [1, n]_{\mathbb{N}}, \forall y \in Y, a_k \leq y$
- $\exists K \in [1, n]_{\mathbb{N}}, \forall y \in Y, y \leq a_K$

Then

- $\forall x \in X, \min\{a_k, a_{n+1}\} \leq x$
- $\forall x \in X, x \leq \max\{a_K, a_{n+1}\}$

Therefore, $P(n+1)$.

By [theorem 1.1.2](#),

$$\forall n \in \mathbb{N}, P(n)$$

□

Lemma 2.2.10.

- (A) $\forall n, m \in [0, \infty)_{\mathbb{N}}, (\exists f : [1, n]_{\mathbb{N}} \hookrightarrow [1, m]_{\mathbb{N}}) \implies n \leq m$
- (B) $\forall n, m \in [0, \infty)_{\mathbb{N}}, (\exists f : [1, n]_{\mathbb{N}} \twoheadrightarrow [1, m]_{\mathbb{N}}) \implies n \geq m \geq 1 \vee n = m = 0$
- (C) $\forall n, m \in [0, \infty)_{\mathbb{N}}, (\exists f : [1, n]_{\mathbb{N}} \xrightarrow{\sim} [1, m]_{\mathbb{N}}) \implies n = m$

Proof. Suppose $n, m \in [0, \infty)_{\mathbb{N}}$. Then

(A) Define

$$\forall n \in [0, \infty)_{\mathbb{N}}, P(n) \iff \forall m \in [0, \infty)_{\mathbb{N}}, (\exists f_n : [1, n]_{\mathbb{N}} \hookrightarrow [1, m]_{\mathbb{N}}) \implies n \leq m$$

Then

• **Base case:**

$$\begin{aligned} \forall m \in [0, \infty)_{\mathbb{N}}, 0 \leq m \\ \Downarrow \\ P(0) \end{aligned}$$

• **Inductive step:** Suppose

- $P(n)$
- $\exists f_{n+1} : [1, n+1]_{\mathbb{N}} \hookrightarrow [1, m]_{\mathbb{N}}$

Then $m \geq 1$. Let

$$f_{n+1}(n+1) = x \in [1, m]_{\mathbb{N}}$$

Define $g_m : [1, m]_{\mathbb{N}} \rightarrow [1, m]_{\mathbb{N}}$ by

$$\forall y \in [1, m]_{\mathbb{N}}, g_m(y) := \begin{cases} x, & \text{if } y = m \\ m, & \text{if } y = x \\ y, & \text{otherwise} \end{cases}$$

Then

– **Injectivity:** Suppose

- * $y_1, y_2 \in [1, m]_{\mathbb{N}}$
- * $g_m(y_1) = g_m(y_2)$

Then

- * If $y_1 = m$, then

$$\begin{aligned} g_m(y_1) &= x \\ \Downarrow \\ g_m(y_2) &= x \\ \Downarrow \\ y_2 &= m \end{aligned}$$

- * If $y_1 = x$, then

$$\begin{aligned} g_m(y_1) &= m \\ \Downarrow \\ g_m(y_2) &= m \\ \Downarrow \\ y_2 &= x \end{aligned}$$

- * Otherwise,

$$\begin{aligned} g_m(y_1) &= y_1 \\ \Downarrow \\ g_m(y_2) &= y_1 \\ \Downarrow \\ g_m(y_2) &= y_2 \\ \Downarrow \\ y_2 &= y_1 \end{aligned}$$

$$y_1 \neq m \wedge y_1 \neq x$$

– **Surjectivity:** Suppose $k \in [1, m]_{\mathbb{N}}$. Then

* If $k = m$, then

$$g_m(x) = k$$

* If $k = x$, then

$$g_m(m) = k$$

* Otherwise,

$$g_m(k) = k$$

Therefore,

$$g_m : [1, m]_{\mathbb{N}} \xrightarrow{\sim} [1, m]_{\mathbb{N}}$$

Also,

$$- g_m \circ f_{n+1}(n+1) = m$$

$$- (g_m \circ f_{n+1})([1, n]_{\mathbb{N}}) \subseteq [1, m-1]_{\mathbb{N}}$$

Define $f_n : [1, n]_{\mathbb{N}} \rightarrow [1, m-1]_{\mathbb{N}}$ by

$$f_n \equiv (g_m \circ f_{n+1})|_{[1, n]_{\mathbb{N}}}^{[1, m-1]_{\mathbb{N}}}$$

By [theorems 2.1.7](#) and [2.1.9](#),

$$f_n : [1, n]_{\mathbb{N}} \hookrightarrow [1, m-1]_{\mathbb{N}}$$

$$\downarrow$$

$$n \leq m-1$$

$$\downarrow$$

$$n+1 \leq m$$

$P(n)$

Therefore, $P(n+1)$.

By [theorem 1.1.2](#),

$$\forall n \in [0, \infty)_{\mathbb{N}}, P(n)$$

(B) Define

$$\forall m \in [0, \infty)_{\mathbb{N}}, Q(m) \iff \forall n \in [0, \infty)_{\mathbb{N}},$$

$$(\exists f_m : [1, n]_{\mathbb{N}} \twoheadrightarrow [1, m]_{\mathbb{N}}) \implies n \geq m \geq 1 \vee n = m = 0$$

Then

• **Base case:**

– ($m = 0$) Suppose $n \in [0, \infty)_{\mathbb{N}}$.

$$\begin{aligned} & \neg(m \geq 1) \\ & \quad \downarrow \\ n \geq m \geq 1 \vee n = m = 0 & \iff n = m = 0 \iff n = 0 \end{aligned}$$

Then

$$\begin{aligned} n \neq 0 & \implies \nexists f_m : [1, n]_{\mathbb{N}} \rightarrow [1, m]_{\mathbb{N}} \\ & \quad \downarrow \\ & Q(0) \end{aligned}$$

– ($m = 1$) Suppose $n \in [0, \infty)_{\mathbb{N}}$.

$$\begin{aligned} & \neg(m = 0) \\ & \quad \downarrow \\ n \geq m \geq 1 \vee n = m = 0 & \iff n \geq m \geq 1 \iff n \geq 1 \end{aligned}$$

Then

$$\begin{aligned} n < 1 & \implies \nexists f_m : [1, n]_{\mathbb{N}} \rightarrow [1, m]_{\mathbb{N}} \\ & \quad \downarrow \\ & Q(1) \end{aligned}$$

• **Inductive step:** Let $m \geq 1$. Suppose

- $Q(m)$
- $\exists f_{m+1} : [1, n]_{\mathbb{N}} \rightarrow [1, m+1]_{\mathbb{N}}$

Then $n \geq 1$. Let

$$f_{m+1}(n) = x \in [1, m+1]_{\mathbb{N}}$$

Define $g_{m+1} : [1, m+1]_{\mathbb{N}} \rightarrow [1, m+1]_{\mathbb{N}}$ by

$$\forall y \in [1, m+1]_{\mathbb{N}}, g_{m+1}(y) := \begin{cases} x, & \text{if } y = m+1 \\ m+1, & \text{if } y = x \\ y, & \text{otherwise} \end{cases}$$

Then

– **Injectivity:** Suppose

- * $y_1, y_2 \in [1, m+1]_{\mathbb{N}}$
- * $g_{m+1}(y_1) = g_{m+1}(y_2)$

Then

* If $y_1 = m + 1$, then

$$\begin{aligned} g_{m+1}(y_1) &= x \\ \Downarrow \\ g_{m+1}(y_2) &= x \\ \Downarrow \\ y_2 &= m + 1 \end{aligned}$$

* If $y_1 = x$, then

$$\begin{aligned} g_{m+1}(y_1) &= m + 1 \\ \Downarrow \\ g_{m+1}(y_2) &= m + 1 \\ \Downarrow \\ y_2 &= x \end{aligned}$$

* Otherwise,

$$\begin{aligned} g_{m+1}(y_1) &= y_1 \\ \Downarrow \\ g_{m+1}(y_2) &= y_1 \\ \Downarrow \\ g_{m+1}(y_2) &= y_2 \\ \Downarrow \\ y_2 &= y_1 \end{aligned} \quad y_1 \neq m + 1 \wedge y_1 \neq x$$

– **Surjectivity:** Suppose $k \in [1, m + 1]_{\mathbb{N}}$. Then

* If $k = m + 1$, then

$$g_{m+1}(x) = k$$

* If $k = x$, then

$$g_{m+1}(m + 1) = k$$

* Otherwise,

$$g_{m+1}(k) = k$$

Therefore,

$$g_{m+1} : [1, m + 1]_{\mathbb{N}} \xrightarrow{\sim} [1, m + 1]_{\mathbb{N}}$$

Also,

$$(g_{m+1} \circ f_{m+1})(n) = m + 1$$

Define

$$K_{m+1} := (g_{m+1} \circ f_{m+1})^{-1}(\{m + 1\}) \subseteq [1, n]_{\mathbb{N}}$$

Then

$$(g_{m+1} \circ f_{m+1})([1, n-1]_{\mathbb{N}} \setminus K_{m+1}) = [1, m]_{\mathbb{N}}$$

Define $f_m : [1, n-1]_{\mathbb{N}} \rightarrow [1, m]_{\mathbb{N}}$ by

$$\forall y \in [1, n-1]_{\mathbb{N}}, f_m(y) := \begin{cases} m, & \text{if } y \in K_{m+1} \\ (g_{m+1} \circ f_{m+1})(y), & \text{otherwise} \end{cases}$$

By [theorems 2.1.7](#) and [2.1.9](#),

$$\begin{array}{ccc} (g_{m+1} \circ f_{m+1})|_{[1, n-1]_{\mathbb{N}} \setminus K_{m+1}}^{[1, m]_{\mathbb{N}}} : [1, n-1]_{\mathbb{N}} \setminus K_{m+1} & \twoheadrightarrow & [1, m]_{\mathbb{N}} \\ \downarrow & & \\ f_m : [1, n-1]_{\mathbb{N}} & \twoheadrightarrow & [1, m]_{\mathbb{N}} \\ \downarrow & & \\ n-1 & \geq & m \\ \downarrow & & \\ n & \geq & m+1 \geq 1 \end{array} \quad Q(m)$$

Therefore, $Q(m+1)$.

By [theorem 1.1.2](#),

$$\forall m \in [0, \infty)_{\mathbb{N}}, Q(m)$$

(C) follows from (A) and (B).

□

Corollary 2.2.11.

$$\forall X, |X| < \aleph_0 \implies \exists! n \in [0, \infty)_{\mathbb{N}}, |X| = n$$

Proof. Suppose $|X| < \aleph_0$. Then

$$\exists n \in [0, \infty)_{\mathbb{N}}, |X| = n$$

Suppose $|X| = m \in [0, \infty)_{\mathbb{N}}$. Then

$$\begin{array}{ccc} [1, n]_{\mathbb{N}} \sim X \wedge [1, m]_{\mathbb{N}} \sim X & & \\ \downarrow & & \\ [1, n]_{\mathbb{N}} \sim [1, m]_{\mathbb{N}} & & \\ \downarrow & & \\ n = m & & \end{array}$$

[Theorem 2.2.1](#)

[Lemma 2.2.10](#)

□

Lemma 2.2.12.

$$(A) \quad \forall n, m \in [0, \infty)_{\mathbb{N}}, n \leq m \implies \exists f : [1, n]_{\mathbb{N}} \hookrightarrow [1, m]_{\mathbb{N}}$$

$$(B) \quad \forall n, m \in [0, \infty)_{\mathbb{N}}, n \geq m \geq 1 \vee n = m = 0 \implies \exists f : [1, n]_{\mathbb{N}} \twoheadrightarrow [1, m]_{\mathbb{N}}$$

$$(C) \quad \forall n, m \in [0, \infty)_{\mathbb{N}}, n = m \implies \exists f : [1, n]_{\mathbb{N}} \xrightarrow{\sim} [1, m]_{\mathbb{N}}$$

Proof. Suppose $n, m \in [0, \infty)_{\mathbb{N}}$. Then

(A) Suppose $n \leq m$. Define $f : [1, n]_{\mathbb{N}} \rightarrow [1, m]_{\mathbb{N}}$ by

$$\forall i \in [1, n]_{\mathbb{N}}, f(i) := i$$

Suppose

- $i, j \in [1, n]_{\mathbb{N}}$
- $f(i) = f(j)$

Then

$$\begin{aligned} f(i) &= f(j) \\ \Downarrow \\ i &= j \end{aligned}$$

(B) • Suppose $n = m = 0$. Define $f : [1, n]_{\mathbb{N}} \rightarrow [1, m]_{\mathbb{N}}$ by

$$\forall i \in [1, n]_{\mathbb{N}}, f(i) := i$$

Then

$$\begin{aligned} [1, m]_{\mathbb{N}} &= \emptyset \\ \Downarrow \\ f : [1, n]_{\mathbb{N}} &\twoheadrightarrow [1, m]_{\mathbb{N}} \end{aligned}$$

• Suppose $n \geq m \geq 1$. Define $f : [1, n]_{\mathbb{N}} \rightarrow [1, m]_{\mathbb{N}}$ by

$$\forall i \in [1, n]_{\mathbb{N}}, f(i) := \begin{cases} i, & \text{if } i < m \\ m, & \text{otherwise} \end{cases}$$

Suppose $j \in [1, m]_{\mathbb{N}}$. Then

– If $j = m$, then

$$f(n) = j$$

– Otherwise,

$$f(j) = j$$

(C) Suppose $n = m$. Define $f : [1, n]_{\mathbb{N}} \rightarrow [1, m]_{\mathbb{N}}$ by

$$\forall i \in [1, n]_{\mathbb{N}}, f(i) := i$$

Then

- **Injectivity:** Suppose
 - $i, j \in [1, m]_{\mathbb{N}}$
 - $f(i) = f(j)$

Then

$$i = j$$

- **Surjectivity:** Suppose $j \in [1, m]_{\mathbb{N}}$. Then

$$f(j) = j$$

Therefore,

$$f : [1, n]_{\mathbb{N}} \xrightarrow{\sim} [1, m]_{\mathbb{N}}$$

□

Theorem 2.2.13.

$$(A) \quad \forall X \forall Y, |X| < \aleph_0 \wedge |Y| < \aleph_0 \implies (|X| \leq |Y| \iff \exists f : X \hookrightarrow Y)$$

$$(B) \quad \forall X \forall Y, |X| < \aleph_0 \wedge |Y| < \aleph_0 \implies$$

$$(|X| \geq |Y| > 0 \vee |X| = |Y| = 0 \iff \exists f : X \twoheadrightarrow Y)$$

$$(C) \quad \forall X \forall Y, |X| < \aleph_0 \wedge |Y| < \aleph_0 \implies (|X| = |Y| \iff \exists f : X \xrightarrow{\sim} Y)$$

Proof. Suppose

- $|X| = n \in [0, \infty)_{\mathbb{N}}$
- $|Y| = m \in [0, \infty)_{\mathbb{N}}$

Then

- $\exists f_X : [1, n]_{\mathbb{N}} \xrightarrow{\sim} X$
- $\exists f_Y : [1, m]_{\mathbb{N}} \xrightarrow{\sim} Y$

This implies

- ($\Leftarrow$) Suppose $f : X \rightarrow Y$. Define $F : [1, n]_{\mathbb{N}} \rightarrow [1, m]_{\mathbb{N}}$ by

$$F \equiv (f_Y)^{-1} \circ f \circ f_X$$

By [theorems 2.1.5](#) and [2.1.9](#),

- $f : X \hookrightarrow Y \implies F : [1, n]_{\mathbb{N}} \hookrightarrow [1, m]_{\mathbb{N}}$
- $f : X \twoheadrightarrow Y \implies F : [1, n]_{\mathbb{N}} \twoheadrightarrow [1, m]_{\mathbb{N}}$

By [lemma 2.2.10](#),

- (A) Suppose $\exists f : X \hookrightarrow Y$. Then

$$\begin{array}{c} \exists F : [1, n]_{\mathbb{N}} \hookrightarrow [1, m]_{\mathbb{N}} \\ \Downarrow \\ n \leq m \end{array}$$

- (B) Suppose $\exists f : X \twoheadrightarrow Y$. Then

$$\begin{array}{c} \exists F : [1, n]_{\mathbb{N}} \twoheadrightarrow [1, m]_{\mathbb{N}} \\ \Downarrow \\ n \geq m \geq 1 \vee n = m = 0 \end{array}$$

- (C) Suppose $\exists f : X \xrightarrow{\sim} Y$. Then

$$\begin{array}{c} \exists F : [1, n]_{\mathbb{N}} \xrightarrow{\sim} [1, m]_{\mathbb{N}} \\ \Downarrow \\ n = m \end{array}$$

- ($\implies$) Suppose $G : [1, n]_{\mathbb{N}} \rightarrow [1, m]_{\mathbb{N}}$. Define $f : X \rightarrow Y$ by

$$f \equiv f_Y \circ G \circ (f_X)^{-1}$$

By [theorems 2.1.5](#) and [2.1.9](#),

- $G : [1, n]_{\mathbb{N}} \hookrightarrow [1, m]_{\mathbb{N}} \implies f : X \hookrightarrow Y$
- $G : [1, n]_{\mathbb{N}} \twoheadrightarrow [1, m]_{\mathbb{N}} \implies f : X \twoheadrightarrow Y$

By [lemma 2.2.12](#),

- (A) Suppose $n \leq m$. Then

$$\begin{array}{c} \exists G : [1, n]_{\mathbb{N}} \hookrightarrow [1, m]_{\mathbb{N}} \\ \Downarrow \\ \exists f : X \hookrightarrow Y \end{array}$$

(B) Suppose $n \geq m \geq 1 \vee n = m = 0$. Then

$$\begin{array}{c} \exists G : [1, n]_{\mathbb{N}} \twoheadrightarrow [1, m]_{\mathbb{N}} \\ \Downarrow \\ \exists f : X \twoheadrightarrow Y \end{array}$$

(C) Suppose $n = m$. Then

$$\begin{array}{c} \exists G : [1, n]_{\mathbb{N}} \xrightarrow{\sim} [1, m]_{\mathbb{N}} \\ \Downarrow \\ \exists f : X \xrightarrow{\sim} Y \end{array}$$

□

Lemma 2.2.14.

$$\forall X \forall Y, X \cap Y = \emptyset \wedge |X| < \aleph_0 \wedge |Y| < \aleph_0 \implies |X \cup Y| < \aleph_0$$

Proof. Suppose

- $X \cap Y = \emptyset$
- $|X| < \aleph_0$
- $|Y| < \aleph_0$

Then

- $\exists n \in [0, \infty)_{\mathbb{N}}, \exists f : [1, n]_{\mathbb{N}} \xrightarrow{\sim} X$
- $\exists m \in [0, \infty)_{\mathbb{N}}, \exists g : [1, m]_{\mathbb{N}} \xrightarrow{\sim} Y$

Define $h : [1, m + n]_{\mathbb{N}} \rightarrow X \cup Y$ by

$$\forall i \in [1, m + n]_{\mathbb{N}}, h(i) := \begin{cases} f(i), & \text{if } i \leq n \\ g(i - n), & \text{otherwise} \end{cases}$$

Then

- **Injectivity:** Suppose
 - $i, j \in [1, m + n]_{\mathbb{N}}$
 - $h(i) = h(j)$

Then

– If $i \leq n$, then

$$\begin{aligned}
 h(i) &= f(i) \\
 \Downarrow \\
 f(i) &= h(j) \in X \\
 \Downarrow \\
 j &\leq n \\
 \Downarrow \\
 h(j) &= f(j) \\
 \Downarrow \\
 f(i) &= f(j) \\
 \Downarrow \\
 i &= j
 \end{aligned}$$

$$X \cap Y = \emptyset$$

$$f : [1, n]_{\mathbb{N}} \xrightarrow{\sim} X$$

– Otherwise,

$$\begin{aligned}
 h(i) &= g(i - n) \\
 \Downarrow \\
 g(i - n) &= h(j) \in Y \\
 \Downarrow \\
 j &> n \\
 \Downarrow \\
 h(j) &= g(j - n) \\
 \Downarrow \\
 g(i - n) &= g(j - n) \\
 \Downarrow \\
 i - n &= j - n \\
 \Downarrow \\
 i &= j
 \end{aligned}$$

$$X \cap Y = \emptyset$$

$$g : [1, m]_{\mathbb{N}} \xrightarrow{\sim} Y$$

• **Surjectivity:** Suppose $k \in X \cup Y$. Then

– If $k \in X$, then

$$\begin{aligned}
 k &\in X \\
 \Downarrow \\
 \exists i \in [1, n]_{\mathbb{N}}, f(i) &= k \\
 \Downarrow \\
 h(i) &= k
 \end{aligned}$$

$$f : [1, n]_{\mathbb{N}} \xrightarrow{\sim} X$$

– Otherwise,

$$\begin{aligned}
 k &\in Y \\
 \Downarrow \\
 \exists j \in [1, m]_{\mathbb{N}}, g(j) &= k \\
 \Downarrow \\
 h(j + n) &= k
 \end{aligned}$$

$$g : [1, m]_{\mathbb{N}} \xrightarrow{\sim} Y$$

Therefore,

$$\begin{aligned} h : [1, m+n]_{\mathbb{N}} &\xrightarrow{\sim} X \cup Y \\ &\downarrow \\ |X \cup Y| &< \aleph_0 \end{aligned}$$

□

Lemma 2.2.15.

$$\forall X \forall Y, |X| < \aleph_0 \wedge |Y| < \aleph_0 \implies |X \cup Y| < \aleph_0$$

Proof. Suppose

- $|X| < \aleph_0$
- $|Y| < \aleph_0$

Define

- $A := X \setminus Y \subseteq X$
- $B := Y \setminus X \subseteq Y$
- $C := X \cap Y \subseteq X$

By [theorem 2.2.5](#),

- $|A| < \aleph_0$
- $|B| < \aleph_0$
- $|C| < \aleph_0$

Also,

- $A \cap B = B \cap C = C \cap A = \emptyset$
- $X = (X \setminus Y) \cup (X \cap Y) = A \cup C$
- $Y = (Y \setminus X) \cup (X \cap Y) = B \cup C$
- $X \cup Y = (X \setminus Y) \cup (Y \setminus X) \cup (X \cap Y) = A \cup B \cup C$

By [lemma 2.2.14](#),

$$\begin{aligned} |A \cup B| &< \aleph_0 \\ &\downarrow \\ |(A \cup B) \cup C| &< \aleph_0 \\ &\downarrow \\ |X \cup Y| &< \aleph_0 \end{aligned}$$

□

Theorem 2.2.16.

$$\forall n \in [0, \infty)_{\mathbb{N}}, \forall \{X_k\}_{k=1}^n, (\forall k \in [1, n]_{\mathbb{N}}, |X_k| < \aleph_0) \implies \left| \bigcup_{k=1}^n X_k \right| < \aleph_0$$

Proof. Define

$$\forall n \in [0, \infty)_{\mathbb{N}}, P(n) \iff \forall \{X_k\}_{k=1}^n, (\forall k \in [1, n]_{\mathbb{N}}, |X_k| < \aleph_0) \implies \left| \bigcup_{k=1}^n X_k \right| < \aleph_0$$

Then

• **Base case:**

$$\begin{array}{c} \bigcup_{k=1}^0 X_k = \emptyset \\ \Downarrow \\ \left| \bigcup_{k=1}^0 X_k \right| < \aleph_0 \\ \Downarrow \\ P(0) \end{array}$$

• **Inductive step:** Suppose

- $P(n)$
- $\forall k \in [1, n+1]_{\mathbb{N}}, |X_k| < \aleph_0$

Since $P(n)$,

$$\left| \bigcup_{k=1}^n X_k \right| < \aleph_0$$

Then

$$\begin{array}{c} \bigcup_{k=1}^{n+1} X_k = \left(\bigcup_{k=1}^n X_k \right) \cup X_{n+1} \\ \Downarrow \\ \left| \bigcup_{k=1}^{n+1} X_k \right| < \aleph_0 \\ \Downarrow \\ P(n+1) \end{array}$$

Lemma 2.2.15

By [theorem 1.1.2](#),

$$\forall n \in [0, \infty)_{\mathbb{N}}, P(n)$$

□

Lemma 2.2.17.

$$\forall X \forall Y, |X| < \aleph_0 \wedge |Y| < \aleph_0 \implies |X \times Y| < \aleph_0$$

Proof. Suppose

- $|X| < \aleph_0$
- $|Y| < \aleph_0$

Then

- $\exists n \in [0, \infty)_{\mathbb{N}}, \exists f : [1, n]_{\mathbb{N}} \xrightarrow{\sim} X$
- $\exists m \in [0, \infty)_{\mathbb{N}}, \exists g : [1, m]_{\mathbb{N}} \xrightarrow{\sim} Y$

This implies

- If $n = 0$ or $m = 0$, then

$$\begin{aligned} X \times Y &= \emptyset \\ \Downarrow \\ |X \times Y| &< \aleph_0 \end{aligned}$$

- Otherwise, define $h : [1, nm]_{\mathbb{N}} \rightarrow X \times Y$ by

$$\forall k \in [1, nm]_{\mathbb{N}}, h(k) := \left(f\left(\left\lceil \frac{k}{m} \right\rceil\right), g\left(k - m\left(\left\lceil \frac{k}{m} \right\rceil - 1\right)\right) \right)$$

Suppose

- $k \in [1, nm]_{\mathbb{N}}$
- $j_k = \left\lceil \frac{k}{m} \right\rceil \in \mathbb{Z}$

Then

- Since $k \in [1, nm]_{\mathbb{N}}$,

$$\begin{aligned} 1 &\leq k \leq nm \\ &\Downarrow \\ 1 &\leq \left\lceil \frac{k}{m} \right\rceil \leq n \\ &\Downarrow \\ j_k &\in [1, n]_{\mathbb{N}} \end{aligned}$$

– By [definition 1.6.6](#),

$$\begin{aligned}
 j_k &= \min \left\{ i \in \mathbb{Z} \mid i \geq \frac{k}{m} \right\} \\
 &\Downarrow \\
 j_k - 1 &< \frac{k}{m} \leq j_k \\
 &\Downarrow \\
 0 &< k - m(j_k - 1) \leq m \\
 &\Downarrow \\
 k - m(j_k - 1) &\in [1, m]_{\mathbb{N}}
 \end{aligned}$$

Therefore, h is well-defined. Then

– **Injectivity:** Suppose

$$* \quad k_1, k_2 \in [1, nm]_{\mathbb{N}}$$

$$* \quad h(k_1) = h(k_2)$$

Then

$$\begin{aligned}
 f(j_{k_1}) &= f(j_{k_2}) \\
 &\Downarrow \\
 j_{k_1} &= j_{k_2}
 \end{aligned}$$

$$f : [1, n]_{\mathbb{N}} \xrightarrow{\sim} X$$

Therefore,

$$\begin{aligned}
 g(k_1 - m(j_{k_1} - 1)) &= g(k_2 - m(j_{k_2} - 1)) \\
 &\Downarrow \\
 k_1 - m(j_{k_1} - 1) &= k_2 - m(j_{k_2} - 1) \\
 &\Downarrow \\
 k_1 &= k_2
 \end{aligned}$$

$$g : [1, m]_{\mathbb{N}} \xrightarrow{\sim} Y$$

$$j_{k_1} = j_{k_2}$$

– **Surjectivity:** Suppose $(x, y) \in X \times Y$. Then

$$\begin{aligned}
 x &\in X \\
 &\Downarrow \\
 \exists j \in [1, n]_{\mathbb{N}}, & f(j) = x
 \end{aligned}$$

$$f : [1, n]_{\mathbb{N}} \xrightarrow{\sim} X$$

Also,

$$\begin{aligned}
 y &\in Y \\
 &\Downarrow \\
 \exists i \in [1, m]_{\mathbb{N}}, & g(i) = y
 \end{aligned}$$

$$g : [1, m]_{\mathbb{N}} \xrightarrow{\sim} Y$$

Define

$$k := m(j - 1) + i \in [1, nm]_{\mathbb{N}}$$

Then

$$h(k) = (f(j), g(i)) = (x, y)$$

Therefore,

$$\begin{aligned} h : [1, nm]_{\mathbb{N}} &\xrightarrow{\sim} X \times Y \\ &\downarrow \\ |X \times Y| &< \aleph_0 \end{aligned}$$

□

Lemma 2.2.18.

$$\forall n, m \in [0, \infty)_{\mathbb{N}}, \forall \{X_k\}_{k=1}^{n+m}, \prod_{k=1}^{n+m} X_k \sim \left(\prod_{k=1}^n X_k \right) \times \left(\prod_{k=n+1}^{n+m} X_k \right)$$

Proof. Suppose $n, m \in [0, \infty)_{\mathbb{N}}$. Define

$$h : \prod_{k=1}^{n+m} X_k \rightarrow \left(\prod_{k=1}^n X_k \right) \times \left(\prod_{k=n+1}^{n+m} X_k \right)$$

by

$$h(x_1, x_2, \dots, x_{n+m}) = ((x_1, x_2, \dots, x_n), (x_{n+1}, x_{n+2}, \dots, x_{n+m}))$$

Then

• **Injectivity:** Suppose

$$\begin{aligned} - (x_1, x_2, \dots, x_{n+m}), (y_1, y_2, \dots, y_{n+m}) &\in \prod_{k=1}^{n+m} X_k \\ - h(x_1, x_2, \dots, x_{n+m}) &= h(y_1, y_2, \dots, y_{n+m}) \end{aligned}$$

Then

$$\begin{aligned} &((x_1, x_2, \dots, x_n), (x_{n+1}, x_{n+2}, \dots, x_{n+m})) \\ &= ((y_1, y_2, \dots, y_n), (y_{n+1}, y_{n+2}, \dots, y_{n+m})) \end{aligned}$$

This implies

$$\begin{aligned} - (x_1, x_2, \dots, x_n) &= (y_1, y_2, \dots, y_n) \\ - (x_{n+1}, x_{n+2}, \dots, x_{n+m}) &= (y_{n+1}, y_{n+2}, \dots, y_{n+m}) \end{aligned}$$

Therefore,

$$(x_1, x_2, \dots, x_{n+m}) = (y_1, y_2, \dots, y_{n+m})$$

- **Surjectivity:** Suppose

$$((x_1, x_2, \dots, x_n), (x_{n+1}, x_{n+2}, \dots, x_{n+m})) \in \left(\prod_{k=1}^n X_k \right) \times \left(\prod_{k=n+1}^{n+m} X_k \right)$$

Then

$$h(x_1, x_2, \dots, x_{n+m}) = ((x_1, x_2, \dots, x_n), (x_{n+1}, x_{n+2}, \dots, x_{n+m}))$$

Therefore,

$$\begin{aligned} h : \prod_{k=1}^{n+m} X_k &\xrightarrow{\sim} \left(\prod_{k=1}^n X_k \right) \times \left(\prod_{k=n+1}^{n+m} X_k \right) \\ &\Downarrow \\ \prod_{k=1}^{n+m} X_k &\sim \left(\prod_{k=1}^n X_k \right) \times \left(\prod_{k=n+1}^{n+m} X_k \right) \end{aligned}$$

□

Theorem 2.2.19.

$$\forall n \in [0, \infty)_{\mathbb{N}}, \forall \{X_k\}_{k=1}^n, (\forall k \in [1, n]_{\mathbb{N}}, |X_k| < \aleph_0) \implies \left| \prod_{k=1}^n X_k \right| < \aleph_0$$

Proof. Define

$$\forall n \in [0, \infty)_{\mathbb{N}}, P(n) \iff \forall \{X_k\}_{k=1}^n, (\forall k \in [1, n]_{\mathbb{N}}, |X_k| < \aleph_0) \implies \left| \prod_{k=1}^n X_k \right| < \aleph_0$$

Then

- **Base case:**

$$\begin{aligned} \prod_{k=1}^0 X_k &= \{()\} \\ &\Downarrow \\ \left| \prod_{k=1}^0 X_k \right| &< \aleph_0 \\ &\Downarrow \\ P(0) & \end{aligned}$$

- **Inductive step:** Suppose

- $P(n)$
- $\forall k \in [1, n+1]_{\mathbb{N}}, |X_k| < \aleph_0$

Since $P(n)$,

$$\left| \prod_{k=1}^n X_k \right| < \aleph_0$$

By lemma 2.2.18,

$$\begin{aligned} \prod_{k=1}^{n+1} X_k &\sim \left(\prod_{k=1}^n X_k \right) \times X_{n+1} \\ &\Downarrow \\ \left| \prod_{k=1}^{n+1} X_k \right| &< \aleph_0 \\ &\Downarrow \\ &P(n+1) \end{aligned}$$

Lemma 2.2.17

By theorem 1.1.2,

$$\forall n \in [0, \infty)_{\mathbb{N}}, P(n)$$

□

Theorem 2.2.20.

$$\forall X, \nexists f : X \rightarrow \mathcal{P}(X)$$

*This theorem is called **Cantor's theorem**.*

Proof. Assume

$$\exists X, \exists f : X \rightarrow \mathcal{P}(X)$$

Define

$$S := \{x \in X \mid x \notin f(x)\} \in \mathcal{P}(X)$$

Then

$$\exists x \in X, f(x) = S$$

This implies

- If $x \in S$, then

$$\begin{aligned} x &\notin f(x) = S \\ &\Downarrow \\ &\perp \end{aligned}$$

- Otherwise,

$$\begin{array}{c} x \in f(x) = S \\ \Downarrow \\ \perp \end{array}$$

Therefore,

$$\forall X, \nexists f : X \rightarrow \mathcal{P}(X)$$

□

2.3 Countable and Uncountable Sets

Theorem 2.3.1.

$$\forall X \forall Y, |X| \geq \aleph_0 \wedge |Y| < \aleph_0 \implies |X \setminus Y| \geq \aleph_0$$

Proof. Suppose

- $|X| \geq \aleph_0$
- $|Y| < \aleph_0$

Assume $|X \setminus Y| < \aleph_0$. By [lemma 2.2.15](#),

$$|(X \setminus Y) \cup Y| < \aleph_0$$

Then

$$\begin{array}{c} X = (X \setminus Y) \cup (X \cap Y) \\ \Downarrow \\ X \subseteq (X \setminus Y) \cup Y \\ \Downarrow \\ |X| < \aleph_0 \\ \Downarrow \\ \perp \end{array}$$

[Theorem 2.2.5](#)

Therefore,

$$|X \setminus Y| \geq \aleph_0$$

□

Theorem 2.3.2.

$$\forall X \forall Y, |X| \geq \aleph_0 \wedge \exists f : X \hookrightarrow Y \implies |Y| \geq \aleph_0$$

Proof. Suppose

- $|X| \geq \aleph_0$
- $\exists f : X \hookrightarrow Y$

Assume $|Y| < \aleph_0$. By [theorem 2.1.7](#),

$$f|^{f(X)} : X \xrightarrow{\sim} f(X)$$

Define $h : f(X) \rightarrow X$ by

$$h \equiv \left(f|^{f(X)}\right)^{-1}$$

Define $g : Y \rightarrow X$ by

$$\forall y \in Y, g(y) := \begin{cases} h(y), & \text{if } y \in f(X) \\ h(y_0), & \text{otherwise} \end{cases}$$

where $y_0 \in f(X)$ is fixed. By [theorem 2.1.5](#),

$$\begin{array}{c} h : f(X) \xrightarrow{\sim} X \\ \downarrow \\ g : Y \twoheadrightarrow X \\ \downarrow \\ |X| < \aleph_0 \\ \downarrow \\ \perp \end{array}$$

[Theorem 2.2.8](#)

Therefore,

$$|Y| \geq \aleph_0$$

□

Lemma 2.3.3.

$$|\mathbb{N}| \geq \aleph_0$$

Proof. Assume $|\mathbb{N}| < \aleph_0$. By [theorem 2.2.9](#),

$$\exists M \in \mathbb{N}, M = \max \mathbb{N}$$

Then

$$\begin{array}{c} M < M + 1 \\ \downarrow \\ M + 1 \notin \mathbb{N} \\ \downarrow \\ \perp \end{array}$$

[Axiom 1.1.1](#)

Therefore,

$$|\mathbb{N}| \geq \aleph_0$$

□

Theorem 2.3.4.

$$\forall X, |X| = \aleph_0 \implies |X| \geq \aleph_0$$

Proof. Suppose $|X| = \aleph_0$. Then

$$\begin{array}{c} \exists f : \mathbb{N} \xrightarrow{\sim} X \\ \downarrow \\ |X| \geq \aleph_0 \end{array}$$

Theorem 2.3.2 and lemma 2.3.3

□

Theorem 2.3.5.

$$|\mathbb{Z}| = \aleph_0$$

Proof. Define $f : \mathbb{N} \rightarrow \mathbb{Z}$ by

$$\forall n \in \mathbb{N}, f(n) := \begin{cases} -\frac{n}{2}, & \text{if } n \equiv 0 \pmod{2} \\ \frac{n-1}{2}, & \text{if } n \equiv 1 \pmod{2} \end{cases}$$

Then

• **Injectivity:** Suppose

- $n_1, n_2 \in \mathbb{N}$
- $f(n_1) = f(n_2)$

Then

- $n \equiv 0 \pmod{2} \implies f(n) = -\frac{n}{2} < 0$
- $n \equiv 1 \pmod{2} \implies f(n) = \frac{n-1}{2} \geq 0$

This implies $n_1 \equiv n_2 \pmod{2}$.

– If $n_1 \equiv n_2 \equiv 0 \pmod{2}$, then

$$\begin{aligned} -\frac{n_1}{2} &= -\frac{n_2}{2} \\ \Downarrow \\ n_1 &= n_2 \end{aligned}$$

– If $n_1 \equiv n_2 \equiv 1 \pmod{2}$, then

$$\begin{aligned} \frac{n_1 - 1}{2} &= \frac{n_2 - 1}{2} \\ \Downarrow \\ n_1 &= n_2 \end{aligned}$$

• **Surjectivity:** Suppose $k \in \mathbb{Z}$.

– If $k \geq 0$, then

$$\begin{aligned} 2k + 1 &\in \mathbb{N} \\ \Downarrow \\ f(2k + 1) &= \frac{(2k + 1) - 1}{2} = \frac{2k}{2} = k \end{aligned}$$

– Otherwise,

$$\begin{aligned} -2k &\in \mathbb{N} \\ \Downarrow \\ f(-2k) &= -\frac{-2k}{2} = k \end{aligned}$$

Therefore,

$$\begin{aligned} f : \mathbb{N} &\xrightarrow{\sim} \mathbb{Z} \\ \Downarrow \\ |\mathbb{Z}| &= \aleph_0 \end{aligned}$$

□

Lemma 2.3.6.

$$\forall X \subseteq \mathbb{N}, |X| \geq \aleph_0 \implies |X| = \aleph_0$$

Proof. Suppose

- $X \subseteq \mathbb{N}$
- $|X| \geq \aleph_0$

Define

- $(X_n)_{n=1}^{\infty}$ in $\mathcal{P}(X)$
- $f : \mathbb{N} \rightarrow X$

by

- $X_1 := X$
- $\forall n \in \mathbb{N}, X_{n+1} := X_n \setminus \{f(n)\}$
- $\forall n \in \mathbb{N}, f(n) := \min X_n$

By [theorem 2.3.1](#),

$$\begin{aligned} \forall n \in \mathbb{N}, |X_n| &\geq \aleph_0 \\ \Downarrow \\ \forall n \in \mathbb{N}, \exists f(n) \in X_n, f(n) &= \min X_n \end{aligned}$$

[Theorem 1.1.4](#)

Therefore, f is well-defined.

- **Injectivity:** Suppose
 - $n_1, n_2 \in \mathbb{N}$
 - $n_1 \neq n_2$ (WLOG $n_1 < n_2$)

Then

$$\begin{aligned} X_{n_2} &\subseteq X_{n_1} \setminus \{f(n_1)\} \\ \Downarrow \\ f(n_2) &\neq f(n_1) \end{aligned}$$

- **Surjectivity:** Suppose $x \in X$. Define

$$S := \{n \in X \mid n < x\} \subseteq [1, x-1]_{\mathbb{N}}$$

By [lemma 2.2.4](#),

$$\begin{aligned} |S| &< \aleph_0 \\ \Downarrow \\ \exists k \in [0, \infty)_{\mathbb{N}}, |S| &= k \end{aligned}$$

- If $k = 0$, then
 - * $S = \emptyset$
 - * $x = \min X_1 = f(1)$
- Otherwise, let
 - * $S = \{s_1, s_2, \dots, s_k\}$
 - * $s_1 < s_2 < \dots < s_k$

Then

$$\begin{array}{ll}
 \min X_1 = \min S & X_2 = X \setminus \{s_1\} \\
 \min X_2 = \min S \setminus \{s_1\} & X_3 = X \setminus \{s_1, s_2\} \\
 \vdots & \vdots \\
 \min X_k = \min S \setminus \{s_1, s_2, \dots, s_{k-1}\} & X_{k+1} = X \setminus \{s_1, s_2, \dots, s_k\} = X \setminus S
 \end{array}$$

Therefore,

$$\begin{array}{c}
 \forall n \in X_{k+1}, n \geq x \\
 \Downarrow \\
 x = \min X_{k+1} \\
 \Downarrow \\
 f(k+1) = x
 \end{array}$$

Therefore,

$$\begin{array}{c}
 f : \mathbb{N} \xrightarrow{\sim} X \\
 \Downarrow \\
 |X| = \aleph_0
 \end{array}$$

□

Theorem 2.3.7.

$$\forall X \forall Y, Y \subseteq X \wedge |Y| \geq \aleph_0 \wedge |X| = \aleph_0 \implies |Y| = \aleph_0$$

Proof. Suppose

- $Y \subseteq X$
- $|X| = \aleph_0$
- $|Y| \geq \aleph_0$

Then

$$\begin{array}{c}
 \exists f : \mathbb{N} \xrightarrow{\sim} X \\
 \Downarrow \\
 f|_{f^{-1}(Y)} : f^{-1}(Y) \xrightarrow{\sim} Y
 \end{array}$$

Theorem 2.1.7

Assume $|f^{-1}(Y)| < \aleph_0$. By [theorem 2.2.1](#),

$$\begin{array}{c}
 |Y| < \aleph_0 \\
 \Downarrow \\
 \perp
 \end{array}$$

Therefore,

$$|f^{-1}(Y)| \geq \aleph_0$$

By lemma 2.3.6,

$$\begin{aligned} |f^{-1}(Y)| &= \aleph_0 \\ \downarrow \\ |Y| &= \aleph_0 \end{aligned}$$

Theorem 2.2.1

□

Corollary 2.3.8.

$$\forall X \forall Y, Y \subseteq X \wedge |Y| > \aleph_0 \implies |X| > \aleph_0$$

Proof. Suppose

- $Y \subseteq X$
- $|Y| > \aleph_0$

By corollary 2.2.6,

$$|X| \geq \aleph_0$$

Suppose $|X| = \aleph_0$. By theorem 2.3.7,

$$\begin{aligned} |Y| &= \aleph_0 \\ \downarrow \\ &\perp \end{aligned}$$

Therefore,

$$|X| > \aleph_0$$

□

Theorem 2.3.9.

$$\forall X, |X| \geq \aleph_0 \wedge (\exists f : \mathbb{N} \rightarrow X) \implies |X| = \aleph_0$$

Proof. Suppose

- $|X| \geq \aleph_0$
- $\exists f : \mathbb{N} \rightarrow X$

Define

$$\forall k \in X, S_k := f^{-1}(\{k\}) \neq \emptyset$$

By [theorem 1.1.4](#),

$$\forall k \in X, \exists m_k \in S_k, m_k = \min S_k$$

Define

$$T := \{m_k \mid k \in X\} \subseteq \mathbb{N}$$

Define $g : T \rightarrow X$ by

$$g \equiv f|_T^X$$

Then

• **Injectivity:** Suppose

- $m_{k_1}, m_{k_2} \in T$
- $g(m_{k_1}) = g(m_{k_2})$

Then

$$\begin{aligned} f(m_{k_1}) &= f(m_{k_2}) \\ \Downarrow \\ k_1 &= k_2 \\ \Downarrow \\ S_{k_1} &= S_{k_2} \\ \Downarrow \\ m_{k_1} &= m_{k_2} \end{aligned}$$

• **Surjectivity:** Suppose $k \in X$. Then

$$g(m_k) = k$$

Therefore,

$$\begin{aligned} g : T &\xrightarrow{\sim} X \\ \Downarrow \\ |T| &\geq \aleph_0 \\ \Downarrow \\ |T| &= \aleph_0 \\ \Downarrow \\ |X| &= \aleph_0 \end{aligned}$$

[Theorems 2.2.1 and 2.3.2](#)

[Lemma 2.3.6](#)

[Theorem 2.2.1](#)

□

Theorem 2.3.10.

$$\forall \{X_n\}_{n=1}^{\infty}, (\forall n \in \mathbb{N}, |X_n| < \aleph_0) \implies \left| \bigcup_{n=1}^{\infty} X_n \right| \leq \aleph_0$$

Proof. Suppose

$$\forall n \in \mathbb{N}, |X_n| < \aleph_0$$

Then

$$\forall n \in \mathbb{N}, \exists m_n \in [0, \infty)_{\mathbb{N}}, \exists g_n : [1, m_n]_{\mathbb{N}} \xrightarrow{\sim} X_n$$

Suppose $\left| \bigcup_{n=1}^{\infty} X_n \right| \geq \aleph_0$. Define

- $s_0 := 0$
- $\forall n \in \mathbb{N}, s_n := \sum_{k=1}^n m_k$

Then

$$\forall n, m \in \mathbb{N}, n \leq m \implies s_n \leq s_m$$

Assume

$$\exists N \in \mathbb{N}, \forall n \in \mathbb{N}, s_n \leq N$$

Then

$$\begin{aligned} \{s_n\}_{n=1}^{\infty} &\subseteq [0, N]_{\mathbb{N}} \\ &\Downarrow \\ \exists n_0 \in \mathbb{N}, s_{n_0} &= \max\{s_n\}_{n=1}^{\infty} \\ &\Downarrow \\ \forall n \in [n_0 + 1, \infty)_{\mathbb{N}}, m_n &= 0 \\ &\Downarrow \\ \bigcup_{n=1}^{\infty} X_n &= \bigcup_{n=1}^{n_0} X_n \\ &\Downarrow \\ \left| \bigcup_{n=1}^{\infty} X_n \right| &< \aleph_0 \\ &\Downarrow \\ &\perp \end{aligned}$$

Theorem 1.4.5

Theorem 2.2.16

Therefore,

$$\forall N \in \mathbb{N}, \exists n \in \mathbb{N}, s_n > N$$

First, we will show

$$\mathbb{N} = \bigcup_{n=1}^{\infty} [s_{n-1} + 1, s_n]_{\mathbb{N}}$$

- ($\subseteq$) Suppose $k \in \mathbb{N}$. Define

$$S_k := \{n \in \mathbb{N} \mid s_n > k\} \neq \emptyset$$

By theorem 1.1.4,

$$\exists n_k \in S_k, n_k = \min S_k$$

Then

$$\begin{aligned} s_{n_k-1} &\leq k < s_{n_k} \\ &\Downarrow \\ k &\in [s_{n_k-1} + 1, s_{n_k}]_{\mathbb{N}} \end{aligned}$$

- ($\supseteq$) Suppose $k \in \bigcup_{n=1}^{\infty} [s_{n-1} + 1, s_n]_{\mathbb{N}}$. Then

$$\begin{aligned} \exists n \in \mathbb{N}, k &\in [s_{n-1} + 1, s_n]_{\mathbb{N}} \\ &\Downarrow \\ k &\in \mathbb{N} \end{aligned}$$

Next, we will show

$$\forall i, j \in \mathbb{N}, i \neq j \implies [s_{i-1} + 1, s_i]_{\mathbb{N}} \cap [s_{j-1} + 1, s_j]_{\mathbb{N}} = \emptyset$$

Suppose

- $i, j \in \mathbb{N}$
- $i \neq j$ (WLOG $i < j$)

Then

$$\begin{aligned} s_i &\leq s_{j-1} \\ &\Downarrow \\ [s_{i-1} + 1, s_i]_{\mathbb{N}} &\cap [s_{j-1} + 1, s_j]_{\mathbb{N}} = \emptyset \end{aligned}$$

Therefore,

$$\forall k \in \mathbb{N}, \exists! n \in \mathbb{N}, \exists! m \in [1, m_n]_{\mathbb{N}}, k = s_{n-1} + m$$

Define $h : \mathbb{N} \rightarrow \bigcup_{n=1}^{\infty} X_n$ by

$$\forall n \in \mathbb{N}, \forall m \in [1, m_n]_{\mathbb{N}}, h(s_{n-1} + m) := g_n(m)$$

Suppose $x \in \bigcup_{n=1}^{\infty} X_n$. Then

$$\begin{aligned} \exists n \in \mathbb{N}, x &\in X_n \\ &\Downarrow \\ \exists m \in [1, m_n]_{\mathbb{N}}, g_n(m) &= x \\ &\Downarrow \\ h(s_{n-1} + m) &= x \end{aligned} \qquad g_n : [1, m_n]_{\mathbb{N}} \xrightarrow{\sim} X_n$$

Since $x \in \bigcup_{n=1}^{\infty} X_n$ was arbitrary,

$$\begin{array}{c} h : \mathbb{N} \rightarrow \bigcup_{n=1}^{\infty} X_n \\ \downarrow \\ \left| \bigcup_{n=1}^{\infty} X_n \right| = \aleph_0 \end{array}$$

Theorem 2.3.9

□

Remark. Strictly speaking, [theorem 2.3.10](#) requires the **axiom of choice** to prove. In this textbook, the **axiom of choice** is not discussed, so we will just assume that the proof is valid. Similar issues arise for [theorems 2.3.11](#) and [2.3.14](#).

Theorem 2.3.11.

$$\forall \{X_n\}_{n=1}^{\infty}, (\forall n \in \mathbb{N}, |X_n| = \aleph_0) \implies \left| \bigcup_{n=1}^{\infty} X_n \right| = \aleph_0$$

Proof. Suppose

$$\forall n \in \mathbb{N}, |X_n| = \aleph_0$$

Then

$$\forall n \in \mathbb{N}, \exists f_n : \mathbb{N} \xrightarrow{\sim} X_n$$

Assume $\left| \bigcup_{n=1}^{\infty} X_n \right| < \aleph_0$. Then

$$\begin{array}{c} X_1 \subseteq \bigcup_{n=1}^{\infty} X_n \\ \downarrow \\ |X_1| < \aleph_0 \\ \downarrow \\ \perp \end{array}$$

Theorem 2.2.5

Theorem 2.3.4

Therefore,

$$\left| \bigcup_{n=1}^{\infty} X_n \right| \geq \aleph_0$$

Define

$$\forall n \in \mathbb{N}, T_n := \left\{ k \in \mathbb{N} \mid n \leq \frac{k(k+1)}{2} \right\} \subseteq \mathbb{N}$$

Then

$$\begin{aligned} \forall n \in \mathbb{N}, 2n \in T_n \\ \Downarrow \\ \forall n \in \mathbb{N}, \exists t_n \in T_n, t_n = \min T_n \end{aligned}$$

Theorem 1.1.4

Define

$$\forall n \in \mathbb{N}, s_n := n - \frac{t_n(t_n - 1)}{2}$$

Then

$$\begin{aligned} t_n - 1 \notin T_n \\ \Downarrow \\ n > \frac{t_n(t_n - 1)}{2} \\ \Downarrow \\ s_n \in \mathbb{N} \end{aligned}$$

Also,

$$\begin{aligned} t_n - s_n + 1 &= \frac{t_n(t_n + 1)}{2} - n + 1 \\ \Downarrow \\ t_n - s_n + 1 &\geq 1 \\ \Downarrow \\ t_n - s_n + 1 &\in \mathbb{N} \end{aligned} \quad t_n \in T_n$$

Define $g : \mathbb{N} \rightarrow \bigcup_{n=1}^{\infty} X_n$ by

$$\forall n \in \mathbb{N}, g(n) := f_{s_n}(t_n - s_n + 1)$$

Suppose $x \in \bigcup_{n=1}^{\infty} X_n$. Then

$$\begin{aligned} \exists m \in \mathbb{N}, x \in X_m \\ \Downarrow \\ \exists l \in \mathbb{N}, f_m(l) = x \end{aligned}$$

$$f_m : \mathbb{N} \xrightarrow{\sim} X_m$$

Define

$$k := \frac{(l + m - 1)(l + m - 2)}{2} + m \in \mathbb{N}$$

Then

- $k - \frac{(l + m - 2)(l + m - 1)}{2} = m > 0$
- $k - \frac{(l + m - 1)(l + m)}{2} = 1 - l \leq 0$

This implies

$$\begin{aligned}
 l + m - 2 &\notin T_k \wedge l + m - 1 \in T_k \\
 &\Downarrow \\
 t_k &= l + m - 1 \\
 &\Downarrow \\
 s_k &= m \\
 &\Downarrow \\
 g(k) &= f_m(l) = x
 \end{aligned}$$

Since $x \in \bigcup_{n=1}^{\infty} X_n$ was arbitrary,

$$\begin{aligned}
 g : \mathbb{N} &\rightarrow \bigcup_{n=1}^{\infty} X_n \\
 &\Downarrow \\
 \left| \bigcup_{n=1}^{\infty} X_n \right| &= \aleph_0
 \end{aligned}$$

Theorem 2.3.9

□

Corollary 2.3.12.

$$\forall n \in \mathbb{N}, \forall \{X_k\}_{k=1}^n, (\forall k \in [1, n]_{\mathbb{N}}, |X_k| = \aleph_0) \implies \left| \bigcup_{k=1}^n X_k \right| = \aleph_0$$

Proof. Suppose

$$\forall k \in [1, n]_{\mathbb{N}}, |X_k| = \aleph_0$$

Define

$$\forall k \in [n + 1, \infty)_{\mathbb{N}}, X_k := X_n$$

Then

$$\begin{aligned}
 \forall n \in \mathbb{N}, |X_n| &= \aleph_0 \\
 &\Downarrow \\
 \left| \bigcup_{n=1}^{\infty} X_n \right| &= \aleph_0 \\
 &\Downarrow \\
 \left| \bigcup_{k=1}^n X_k \right| &= \aleph_0
 \end{aligned}$$

Theorem 2.3.11

□

Lemma 2.3.13.

$$\forall X \forall Y, |X| < \aleph_0 \wedge |Y| = \aleph_0 \implies |X \cup Y| = \aleph_0$$

Proof. Suppose

- $|X| < \aleph_0$
- $|Y| = \aleph_0$

Then

- $\exists f : \mathbb{N} \xrightarrow{\sim} Y$
- $\exists m \in [0, \infty)_{\mathbb{N}}, \exists g : [1, m]_{\mathbb{N}} \xrightarrow{\sim} X$

Also,

$$\begin{array}{c} Y \subseteq X \cup Y \\ \downarrow \\ |X \cup Y| \geq \aleph_0 \end{array} \quad \text{Corollary 2.2.6 and theorem 2.3.4}$$

Define $h : \mathbb{N} \rightarrow X \cup Y$ by

- $\forall n \in [1, m]_{\mathbb{N}}, h(n) := g(n)$
- $\forall n \in [m + 1, \infty)_{\mathbb{N}}, h(n) := f(n - m)$

Suppose $x \in X \cup Y$. Then

- If $x \in X$, then

$$\exists n \in [1, m]_{\mathbb{N}}, g(n) = x$$

This implies

$$h(n) = x$$

- Otherwise,

$$\exists n \in \mathbb{N}, f(n) = x$$

This implies

$$h(n + m) = x$$

Since $x \in X \cup Y$ was arbitrary,

$$\begin{array}{c} h : \mathbb{N} \twoheadrightarrow X \cup Y \\ \downarrow \\ |X \cup Y| = \aleph_0 \end{array} \quad \text{Theorem 2.3.9}$$

□

Theorem 2.3.14.

$$\forall Y, \forall (X_\alpha)_{\alpha \in Y}, |Y| \leq \aleph_0 \wedge (\forall \alpha \in Y, |X_\alpha| \leq \aleph_0) \implies \left| \bigcup_{\alpha \in Y} X_\alpha \right| \leq \aleph_0$$

Proof. Suppose

- $|Y| \leq \aleph_0$
- $\forall \alpha \in Y, |X_\alpha| \leq \aleph_0$

Then

- If

$$\forall \alpha \in Y, |X_\alpha| < \aleph_0$$

then

- If $|Y| < \aleph_0$, then

$$\exists n \in [0, \infty)_{\mathbb{N}}, \exists f : [1, n]_{\mathbb{N}} \xrightarrow{\sim} Y$$

This implies

$$\begin{aligned} \bigcup_{\alpha \in Y} X_\alpha &= \bigcup_{k=1}^n X_{f(k)} \\ &\downarrow \\ \left| \bigcup_{\alpha \in Y} X_\alpha \right| &< \aleph_0 \end{aligned}$$

Theorem 2.2.16

- Otherwise,

$$\exists g : \mathbb{N} \xrightarrow{\sim} Y$$

This implies

$$\begin{aligned} \bigcup_{\alpha \in Y} X_\alpha &= \bigcup_{n=1}^{\infty} X_{g(n)} \\ &\downarrow \\ \left| \bigcup_{\alpha \in Y} X_\alpha \right| &\leq \aleph_0 \end{aligned}$$

Theorem 2.3.10

- Otherwise,

$$\exists \beta \in Y, |X_\beta| = \aleph_0$$

Define

$$\forall \alpha \in Y, Y_\alpha := X_\alpha \cup X_\beta$$

By [corollary 2.3.12](#) and [lemma 2.3.13](#),

$$\forall \alpha \in Y, |Y_\alpha| = \aleph_0$$

Then

- If $|Y| < \aleph_0$, then

$$\exists n \in \mathbb{N}, \exists f : [1, n]_{\mathbb{N}} \xrightarrow{\sim} Y$$

This implies

$$\begin{aligned} \bigcup_{\alpha \in Y} X_\alpha &= \bigcup_{\alpha \in Y} Y_\alpha = \bigcup_{k=1}^n Y_{f(k)} \\ &\downarrow \\ \left| \bigcup_{\alpha \in Y} X_\alpha \right| &= \aleph_0 \end{aligned}$$

[Corollary 2.3.12](#)

- Otherwise,

$$\exists g : \mathbb{N} \xrightarrow{\sim} Y$$

This implies

$$\begin{aligned} \bigcup_{\alpha \in Y} X_\alpha &= \bigcup_{\alpha \in Y} Y_\alpha = \bigcup_{n=1}^{\infty} Y_{g(n)} \\ &\downarrow \\ \left| \bigcup_{\alpha \in Y} X_\alpha \right| &= \aleph_0 \end{aligned}$$

[Theorem 2.3.11](#)

□

Lemma 2.3.15.

$$|\mathbb{N} \times \mathbb{N}| = \aleph_0$$

Proof. Define $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ by

$$\forall m, n \in \mathbb{N}, f(m, n) := \frac{(m+n-1)(m+n-2)}{2} + m$$

Then

- **Injectivity:** Suppose

- $(m_1, n_1), (m_2, n_2) \in \mathbb{N} \times \mathbb{N}$
- $(m_1, n_1) \neq (m_2, n_2)$

Then

- If $m_1 + n_1 \neq m_2 + n_2$ (**WLOG** $m_1 + n_1 < m_2 + n_2$), then

$$\begin{aligned}
 f(m_1, n_1) &= \frac{(m_1 + n_1 - 1)(m_1 + n_1 - 2)}{2} + m_1 \\
 &\leq \frac{(m_1 + n_1 - 1)(m_1 + n_1 - 2)}{2} + (m_1 + n_1 - 1) \\
 &= \frac{(m_1 + n_1)(m_1 + n_1 - 1)}{2} \\
 &\leq \frac{(m_2 + n_2 - 1)(m_2 + n_2 - 2)}{2} \\
 &< \frac{(m_2 + n_2 - 1)(m_2 + n_2 - 2)}{2} + m_2 = f(m_2, n_2)
 \end{aligned}$$

- Otherwise,

$$\begin{aligned}
 m_1 &\neq m_2 \\
 \Downarrow \\
 f(m_1, n_1) &\neq f(m_2, n_2)
 \end{aligned}$$

- **Surjectivity:** Suppose $k \in \mathbb{N}$. Define t_k, s_k as [theorem 2.3.11](#). Then

$$f(s_k, t_k - s_k + 1) = k$$

Therefore,

$$\begin{aligned}
 f : \mathbb{N} \times \mathbb{N} &\xrightarrow{\sim} \mathbb{N} \\
 \Downarrow \\
 |\mathbb{N} \times \mathbb{N}| &= \aleph_0
 \end{aligned}$$

[Theorem 2.2.1](#)

□

Theorem 2.3.16.

$$\forall X \forall Y, |X| = \aleph_0 \wedge |Y| = \aleph_0 \implies |X \times Y| = \aleph_0$$

Proof. Suppose

- $|X| = \aleph_0$
- $|Y| = \aleph_0$

Then

- $\exists f : \mathbb{N} \xrightarrow{\sim} X$
- $\exists g : \mathbb{N} \xrightarrow{\sim} Y$

Define $h : \mathbb{N} \times \mathbb{N} \rightarrow X \times Y$ by

$$\forall m, n \in \mathbb{N}, h(m, n) := (f(m), g(n))$$

Then

• **Injectivity:** Suppose

- $(m_1, n_1), (m_2, n_2) \in \mathbb{N} \times \mathbb{N}$
- $h(m_1, n_1) = h(m_2, n_2)$

Then

$$\begin{aligned} f(m_1) = f(m_2) \wedge g(n_1) = g(n_2) \\ \Downarrow \\ m_1 = m_2 \wedge n_1 = n_2 \\ \Downarrow \\ (m_1, n_1) = (m_2, n_2) \end{aligned}$$

• **Surjectivity:** Suppose $(x, y) \in X \times Y$. Then

$$\begin{aligned} \exists m, n \in \mathbb{N}, f(m) = x \wedge g(n) = y \\ \Downarrow \\ h(m, n) = (x, y) \end{aligned}$$

Therefore,

$$\begin{aligned} h : \mathbb{N} \times \mathbb{N} &\xrightarrow{\sim} X \times Y \\ \Downarrow & \text{Theorem 2.2.1 and lemma 2.3.15} \\ |X \times Y| &= \aleph_0 \end{aligned}$$

□

Theorem 2.3.17.

$$\forall n \in \mathbb{N}, \forall \{X_k\}_{k=1}^n, (\forall k \in [1, n]_{\mathbb{N}}, |X_k| = \aleph_0) \implies \left| \prod_{k=1}^n X_k \right| = \aleph_0$$

Proof. Define

$$\forall n \in \mathbb{N}, P(n) \iff \forall \{X_k\}_{k=1}^n, (\forall k \in [1, n]_{\mathbb{N}}, |X_k| = \aleph_0) \implies \left| \prod_{k=1}^n X_k \right| = \aleph_0$$

Then

- **Base case:** Suppose $|X_1| = \aleph_0$. Then

$$\begin{array}{c} X_1 = \prod_{k=1}^1 X_k \\ \Downarrow \\ \left| \prod_{k=1}^1 X_k \right| = \aleph_0 \\ \Downarrow \\ P(1) \end{array}$$

- **Inductive step:** Suppose

- $P(n)$
- $\forall k \in [1, n+1]_{\mathbb{N}}, |X_k| = \aleph_0$

Since $P(n)$,

$$\left| \prod_{k=1}^n X_k \right| = \aleph_0$$

By [lemma 2.2.18](#),

$$\begin{array}{c} \prod_{k=1}^{n+1} X_k \sim \left(\prod_{k=1}^n X_k \right) \times X_{n+1} \\ \Downarrow \\ \left| \prod_{k=1}^{n+1} X_k \right| = \aleph_0 \\ \Downarrow \\ P(n+1) \end{array}$$

[Theorem 2.3.16](#)

By [theorem 1.1.2](#),

$$\forall n \in \mathbb{N}, P(n)$$

□

Theorem 2.3.18.

$$|\mathbb{Q}| = \aleph_0$$

Proof. Define $f : \mathbb{Z} \times \mathbb{N} \rightarrow \mathbb{Q}$ by

$$\forall (m, n) \in \mathbb{Z} \times \mathbb{N}, f(m, n) := \frac{m}{n}$$

Then

$$f : \mathbb{Z} \times \mathbb{N} \rightarrow \mathbb{Q}$$

By [theorems 2.3.5](#) and [2.3.16](#),

$$\begin{aligned} |\mathbb{Z} \times \mathbb{N}| &= \aleph_0 \\ \Downarrow \\ \exists g : \mathbb{N} &\xrightarrow{\sim} \mathbb{Z} \times \mathbb{N} \end{aligned}$$

Also,

$$\begin{aligned} \mathbb{N} &\subseteq \mathbb{Q} \\ \Downarrow \\ |\mathbb{Q}| &\geq \aleph_0 \end{aligned}$$

[Corollary 2.2.6](#) and [lemma 2.3.3](#)

By [theorem 2.1.9](#),

$$\begin{aligned} f \circ g : \mathbb{N} &\rightarrow \mathbb{Q} \\ \Downarrow \\ |\mathbb{Q}| &= \aleph_0 \end{aligned}$$

[Theorem 2.3.9](#)

□

Lemma 2.3.19.

$$\left| \prod_{n=1}^{\infty} \{0, 1\} \right| > \aleph_0$$

Proof. Define

$$\forall n, m \in \mathbb{N}, e_{nm} := \begin{cases} 1, & \text{if } m = n \\ 0, & \text{otherwise} \end{cases}$$

Define $f : \mathbb{N} \rightarrow \prod_{n=1}^{\infty} \{0, 1\}$ by

$$\forall n \in \mathbb{N}, f(n) := (e_{nm})_{m=1}^{\infty}$$

Suppose

- $n_1, n_2 \in \mathbb{N}$
- $f(n_1) = f(n_2)$

Then

$$\begin{aligned} \forall m \in \mathbb{N}, e_{n_1 m} &= e_{n_2 m} \\ \Downarrow \\ e_{n_1 n_1} &= e_{n_2 n_1} = 1 \\ \Downarrow \\ n_2 &= n_1 \end{aligned}$$

Therefore,

$$f : \mathbb{N} \hookrightarrow \prod_{n=1}^{\infty} \{0, 1\}$$

By [theorem 2.3.2](#) and [lemma 2.3.3](#),

$$\left| \prod_{n=1}^{\infty} \{0, 1\} \right| \geq \aleph_0$$

Assume $\left| \prod_{n=1}^{\infty} \{0, 1\} \right| = \aleph_0$. Then

$$\exists f : \mathbb{N} \xrightarrow{\sim} \prod_{n=1}^{\infty} \{0, 1\}$$

Define

- $\forall n \in \mathbb{N}, f(n) = (a_{nm})_{m=1}^{\infty}$
- $\forall m \in \mathbb{N}, b_m := 1 - a_{mm}$

Then

$$(b_m)_{m=1}^{\infty} \in \prod_{n=1}^{\infty} \{0, 1\}$$

Also,

$$\forall n \in \mathbb{N}, (b_m)_{m=1}^{\infty} \neq f(n)$$

$$\Downarrow$$

$$\perp$$

$$f : \mathbb{N} \xrightarrow{\sim} \prod_{n=1}^{\infty} \{0, 1\}$$

Therefore,

$$\left| \prod_{n=1}^{\infty} \{0, 1\} \right| > \aleph_0$$

□

Theorem 2.3.20.

$$|\mathbb{R}| > \aleph_0$$

Proof. Define $f : \mathbb{N} \rightarrow [0, 1)$ by

$$\forall n \in \mathbb{N}, f(n) := \frac{1}{n+1}$$

Suppose

- $n_1, n_2 \in \mathbb{N}$
- $f(n_1) = f(n_2)$

Then

$$\begin{aligned} \frac{1}{n_1+1} &= \frac{1}{n_2+1} \\ \Downarrow \\ n_1 &= n_2 \end{aligned}$$

Therefore,

$$f : \mathbb{N} \hookrightarrow [0, 1)$$

By [lemma 2.3.3](#) and [theorem 2.3.2](#),

$$|[0, 1)| \geq \aleph_0$$

Define $g : [0, 1) \rightarrow \prod_{n=1}^{\infty} \{0, 1\}$ by

$$\forall x \in [0, 1), g(x) := \mathcal{G}_2(x)$$

By [theorem 1.6.13](#), g is well-defined. By [theorem 1.6.14](#),

$$g : [0, 1) \hookrightarrow \prod_{n=1}^{\infty} \{0, 1\}$$

Define

$$Y := g([0, 1)) \subseteq \prod_{n=1}^{\infty} \{0, 1\}$$

By [theorem 2.1.7](#),

$$g|_Y : [0, 1) \xrightarrow{\sim} Y$$

Assume $|[0, 1)| = \aleph_0$. By [theorem 2.2.1](#),

$$|Y| = \aleph_0$$

Define

$$Z := \prod_{n=1}^{\infty} \{0, 1\} \setminus Y$$

By [theorems 1.6.15](#) and [1.6.16](#),

$$Z = \left\{ (a_n)_{n=1}^{\infty} \in \prod_{n=1}^{\infty} \{0, 1\} \mid \exists N \in \mathbb{N}, \forall n \in [N, \infty)_{\mathbb{N}}, a_n = 1 \right\}$$

Define

$$\forall N \in \mathbb{N}, Z_N := \left\{ (a_n)_{n=1}^{\infty} \in \prod_{n=1}^{\infty} \{0, 1\} \mid \forall n \in [N, \infty)_{\mathbb{N}}, a_n = 1 \right\} \neq \emptyset$$

Define

$$\forall N \in \mathbb{N}, F_N : \prod_{n=1}^{N-1} \{0, 1\} \rightarrow Z_N$$

by

$$\begin{aligned} \forall N \in \mathbb{N}, \forall (a_1, \dots, a_{N-1}) \in \prod_{n=1}^{N-1} \{0, 1\}, \\ F_N(a_1, \dots, a_{N-1}) = (a_1, \dots, a_{N-1}, 1, 1, 1, \dots) \end{aligned}$$

Suppose

- $N \in \mathbb{N}$
- $(a_1, \dots, a_{N-1}, 1, 1, 1, \dots) \in Z_N$

Then

$$F_N(a_1, \dots, a_{N-1}) = (a_1, \dots, a_{N-1}, 1, 1, 1, \dots)$$

Therefore,

$$\forall N \in \mathbb{N}, F_N : \prod_{n=1}^{N-1} \{0, 1\} \twoheadrightarrow Z_N$$

By [theorem 2.2.19](#),

$$\begin{aligned} \forall N \in \mathbb{N}, \left| \prod_{n=1}^{N-1} \{0, 1\} \right| < \aleph_0 \\ \downarrow \\ \forall N \in \mathbb{N}, |Z_N| < \aleph_0 \end{aligned}$$

[Theorem 2.2.8](#)

This implies

$$\begin{aligned} Z = \bigcup_{N=1}^{\infty} Z_N \\ \downarrow \\ |Z| \leq \aleph_0 \end{aligned}$$

[Theorem 2.3.10](#)

Then

$$\begin{array}{ccc} \prod_{n=1}^{\infty} \{0, 1\} = Y \cup Z & & \text{Theorem 2.3.14} \\ \Downarrow & & \\ \left| \prod_{n=1}^{\infty} \{0, 1\} \right| \leq \aleph_0 & & \text{Lemma 2.3.19} \\ \Downarrow & & \\ \perp & & \end{array}$$

Therefore,

$$\begin{array}{ccc} |[0, 1]| > \aleph_0 & & \text{Corollary 2.3.8} \\ \Downarrow & & \\ |\mathbb{R}| > \aleph_0 & & \end{array}$$

□

2.4 Metric Spaces

Definition 2.4.1. A function $d : X \times X \rightarrow [0, \infty)$ is called a **metric** on X if

- (A) $\forall p, q \in X, d(p, q) = 0 \iff p = q$
- (B) $\forall p, q \in X, d(p, q) = d(q, p)$
- (C) $\forall p, q, r \in X, d(p, r) \leq d(p, q) + d(q, r)$

(C) is called the **triangle inequality**. The structure (X, d) is called a **metric space**.

Theorem 2.4.2. Suppose $n \in \mathbb{N}$. Define $d : \mathbb{R}^n \times \mathbb{R}^n \rightarrow [0, \infty)$ by

$$\forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n, d(\mathbf{x}, \mathbf{y}) := \|\mathbf{x} - \mathbf{y}\|$$

Then d is a metric on $\mathbb{R}^n$.

Proof. By [definition 2.4.1](#),

- Suppose $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. By [definition 1.9.3](#),

$$\|\mathbf{x} - \mathbf{y}\| = 0 \iff \forall k \in [1, n]_{\mathbb{N}}, x_k = y_k \iff \mathbf{x} = \mathbf{y}$$

- Suppose $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. By [definition 1.9.3](#),

$$\|\mathbf{x} - \mathbf{y}\| = \sqrt{\sum_{k=1}^n (x_k - y_k)^2} = \sqrt{\sum_{k=1}^n (y_k - x_k)^2} = \|\mathbf{y} - \mathbf{x}\|$$

- Suppose $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$. By [theorem 1.9.4](#),

$$\|\mathbf{x} - \mathbf{z}\| = \|(\mathbf{x} - \mathbf{y}) + (\mathbf{y} - \mathbf{z})\| \leq \|\mathbf{x} - \mathbf{y}\| + \|\mathbf{y} - \mathbf{z}\|$$

□

Remark. Throughout this textbook, **euclidean spaces** use the **standard metric** defined in [theorem 2.4.2](#), unless otherwise specified.

Definition 2.4.3. Suppose

- $p \in |(X, d)|$
- $r \in \mathbb{R}^+$

Then

- The **neighborhood** of p with radius r is defined by

$$N_r(p) := \{q \in X \mid d(p, q) < r\}$$

- The **deleted neighborhood** of p with radius r is defined by

$$\tilde{N}_r(p) := N_r(p) \setminus \{p\} = \{q \in X \mid 0 < d(p, q) < r\}$$

Definition 2.4.4. Suppose

- $E \subseteq |(X, d)|$
- $p \in X$

Then

- p is called an **limit point** of E if

$$\forall r \in \mathbb{R}^+, E \cap \tilde{N}_r(p) \neq \emptyset$$

The set of all limit points of E is denoted by

$$E' := \left\{ p \in X \mid \forall r \in \mathbb{R}^+, E \cap \tilde{N}_r(p) \neq \emptyset \right\}$$

- p is called an **interior point** of E if

$$\exists r \in \mathbb{R}^+, N_r(p) \subseteq E$$

The set of all interior points of E is denoted by

$$E^\circ := \{p \in X \mid \exists r \in \mathbb{R}^+, N_r(p) \subseteq E\}$$

- p is called a **isolated point** of E if

$$\exists r \in \mathbb{R}^+, E \cap N_r(p) = \{p\}$$

The set of all isolated points of E is denoted by

$$E^i := \{p \in X \mid \exists r \in \mathbb{R}^+, E \cap N_r(p) = \{p\}\}$$

Definition 2.4.5. Suppose $E \subseteq |(X, d)|$. Then

- E is said to be **open** if

$$\forall p \in E, p \in E^\circ$$

The set of all open sets in X is denoted by

$$\mathcal{O}(X) := \{E \subseteq X \mid \forall p \in E, p \in E^\circ\}$$

- E is said to be **closed** if

$$E' \subseteq E$$

The set of all closed sets in X is denoted by

$$\mathcal{C}(X) := \{E \subseteq X \mid E' \subseteq E\}$$

Definition 2.4.6. Suppose $E \subseteq |(X, d)|$. The **closure** $\overline{E}$ of E is defined by

$$\overline{E} := E \cup E'$$

Definition 2.4.7. Suppose $E \subseteq |(X, d)|$. Then

- E is said to be **bounded** if

$$X \neq \emptyset \implies \exists p \in X, \exists r \in \mathbb{R}^+, E \subseteq N_r(p)$$

The set of all bounded sets in X is denoted by

$$\mathcal{B}(X) := \{E \subseteq X \mid X \neq \emptyset \implies \exists p \in X, \exists r \in \mathbb{R}^+, E \subseteq N_r(p)\}$$

- E is said to be **perfect** if

$$E = E'$$

- E is said to be **dense** if

$$\overline{E} = X$$

Theorem 2.4.8.

$$\forall p \in |(X, d)|, \forall r \in \mathbb{R}^+, N_r(p) \in \mathcal{O}(X)$$

Proof. Suppose $q \in N_r(p)$. Then

$$\begin{aligned} d(p, q) &< r \\ \Downarrow \\ \varepsilon := r - d(p, q) &> 0 \end{aligned}$$

Suppose $s \in N_\varepsilon(q)$. Then

$$\begin{aligned} d(q, s) &< \varepsilon \\ \Downarrow \\ d(p, s) &\leq d(p, q) + d(q, s) < d(p, q) + \varepsilon = r \\ \Downarrow \\ s &\in N_r(p) \end{aligned}$$

Since $s \in N_\varepsilon(q)$ was arbitrary,

$$\begin{aligned} N_\varepsilon(q) &\subseteq N_r(p) \\ \Downarrow \\ q &\in (N_r(p))^\circ \end{aligned}$$

□

Theorem 2.4.9.

$$\forall E \subseteq |(X, d)|, \forall p \in E', \forall r \in \mathbb{R}^+, |E \cap N_r(p)| \geq \aleph_0$$

Proof. Suppose $E \subseteq X$. Assume

$$\exists p \in E', \exists r \in \mathbb{R}^+, |E \cap N_r(p)| < \aleph_0$$

Define

$$S := E \cap \tilde{N}_r(p) \neq \emptyset$$

Then

$$\begin{aligned} S &\subseteq E \cap N_r(p) \\ &\Downarrow \\ |S| &< \aleph_0 \end{aligned}$$

Theorem 2.2.5

Define

$$T := \{d(p, q) \mid q \in S\} \neq \emptyset$$

Then

$$\begin{aligned} p &\notin S \\ &\Downarrow \\ T &\subseteq \mathbb{R}^+ \end{aligned}$$

Define $f : S \rightarrow T$ by

$$\forall q \in S, f(q) := d(p, q)$$

Then

$$\begin{aligned} \forall t \in T, \exists q \in S, t &= d(p, q) \\ &\Downarrow \\ f : S &\twoheadrightarrow T \\ &\Downarrow \\ |T| &< \aleph_0 \end{aligned}$$

Theorem 2.2.8

Theorem 2.2.9

$$\begin{aligned} &\Downarrow \\ \exists \varepsilon \in \mathbb{R}^+, \varepsilon &= \min T \end{aligned}$$

This implies

$$\begin{aligned} \forall q \in S, d(p, q) &\geq \varepsilon \\ &\Downarrow \\ E \cap \tilde{N}_\varepsilon(p) &= \emptyset \\ &\Downarrow \\ &\perp \end{aligned}$$

Therefore,

$$|E \cap N_r(p)| \geq \aleph_0$$

□

Theorem 2.4.10.

$$\forall E \subseteq |X, d|, E \in \mathcal{O}(X) \iff E^c \in \mathcal{C}(X)$$

Proof. ($\implies$) Suppose

- $E \in \mathcal{O}(X)$
- $p \in (E^c)'$

Assume $p \in E$. Then

$$\begin{aligned} \exists \varepsilon \in \mathbb{R}^+, N_\varepsilon(p) \subseteq E \\ \Downarrow \\ E^c \cap \tilde{N}_\varepsilon(p) = \emptyset \\ \Downarrow \\ \perp \end{aligned}$$

Therefore,

$$p \in E^c$$

Since $p \in (E^c)'$ was arbitrary,

$$\begin{aligned} (E^c)' \subseteq E^c \\ \Downarrow \\ E^c \in \mathcal{C}(X) \end{aligned}$$

($\Leftarrow$) Suppose

- $E^c \in \mathcal{C}(X)$
- $p \in E$

Assume

$$\nexists r \in \mathbb{R}^+, N_r(p) \subseteq E,$$

Then

$$\begin{aligned} p \notin E^c \\ \Downarrow \\ \forall r \in \mathbb{R}^+, E^c \cap \tilde{N}_r(p) \neq \emptyset \\ \Downarrow \\ p \in (E^c)' \\ \Downarrow \\ p \in E^c \\ \Downarrow \\ \perp \end{aligned}$$

$$E^c \in \mathcal{C}(X)$$

Therefore,

$$\begin{aligned} \exists r \in \mathbb{R}^+, N_r(p) \subseteq E \\ \Downarrow \\ p \in E^\circ \end{aligned}$$

□

Theorem 2.4.11.

$$\forall \{E_y\}_{y \in Y} \subseteq \mathcal{O}(|(X, d)|), \bigcup_{y \in Y} E_y \in \mathcal{O}(X)$$

Proof. Suppose

- $\{E_y\}_{y \in Y} \subseteq \mathcal{O}(X)$
- $p \in \bigcup_{y \in Y} E_y$

Then

$$\begin{aligned} & \exists y_0 \in Y, p \in E_{y_0} \\ & \quad \downarrow \\ & \exists r \in \mathbb{R}^+, N_r(p) \subseteq E_{y_0} \\ & \quad \downarrow \\ & N_r(p) \subseteq \bigcup_{y \in Y} E_y \\ & \quad \downarrow \\ & p \in \left(\bigcup_{y \in Y} E_y \right)^\circ \end{aligned}$$

□

Corollary 2.4.12.

$$\forall \{E_y\}_{y \in Y} \subseteq \mathcal{C}(|(X, d)|), \bigcap_{y \in Y} E_y \in \mathcal{C}(X)$$

Proof. Suppose $\{E_y\}_{y \in Y} \subseteq \mathcal{C}(X)$. By [theorem 2.4.10](#),

$$\begin{aligned} & \forall y \in Y, (E_y)^c \in \mathcal{O}(X) \\ & \quad \downarrow \\ & \bigcup_{y \in Y} (E_y)^c \in \mathcal{O}(X) \\ & \quad \downarrow \\ & \bigcap_{y \in Y} E_y = \left(\bigcup_{y \in Y} (E_y)^c \right)^c \in \mathcal{C}(X) \end{aligned}$$

[Theorem 2.4.11](#)

[Theorem 2.4.10](#)

□

Theorem 2.4.13.

$$\forall n \in \mathbb{N}, \forall \{E_k\}_{k=1}^n \subseteq \mathcal{O}(|(X, d)|), \bigcap_{k=1}^n E_k \in \mathcal{O}(X)$$

Proof. Suppose

- $n \in \mathbb{N}$
- $\{E_k\}_{k=1}^n \subseteq \mathcal{O}(X)$
- $p \in \bigcap_{k=1}^n E_k$

Then

$$\begin{aligned} & \forall j \in [1, n]_{\mathbb{N}}, p \in E_j \\ & \quad \downarrow \\ & \forall j \in [1, n]_{\mathbb{N}}, \exists r_j \in \mathbb{R}^+, N_{r_j}(p) \subseteq E_j \end{aligned}$$

Define

$$R := \{r_j \mid j \in [1, n]_{\mathbb{N}}\} \subseteq \mathbb{R}^+$$

Define $f : [1, n]_{\mathbb{N}} \rightarrow R$ by

$$\forall j \in [1, n]_{\mathbb{N}}, f(j) := r_j$$

Then

$$\forall \varepsilon \in R, \exists j \in [1, n]_{\mathbb{N}}, \varepsilon = r_j$$

$$\downarrow$$

$$f : [1, n]_{\mathbb{N}} \twoheadrightarrow R$$

$$\downarrow$$

$$|R| < \aleph_0$$

$$\downarrow$$

$$\exists r \in \mathbb{R}^+, r = \min R$$

$$\downarrow$$

$$\forall j \in [1, n]_{\mathbb{N}}, N_r(p) \subseteq E_j$$

$$\downarrow$$

$$N_r(p) \subseteq \bigcap_{k=1}^n E_k$$

$$\downarrow$$

$$p \in \left(\bigcap_{k=1}^n E_k \right)^\circ$$

Lemma 2.2.4 and theorem 2.2.8

Theorem 2.2.9

□

Corollary 2.4.14.

$$\forall n \in \mathbb{N}, \forall \{E_k\}_{k=1}^n \subseteq \mathcal{C}(|(X, d)|), \bigcup_{k=1}^n E_k \in \mathcal{C}(X)$$

Proof. Suppose

- $n \in \mathbb{N}$
- $\{E_k\}_{k=1}^n \subseteq \mathcal{C}(X)$

By [theorem 2.4.10](#),

$$\begin{aligned} \forall k \in [1, n]_{\mathbb{N}}, (E_k)^c &\in \mathcal{O}(X) && \text{Theorem 2.4.13} \\ \Downarrow \\ \bigcap_{k=1}^n (E_k)^c &\in \mathcal{O}(X) \\ \Downarrow &&& \text{Theorem 2.4.10} \\ \bigcup_{k=1}^n E_k &= \left(\bigcap_{k=1}^n (E_k)^c \right)^c \in \mathcal{C}(X) \end{aligned}$$

□

Theorem 2.4.15. Suppose $E \subseteq |(X, d)|$. Then

- (A) $\overline{E} \in \mathcal{C}(X)$
- (B) $E = \overline{E} \iff E \in \mathcal{C}(X)$
- (C) $\forall F \in \mathcal{C}(X), E \subseteq F \implies \overline{E} \subseteq F$

Proof.

(A) Suppose $p \in (\overline{E})^c$. Then

$$\begin{aligned} p &\notin E' \\ \Downarrow \\ \exists r \in \mathbb{R}^+, E \cap \tilde{N}_r(p) &= \emptyset \\ \Downarrow \\ E \cap N_r(p) &= \emptyset \\ \Downarrow \\ N_r(p) &\subseteq E^c \end{aligned} \quad p \notin E$$

Suppose $q \in N_r(p)$. By [theorem 2.4.8](#),

$$\begin{aligned} \exists \varepsilon \in \mathbb{R}^+, N_\varepsilon(q) &\subseteq N_r(p) \\ \Downarrow \\ E \cap N_\varepsilon(q) &= \emptyset \\ \Downarrow \\ E \cap \tilde{N}_\varepsilon(q) &= \emptyset \\ \Downarrow \\ q &\notin E' \end{aligned}$$

Since $q \in N_r(p)$ was arbitrary,

$$\begin{aligned} N_r(p) &\subseteq (E')^c \\ \Downarrow \\ N_r(p) &\subseteq (\overline{E})^c \\ \Downarrow \\ p &\in ((\overline{E})^c)^\circ \end{aligned} \qquad N_r(p) \subseteq E^c$$

Since $p \in (\overline{E})^c$ was arbitrary,

$$\begin{aligned} (\overline{E})^c &\in \mathcal{O}(X) \\ \Downarrow \\ \overline{E} &\in \mathcal{C}(X) \end{aligned} \qquad \text{Theorem 2.4.10}$$

$$(B) \quad E \in \mathcal{C}(X) \iff E' \subseteq E \iff E \cup E' = E \iff \overline{E} = E$$

(C) Suppose

- $F \in \mathcal{C}(X)$
- $E \subseteq F$
- $p \in E'$

Then

$$\begin{aligned} \forall r \in \mathbb{R}^+, E \cap \tilde{N}_r(p) &\neq \emptyset \\ \Downarrow \\ \forall r \in \mathbb{R}^+, F \cap \tilde{N}_r(p) &\neq \emptyset \\ \Downarrow \\ p &\in F' \end{aligned}$$

Since $p \in E'$ was arbitrary,

$$\begin{aligned} E' &\subseteq F' \\ \Downarrow \\ E' &\subseteq F \\ \Downarrow \\ \overline{E} &\subseteq F \end{aligned} \qquad F \in \mathcal{C}(X)$$

□

Theorem 2.4.16. *Suppose*

- $E \subseteq \mathbb{R}$
- $E \neq \emptyset$
- $E^u \neq \emptyset$

Then

(A) $\exists \alpha \in \overline{E}, \alpha = \sup E$

(B) $E \in \mathcal{C}(\mathbb{R}) \implies \exists \alpha \in E, \alpha = \sup E$

Proof.

(A) By [axiom 1.6.1](#),

$$\exists \alpha \in \mathbb{R}, \alpha = \sup E$$

Suppose $\alpha \notin E$. Then

$$\begin{aligned} & \forall x \in \mathbb{R}^+, \alpha - x \notin E^u \\ & \quad \downarrow \\ & \forall x \in \mathbb{R}^+, \exists p_x \in E, \alpha - x < p_x < \alpha \\ & \quad \downarrow \\ & \forall x \in \mathbb{R}^+, p_x \in \tilde{N}_x(\alpha) \cap E \\ & \quad \downarrow \\ & \alpha \in E' \end{aligned}$$

(B) follows from (A).

□

Theorem 2.4.17.

$$\forall E \subseteq |(X, d)|, E^i = E \setminus E'$$

Proof.

- ($\subseteq$) Suppose $p \in E^i$. Then

$$\begin{aligned}
 & \exists r \in \mathbb{R}^+, E \cap N_r(p) = \{p\} \\
 & \quad \Downarrow \\
 & p \in E \wedge E \cap \tilde{N}_r(p) = \emptyset \\
 & \quad \Downarrow \\
 & p \notin E' \\
 & \quad \Downarrow \\
 & p \in E \setminus E'
 \end{aligned}$$

- ($\supseteq$) Suppose $p \in E \setminus E'$. Then

$$\begin{aligned}
 & p \notin E' \\
 & \quad \Downarrow \\
 & \exists r \in \mathbb{R}^+, E \cap \tilde{N}_r(p) = \emptyset \\
 & \quad \Downarrow \\
 & E \cap N_r(p) = \{p\} \\
 & \quad \Downarrow \\
 & p \in E^i
 \end{aligned}
 \qquad p \in E$$

□

Lemma 2.4.18.

$$\forall E \subseteq |(X, d)|, (E^\circ)^c = \overline{E^c}$$

Proof. Suppose $p \in X$. Then

$$\begin{aligned}
 p \in (E^\circ)^c & \iff p \notin E^\circ \\
 & \iff \forall r \in \mathbb{R}^+, N_r(p) \not\subseteq E \\
 & \iff \forall r \in \mathbb{R}^+, N_r(p) \cap E^c \neq \emptyset \\
 & \iff p \in \overline{E^c}
 \end{aligned}$$

□

Definition 2.4.19. Suppose $E \subseteq |(X, d)|$. The **boundary** ∂E of E is defined by

$$\partial E := \overline{E} \cap \overline{E^c}$$

or equivalently, by [lemma 2.4.18](#),

$$\partial E = \overline{E} \setminus E^\circ$$

Lemma 2.4.20.

$$\forall n \in \mathbb{N}, \forall \mathbf{p} \in \mathbb{R}^n, \forall r \in \mathbb{R}^+, \tilde{N}_r(\mathbf{p}) \neq \emptyset$$

Proof. Suppose

- $n \in \mathbb{N}$
- $\mathbf{p} \in \mathbb{R}^n$
- $r \in \mathbb{R}^+$

By [definition 1.9.3](#),

$$\begin{aligned} \left\| \left(\mathbf{p} + \frac{r}{2} \mathbf{e}_1 \right) - \mathbf{p} \right\| &= \left\| \frac{r}{2} \mathbf{e}_1 \right\| = \frac{r}{2} < r \\ &\downarrow \\ \mathbf{p} + \frac{r}{2} \mathbf{e}_1 &\in \tilde{N}_r(\mathbf{p}) \end{aligned}$$

where $\mathbf{e}_1 := (1, 0, 0, \dots, 0) \in \mathbb{R}^n$. □

Theorem 2.4.21.

$$\forall n \in \mathbb{N}, \forall E \subseteq \mathbb{R}^n, \partial E = E^i \cup (E^c)^i \cup (E' \cap (E^c)')$$

Proof. Suppose

- $n \in \mathbb{N}$
- $E \subseteq \mathbb{R}^n$

By [definition 2.4.19](#),

$$\begin{aligned} \partial E &= \overline{E} \cap \overline{E^c} \\ &= (E \cup E') \cap (E^c \cup (E^c)') \\ &= (E \cap E^c) \cup (E \cap (E^c)') \cup (E' \cap E^c) \cup (E' \cap (E^c)') \end{aligned}$$

Consider

- $E = (E \setminus E') \cup (E \cap E')$
- $E^c = (E^c \setminus (E^c)') \cup (E^c \cap (E^c)')$

This implies

- $E \cap (E^c)' = ((E \setminus E') \cap (E^c)') \cup ((E \cap E') \cap (E^c)')$
- $E' \cap E^c = (E' \cap (E^c \setminus (E^c)')) \cup (E' \cap (E^c \cap (E^c)'))$

Consider

- $((E \cap E') \cap (E^c)') \subseteq E' \cap (E^c)'$
- $(E' \cap (E^c \cap (E^c)')) \subseteq E' \cap (E^c)'$

This implies

$$\begin{aligned} \partial E &= \emptyset \cup ((E \setminus E') \cap (E^c)') \cup (E' \cap (E^c \setminus (E^c)')) \cup (E' \cap (E^c)') \\ &= ((E \setminus E') \cap (E^c)') \cup (E' \cap (E^c \setminus (E^c)')) \cup (E' \cap (E^c)') \end{aligned}$$

Suppose $p \in E \setminus E'$. Then

$$\begin{aligned} \exists r \in \mathbb{R}^+, E \cap \tilde{N}_r(p) &= \emptyset \\ \Downarrow \\ \tilde{N}_r(p) &\subseteq E^c \end{aligned}$$

By [lemma 2.4.20](#),

$$\begin{aligned} \forall s \in \mathbb{R}^+, \tilde{N}_s(p) \cap \tilde{N}_r(p) &= \tilde{N}_{\min(s,r)}(p) \neq \emptyset \\ \Downarrow \\ \tilde{N}_s(p) \cap (\tilde{N}_r(p) \cap E^c) &\neq \emptyset \\ \Downarrow \\ E^c \cap \tilde{N}_s(p) &\neq \emptyset \\ \Downarrow \\ p &\in (E^c)' \end{aligned}$$

Since $p \in E \setminus E'$ was arbitrary,

$$\begin{aligned} E \setminus E' &\subseteq (E^c)' \\ \Downarrow \\ (E \setminus E') \cap (E^c)' &= E \setminus E' \end{aligned}$$

Similarly,

$$\begin{aligned} E^c \setminus (E^c)' &\subseteq E' \\ \Downarrow \\ E' \cap (E^c \setminus (E^c)') &= E^c \setminus (E^c)' \end{aligned}$$

Therefore,

$$\begin{aligned} \partial E &= (E \setminus E') \cup (E^c \setminus (E^c)') \cup (E' \cap (E^c)') \\ \Downarrow \\ \partial E &= E^i \cup (E^c)^i \cup (E' \cap (E^c)') \end{aligned}$$

[Theorem 2.4.17](#)

□

Remark. Suppose $Y \subseteq |(X, d)|$. Then one can verify

$$d|_{Y \times Y} : Y \times Y \rightarrow [0, \infty)$$

is a metric on Y . Therefore, the structure

$$(Y, d|_{Y \times Y})$$

is a metric space, which is called the **subspace** of (X, d) .

- (Y, d) denotes $(Y, d|_{Y \times Y})$
- $d(p, q)$ denotes $d|_{Y \times Y}(p, q)$

for the sake of simplicity.

Definition 2.4.22. Suppose $E \subseteq Y \subseteq |(X, d)|$. Then

- E is said to be **open relative to Y** if

$$\forall p \in E, \exists r \in \mathbb{R}^+, Y \cap N_r(p) \subseteq E$$

or equivalently,

$$E \in \mathcal{O}(|(Y, d)|)$$

- E is said to be **closed relative to Y** if

$$E' \cap Y \subseteq E$$

or equivalently,

$$E \in \mathcal{C}(|(Y, d)|)$$

Theorem 2.4.23.

$$\forall E, Y \subseteq |(X, d)|, E \in \mathcal{O}(|(Y, d)|) \iff \exists V \in \mathcal{O}(X), E = V \cap Y$$

Proof. Suppose $E, Y \subseteq X$. Then

- $(\implies)$ Suppose $E \in \mathcal{O}(|(Y, d)|)$. Then

$$\forall p \in E, \exists r_p \in \mathbb{R}^+, Y \cap N_{r_p}(p) \subseteq E$$

Define

$$V := \bigcup_{p \in E} N_{r_p}(p) \subseteq X$$

By [theorems 2.4.8](#) and [2.4.11](#),

$$V \in \mathcal{O}(X)$$

We will show $E = V \cap Y$.

– ($\subseteq$)

$$\begin{aligned} \forall p \in E, p \in N_{r_p}(p) \subseteq V \\ \Downarrow \\ E \subseteq V \end{aligned}$$

– ($\supseteq$)

$$\begin{aligned} \bigcup_{p \in E} (Y \cap N_{r_p}(p)) \subseteq E \\ \Downarrow \\ Y \cap V = Y \cap \bigcup_{p \in E} N_{r_p}(p) \subseteq E \end{aligned}$$

• ($\Leftarrow$) Suppose

- $V \in \mathcal{O}(X)$
- $E = V \cap Y$

Then

$$\begin{aligned} \forall p \in E, \exists r_p \in \mathbb{R}^+, N_{r_p}(p) \subseteq V \\ \Downarrow \\ \forall p \in E, Y \cap N_{r_p}(p) \subseteq V \cap Y = E \end{aligned}$$

□

2.5 Compact Sets

Definition 2.5.1. Suppose $E \subseteq |(X, d)|$.

- A collection $\{G_\alpha\}_{\alpha \in A}$ is called an **open cover** of E if
 - $\forall \alpha \in A, G_\alpha \in \mathcal{O}(X)$
 - $E \subseteq \bigcup_{\alpha \in A} G_\alpha$
- A collection $\{G_\alpha\}_{\alpha \in S}$ is called a **subcover** of $\{G_\alpha\}_{\alpha \in A}$ if
 - $S \subseteq A$
 - $E \subseteq \bigcup_{\alpha \in S} G_\alpha$

Definition 2.5.2. Suppose $E \subseteq (X, d)$. Then E is called **compact** if

$$\forall \{G_\alpha\}_{\alpha \in A} \subseteq \mathcal{O}(X), E \subseteq \bigcup_{\alpha \in A} G_\alpha \implies \exists S \subseteq A, |S| < \aleph_0 \wedge E \subseteq \bigcup_{\alpha \in S} G_\alpha$$

The set of all compact sets in X is denoted by

$$\mathcal{K}(X) := \left\{ E \subseteq X \left| \begin{array}{l} \forall \{G_\alpha\}_{\alpha \in A} \subseteq \mathcal{O}(X), E \subseteq \bigcup_{\alpha \in A} G_\alpha \\ \implies \exists S \subseteq A, |S| < \aleph_0 \wedge E \subseteq \bigcup_{\alpha \in S} G_\alpha \end{array} \right. \right\}$$

Theorem 2.5.3.

$$\forall Y \subseteq (X, d), \forall E \subseteq Y, E \in \mathcal{K}((Y, d)) \iff E \in \mathcal{K}(X)$$

Proof. Suppose $E \subseteq Y \subseteq X$. Then

- ($\implies$) Suppose
 - $E \in \mathcal{K}(Y)$
 - $\{G_\alpha\}_{\alpha \in A} \subseteq \mathcal{O}(X)$
 - $E \subseteq \bigcup_{\alpha \in A} G_\alpha$

Define

$$\forall \alpha \in A, H_\alpha := G_\alpha \cap Y \subseteq Y$$

By [theorem 2.4.23](#),

$$\forall \alpha \in A, H_\alpha \in \mathcal{O}(Y)$$

Then

$$\begin{aligned} E &\subseteq \left(\bigcup_{\alpha \in A} G_\alpha \right) \cap Y = \bigcup_{\alpha \in A} (G_\alpha \cap Y) = \bigcup_{\alpha \in A} H_\alpha \\ &\quad \downarrow \\ &\exists S \subseteq A, |S| < \aleph_0 \wedge E \subseteq \bigcup_{\alpha \in S} H_\alpha \\ &\quad \downarrow \\ &E \subseteq \bigcup_{\alpha \in S} (G_\alpha \cap Y) \subseteq \bigcup_{\alpha \in S} G_\alpha \end{aligned}$$

- ($\impliedby$) Suppose

- $E \in \mathcal{K}(X)$
- $\{H_\alpha\}_{\alpha \in A} \subseteq \mathcal{O}(Y)$
- $E \subseteq \bigcup_{\alpha \in A} H_\alpha$

By [theorem 2.4.23](#),

$$\forall \alpha \in A, \exists G_\alpha \in \mathcal{O}(X), H_\alpha = G_\alpha \cap Y$$

Then

$$\begin{aligned} E &\subseteq \bigcup_{\alpha \in A} (G_\alpha \cap Y) \subseteq \bigcup_{\alpha \in A} G_\alpha \\ &\Downarrow \\ \exists S \subseteq A, |S| < \aleph_0 \wedge E &\subseteq \bigcup_{\alpha \in S} G_\alpha \\ &\Downarrow \\ E &\subseteq \left(\bigcup_{\alpha \in S} G_\alpha \right) \cap Y = \bigcup_{\alpha \in S} (G_\alpha \cap Y) = \bigcup_{\alpha \in S} H_\alpha \end{aligned}$$

□

Theorem 2.5.4.

$$\forall E \in \mathcal{K}(|(X, d)|), E \in \mathcal{C}(X)$$

Proof. Suppose

- $E \in \mathcal{K}(X)$
- $E \neq \emptyset$
- $p \in E^c$

Define

$$\forall q \in E, r_q := \frac{d(p, q)}{2} \in \mathbb{R}^+$$

Then

$$\begin{aligned} \forall q \in E, q &\in N_{r_q}(q) \\ &\Downarrow \\ E &\subseteq \bigcup_{q \in E} N_{r_q}(q) \\ &\Downarrow \\ \exists S \subseteq E, |S| < \aleph_0 \wedge E &\subseteq \bigcup_{q \in S} N_{r_q}(q) \end{aligned}$$

[Theorem 2.4.8](#)

Define

$$T := \{r_q \mid q \in S\} \neq \emptyset$$

Define $f : S \rightarrow T$ by

$$f(q) := r_q$$

Then

$$\forall t \in T, \exists q \in S, r_q = t$$

$$\Downarrow$$

$$f : S \twoheadrightarrow T$$

$$\Downarrow$$

$$|T| < \aleph_0$$

$$\Downarrow$$

$$\exists \varepsilon \in T, \varepsilon = \min T$$

$$\Downarrow$$

$$\forall x \in N_\varepsilon(p), \forall q \in S, d(x, p) < \varepsilon \leq r_q$$

Theorem 2.2.8

Theorem 2.2.9

Assume

$$\exists x \in N_\varepsilon(p), \exists q \in S, x \in N_{r_q}(q)$$

Then

$$d(p, q) \leq d(p, x) + d(x, q) < r_q + r_q = d(p, q)$$

$$\Downarrow$$

$$\perp$$

Therefore,

$$\forall x \in N_\varepsilon(p), \forall q \in S, x \notin N_{r_q}(q)$$

$$\Downarrow$$

$$\forall q \in S, N_\varepsilon(p) \cap N_{r_q}(q) = \emptyset$$

$$\Downarrow$$

$$N_\varepsilon(p) \cap E \subseteq N_\varepsilon(p) \cap \bigcup_{q \in S} N_{r_q}(q) = \emptyset$$

$$\Downarrow$$

$$N_\varepsilon(p) \subseteq E^c$$

Since $p \in E^c$ was arbitrary,

$$E^c \in \mathcal{O}(X)$$

$$\Downarrow$$

$$E \in \mathcal{C}(X)$$

Theorem 2.4.10

□

Theorem 2.5.5.

$$\forall E \in \mathcal{K}(|(X, d)|), E \in \mathcal{B}(X)$$

Proof. Suppose

- $E \in \mathcal{K}(X)$
- $E \neq \emptyset$

Let $p \in E$. By [theorem 1.6.2](#),

$$\begin{aligned}
 & \forall q \in E, \exists n \in \mathbb{N}, d(p, q) < \frac{1}{n} \\
 & \Downarrow \\
 & E \subseteq \bigcup_{n=1}^{\infty} N_n(p) \\
 & \Downarrow \\
 & \exists S \subseteq \mathbb{N}, |S| < \aleph_0 \wedge E \subseteq \bigcup_{n \in S} N_n(p) \\
 & \Downarrow \\
 & \exists m \in S, m = \max S \\
 & \Downarrow \\
 & E \subseteq N_m(p) \\
 & \Downarrow \\
 & E \in \mathcal{B}(X)
 \end{aligned}$$

[Theorem 2.4.8](#)

[Theorem 2.2.9](#)

□

Theorem 2.5.6.

$$\forall E \in \mathcal{K}(|(X, d)|), \forall F \in \mathcal{C}(X), F \subseteq E \implies F \in \mathcal{K}(X)$$

Proof. Suppose

- $E \in \mathcal{K}(X)$
- $F \in \mathcal{C}(X)$
- $F \subseteq E$
- $\{H_\alpha\}_{\alpha \in A} \subseteq \mathcal{O}(X)$
- $F \subseteq \bigcup_{\alpha \in A} H_\alpha$

Then

$$X = F \cup F^c \subseteq \left(\bigcup_{\alpha \in A} H_\alpha \right) \cup F^c$$

Therefore,

$$\begin{aligned} E &\subseteq \left(\bigcup_{\alpha \in A} H_\alpha \right) \cup F^c \\ &\Downarrow \\ \exists B \subseteq A, |B| < \aleph_0 \wedge E &\subseteq \left(\bigcup_{\alpha \in B} H_\alpha \right) \cup F^c \\ &\Downarrow \\ F &\subseteq \left(\bigcup_{\alpha \in B} H_\alpha \right) \cup F^c \\ &\Downarrow \\ F &\subseteq \bigcup_{\alpha \in B} H_\alpha \end{aligned}$$

Theorem 2.4.10

□

Theorem 2.5.7.

$$\forall E \in \mathcal{K}(|(X, d)|), \forall F \subseteq E, |F| \geq \aleph_0 \implies F' \cap E \neq \emptyset$$

Proof. Suppose

- $E \in \mathcal{K}(X)$
- $F \subseteq E$
- $|F| \geq \aleph_0$

Assume $F' \cap E = \emptyset$. Then

$$\begin{aligned} \forall p \in E, \exists r_p \in \mathbb{R}^+, \tilde{N}_{r_p}(p) \cap F &= \emptyset \\ &\Downarrow \\ \forall p \in E, N_{r_p}(p) \cap F &\subseteq \{p\} \end{aligned}$$

This implies

$$\begin{aligned}
 & \forall \alpha \in E, \alpha \in N_{r_\alpha}(\alpha) \\
 & \quad \Downarrow \\
 & E \subseteq \bigcup_{\alpha \in E} N_{r_\alpha}(\alpha) \\
 & \quad \Downarrow \\
 & \exists S \subseteq E, |S| < \aleph_0 \wedge E \subseteq \bigcup_{\alpha \in S} N_{r_\alpha}(\alpha) \quad \text{Theorem 2.4.8} \\
 & \quad \Downarrow \\
 & F \subseteq \bigcup_{\alpha \in S} N_{r_\alpha}(\alpha) \\
 & \quad \Downarrow \\
 & F = \left(\bigcup_{\alpha \in S} N_{r_\alpha}(\alpha) \right) \cap F = \bigcup_{\alpha \in S} (N_{r_\alpha}(\alpha) \cap F) \subseteq \bigcup_{\alpha \in S} \{\alpha\} = S \\
 & \quad \Downarrow \quad \text{Theorem 2.2.5} \\
 & |F| < \aleph_0 \\
 & \quad \Downarrow \\
 & \perp
 \end{aligned}$$

Therefore,

$$F' \cap E \neq \emptyset$$

□

Theorem 2.5.8.

$$\begin{aligned}
 & \forall \{K_\alpha\}_{\alpha \in A} \subseteq \mathcal{K}(|(X, d)|), \\
 & \left(\forall B \subseteq A, |B| < \aleph_0 \implies \bigcap_{\alpha \in B} K_\alpha \neq \emptyset \right) \implies \bigcap_{\alpha \in A} K_\alpha \neq \emptyset
 \end{aligned}$$

Proof. Suppose

- $\{K_\alpha\}_{\alpha \in A} \subseteq \mathcal{K}(X)$
- $\forall B \subseteq A, |B| < \aleph_0 \implies \bigcap_{\alpha \in B} K_\alpha \neq \emptyset$
- $A \neq \emptyset$

By [theorem 2.5.4](#),

$$\begin{aligned} \forall \alpha \in A, K_\alpha &\in \mathcal{C}(X) \\ &\Downarrow \\ \forall \alpha \in A, (K_\alpha)^c &\in \mathcal{O}(X) \end{aligned}$$

[Theorem 2.4.10](#)

Assume $\bigcap_{\alpha \in A} K_\alpha = \emptyset$. Then

$$X = \left(\bigcap_{\alpha \in A} K_\alpha \right)^c = \bigcup_{\alpha \in A} (K_\alpha)^c$$

Let $\beta \in A$. Then

$$\begin{aligned} K_\beta &\subseteq \bigcup_{\alpha \in A} (K_\alpha)^c \\ &\Downarrow \\ \exists B \subseteq A, |B| < \aleph_0 \wedge K_\beta &\subseteq \bigcup_{\alpha \in B} (K_\alpha)^c \\ &\Downarrow \\ K_\beta &\subseteq \left(\bigcap_{\alpha \in B} K_\alpha \right)^c \\ &\Downarrow \\ K_\beta \cap \left(\bigcap_{\alpha \in B} K_\alpha \right) &= \emptyset \\ &\Downarrow \\ \bigcap_{\alpha \in B \cup \{\beta\}} K_\alpha &= \emptyset \end{aligned}$$

By [theorem 2.2.16](#),

$$\begin{aligned} |B \cup \{\beta\}| &< \aleph_0 \\ &\Downarrow \\ &\perp \end{aligned}$$

Therefore,

$$\bigcap_{\alpha \in A} K_\alpha \neq \emptyset$$

□

Lemma 2.5.9.

$$(\forall n \in \mathbb{N}, -\infty < a_n \leq a_{n+1} \leq b_{n+1} \leq b_n < +\infty) \implies \bigcap_{n=1}^{\infty} [a_n, b_n] \neq \emptyset$$

Proof. Suppose

$$\forall n \in \mathbb{N}, -\infty < a_n \leq a_{n+1} \leq b_{n+1} \leq b_n < +\infty$$

Then

- $\forall n, m \in \mathbb{N}, n \leq m \implies a_n \leq a_m \leq b_m$
- $\forall n, m \in \mathbb{N}, n > m \implies a_n \leq b_n \leq b_m$

Therefore,

$$\begin{aligned} & \forall m \in \mathbb{N}, b_m \in (\{a_n\}_{n=1}^\infty)^u \\ & \quad \Downarrow \\ & \exists \alpha \in \mathbb{R}, \alpha = \sup\{a_n\}_{n=1}^\infty \\ & \quad \Downarrow \\ & \forall m \in \mathbb{N}, a_m \leq \alpha \leq b_m \\ & \quad \Downarrow \\ & \alpha \in \bigcap_{n=1}^\infty [a_n, b_n] \end{aligned}$$

Axiom 1.6.1

□

Theorem 2.5.10.

$$\forall k \in \mathbb{N}, (\forall n \in \mathbb{N}, \forall j \in [1, k]_{\mathbb{N}}, -\infty < a_{n,j} \leq a_{n+1,j} \leq b_{n+1,j} \leq b_{n,j} < +\infty)$$

$$\implies \bigcap_{n=1}^\infty \left(\prod_{j=1}^k [a_{n,j}, b_{n,j}] \right) \neq \emptyset$$

Proof. Suppose

- $k \in \mathbb{N}$
- $\forall n \in \mathbb{N}, \forall j \in [1, k]_{\mathbb{N}}, -\infty < a_{n,j} \leq a_{n+1,j} \leq b_{n+1,j} \leq b_{n,j} < +\infty$

By lemma 2.5.9,

$$\begin{aligned} & \forall j \in [1, k]_{\mathbb{N}}, \bigcap_{n=1}^\infty [a_{n,j}, b_{n,j}] \neq \emptyset \\ & \quad \Downarrow \\ & \forall j \in [1, k]_{\mathbb{N}}, \exists \alpha_j \in \bigcap_{n=1}^\infty [a_{n,j}, b_{n,j}] \\ & \quad \Downarrow \\ & (\alpha_1, \alpha_2, \dots, \alpha_k) \in \bigcap_{n=1}^\infty \left(\prod_{j=1}^k [a_{n,j}, b_{n,j}] \right) \end{aligned}$$

□

Theorem 2.5.11.

$$\begin{aligned} & \forall k \in \mathbb{N}, (\forall j \in [1, k]_{\mathbb{N}}, -\infty < a_j \leq b_j < +\infty) \\ & \implies \prod_{j=1}^k [a_j, b_j] \in \mathcal{K}(\mathbb{R}^k) \end{aligned}$$

Proof. Suppose

- $k \in \mathbb{N}$
- $\forall j \in [1, k]_{\mathbb{N}}, -\infty < a_j \leq b_j < +\infty$

Define

- $\forall j \in [1, k]_{\mathbb{N}}, (a_{1,j}, b_{1,j}) := (a_j, b_j)$
- $I_1 := \prod_{j=1}^k [a_{1,j}, b_{1,j}] \subseteq \mathbb{R}^k$
- $\forall j \in [1, k]_{\mathbb{N}}, \forall m \in \{0, 1\}, I_{1,j,m} := \begin{cases} \left[a_{1,j}, \frac{a_{1,j} + b_{1,j}}{2} \right], & \text{if } m = 0 \\ \left[\frac{a_{1,j} + b_{1,j}}{2}, b_{1,j} \right], & \text{otherwise} \end{cases}$
- $S := \prod_{j=1}^k \{0, 1\} = \left\{ \mathbf{m} \in \mathbb{R}^k \mid \forall j \in [1, k]_{\mathbb{N}}, m_j \in \{0, 1\} \right\}$

Then

$$\begin{aligned} & \forall \mathbf{m} \in S, \forall j \in [1, k]_{\mathbb{N}}, I_{1,j,m_j} \subseteq [a_{1,j}, b_{1,j}] \\ & \quad \downarrow \\ & \forall \mathbf{m} \in S, \prod_{j=1}^k I_{1,j,m_j} \subseteq I_1 \end{aligned}$$

Suppose

- $\{G_\alpha\}_{\alpha \in A} \subseteq \mathcal{O}(\mathbb{R}^k)$
- $I_1 \subseteq \bigcup_{\alpha \in A} G_\alpha$

Assume

- $\nexists B \subseteq A, |B| < \aleph_0 \wedge I_1 \subseteq \bigcup_{\alpha \in B} G_\alpha$

$$\bullet \forall \mathbf{m} \in S, \exists A_{1,\mathbf{m}} \subseteq A, |A_{1,\mathbf{m}}| < \aleph_0 \wedge \prod_{j=1}^k I_{1,j,m_j} \subseteq \bigcup_{\alpha \in A_{1,\mathbf{m}}} G_\alpha$$

Define

$$B_1 := \bigcup_{\mathbf{m} \in S} A_{1,\mathbf{m}}$$

By [theorem 2.2.19](#),

$$\begin{aligned} |S| &< \aleph_0 \\ \Downarrow \\ |B_1| &< \aleph_0 \end{aligned}$$

[Theorem 2.2.16](#)

Then

$$\begin{aligned} I_1 &= \bigcup_{\mathbf{m} \in S} \left(\prod_{j=1}^k I_{1,j,m_j} \right) \\ &\Downarrow \\ I_1 &\subseteq \bigcup_{\mathbf{m} \in S} \left(\bigcup_{\alpha \in A_{1,\mathbf{m}}} G_\alpha \right) = \bigcup_{\alpha \in B_1} G_\alpha \\ &\Downarrow \\ &\perp \end{aligned}$$

Therefore,

$$\exists \mathbf{m}_1 \in S, \nexists A_{1,\mathbf{m}_1} \subseteq A, |A_{1,\mathbf{m}_1}| < \aleph_0 \wedge \prod_{j=1}^k I_{1,j,(m_1)_j} \subseteq \bigcup_{\alpha \in A_{1,\mathbf{m}_1}} G_\alpha$$

Define

$$\bullet \forall j \in [1, k]_{\mathbb{N}}, a_{2,j} := \begin{cases} a_{1,j}, & \text{if } (m_1)_j = 0 \\ \frac{a_{1,j} + b_{1,j}}{2}, & \text{otherwise} \end{cases}$$

$$\bullet \forall j \in [1, k]_{\mathbb{N}}, b_{2,j} := \begin{cases} \frac{a_{1,j} + b_{1,j}}{2}, & \text{if } (m_1)_j = 0 \\ b_{1,j}, & \text{otherwise} \end{cases}$$

$$\bullet I_2 := \prod_{j=1}^k [a_{2,j}, b_{2,j}] \subseteq \mathbb{R}^k$$

$$\bullet I_{2,j,m} := \begin{cases} \left[a_{2,j}, \frac{a_{2,j} + b_{2,j}}{2} \right], & \text{if } m = 0 \\ \left[\frac{a_{2,j} + b_{2,j}}{2}, b_{2,j} \right], & \text{otherwise} \end{cases}$$

Then

$$\begin{aligned} \forall j \in [1, k]_{\mathbb{N}}, [a_{2,j}, b_{2,j}] &= I_{1,j,(m_1)_j} \\ &\Downarrow \\ I_2 &\subseteq I_1 \end{aligned}$$

Assume

$$\forall \mathbf{m} \in S, \exists A_{2,\mathbf{m}} \subseteq A, |A_{2,\mathbf{m}}| < \aleph_0 \wedge \prod_{j=1}^k I_{2,j,m_j} \subseteq \bigcup_{\alpha \in A_{2,\mathbf{m}}} G_{\alpha}$$

Define

$$B_2 := \bigcup_{\mathbf{m} \in S} A_{2,\mathbf{m}}$$

By [theorem 2.2.19](#),

$$\begin{aligned} |S| &< \aleph_0 \\ &\Downarrow \\ |B_2| &< \aleph_0 \end{aligned}$$

[Theorem 2.2.16](#)

Then

$$\begin{aligned} I_2 &= \bigcup_{\mathbf{m} \in S} \left(\prod_{j=1}^k I_{2,j,m_j} \right) \\ &\Downarrow \\ I_2 &\subseteq \bigcup_{\mathbf{m} \in S} \left(\bigcup_{\alpha \in A_{2,\mathbf{m}}} G_{\alpha} \right) = \bigcup_{\alpha \in B_2} G_{\alpha} \\ &\Downarrow \\ &\perp \end{aligned}$$

Therefore,

$$\exists \mathbf{m}_2 \in S, \nexists A_{\mathbf{m}_2} \subseteq A, |A_{\mathbf{m}_2}| < \aleph_0 \wedge \prod_{j=1}^k I_{2,j,(m_2)_j} \subseteq \bigcup_{\alpha \in A_{\mathbf{m}_2}} G_{\alpha}$$

By repeating this process,

- $\forall n \in \mathbb{N}, \forall j \in [1, k]_{\mathbb{N}}, a_{n,j} \leq a_{n+1,j} \leq b_{n+1,j} \leq b_{n,j}$
- $\forall n \in \mathbb{N}, \forall j \in [1, k]_{\mathbb{N}}, b_{n+1,j} - a_{n+1,j} = \frac{1}{2}(b_{n,j} - a_{n,j})$
- $\forall n \in \mathbb{N}, \nexists A_n \subseteq A, |A_n| < \aleph_0 \wedge I_n \subseteq \bigcup_{\alpha \in A_n} G_{\alpha}$

Define

$$\delta := \sqrt{\sum_{j=1}^k (b_{1,j} - a_{1,j})^2} \in [0, +\infty)$$

Suppose $\delta \neq 0$. Then

$$\forall n \in \mathbb{N}, \forall \mathbf{x}, \mathbf{y} \in I_n, \|\mathbf{x} - \mathbf{y}\| \leq \sqrt{\sum_{j=1}^k (b_{n,j} - a_{n,j})^2} = \frac{\delta}{2^{n-1}}$$

By [theorem 2.5.10](#),

$$\bigcap_{n=1}^{\infty} I_n \neq \emptyset$$

Let $\beta \in \bigcap_{n=1}^{\infty} I_n$. Then

$$\begin{aligned} \beta &\in \bigcup_{\alpha \in A} G_{\alpha} \\ &\downarrow \\ \exists \alpha_0 \in A, \beta &\in G_{\alpha_0} \\ &\downarrow \\ \exists \varepsilon \in \mathbb{R}^+, N_{\varepsilon}(\beta) &\subseteq G_{\alpha_0} \\ &\downarrow \\ \exists N \in \mathbb{N}, \frac{1}{2^N} &< \frac{\varepsilon}{2\delta} \\ &\downarrow \\ \forall \gamma \in I_N, \|\beta - \gamma\| &\leq \frac{\delta}{2^{N-1}} < \varepsilon \\ &\downarrow \\ I_N &\subseteq N_{\varepsilon}(\beta) \\ &\downarrow \\ I_N &\subseteq \bigcup_{\alpha \in \{\alpha_0\}} G_{\alpha} \\ &\downarrow \\ &\perp \end{aligned}$$

[Theorem 1.6.4](#)

$$\beta \in I_N$$

Therefore,

$$\exists B \subseteq A, |B| < \aleph_0 \wedge I_1 \subseteq \bigcup_{\alpha \in B} G_{\alpha}$$

□

Corollary 2.5.12.

$$\forall a, b \in \mathbb{R}, a \leq b \implies [a, b] \in \mathcal{K}(\mathbb{R})$$

Theorem 2.5.13.

$$\forall n \in \mathbb{N}, \forall E \subseteq \mathbb{R}^n, E \in \mathcal{K}(\mathbb{R}^n) \iff E \in \mathcal{C}(\mathbb{R}^n) \wedge E \in \mathcal{B}(\mathbb{R}^n)$$

This theorem is called the **Heine-Borel Theorem**.

Proof. Suppose $n \in \mathbb{N}$.

- ($\implies$) The proof follows directly from [theorems 2.5.4](#) and [2.5.5](#).
- ($\impliedby$) Suppose
 - $E \in \mathcal{C}(\mathbb{R}^n)$
 - $E \in \mathcal{B}(\mathbb{R}^n)$

Then

$$\exists r \in \mathbb{R}^+, \exists \mathbf{p} \in \mathbb{R}^n, E \subseteq N_r(\mathbf{p})$$

Define

$$I := \prod_{j=1}^n [p_j - r, p_j + r] \subseteq \mathbb{R}^n$$

By [theorem 2.5.11](#),

$$\begin{aligned} I &\in \mathcal{K}(\mathbb{R}^n) \\ &\Downarrow \\ E &\in \mathcal{K}(\mathbb{R}^n) \end{aligned}$$

[Theorem 2.5.6](#)

□

2.6 Connected Sets

Definition 2.6.1. Suppose (X, d) is a metric space and $E \subseteq X$. E is called **connected** if there is no pair of sets $A, B \subseteq X$ such that

- $\emptyset \neq A \subseteq E$ and $\emptyset \neq B \subseteq E$.
- $\overline{A} \cap B = A \cap \overline{B} = \emptyset$.
- $E = A \cup B$.

Such pair of sets A and B is called **separated**. If E is not connected, then E is called **disconnected**.

Theorem 2.6.2. Suppose $E \subseteq \mathbb{R}$. Then E is connected if and only if

$$a < x < b \implies x \in E \quad \text{for all } a, b \in E \text{ and } x \in \mathbb{R}.$$

Proof. ($\implies$) Suppose $E \subseteq \mathbb{R}$ is connected. Assume there exist $a, b \in E$ and $x \in \mathbb{R}$ such that

$$a < x < b \text{ and } x \notin E.$$

Define

- $A := E \cap (-\infty, x) \subseteq E$.
- $B := E \cap (x, \infty) \subseteq E$.

Then we have

- $A \neq \emptyset$ and $B \neq \emptyset$ since $a \in A$ and $b \in B$.
- $\overline{A} \cap B = A \cap \overline{B} = \emptyset$.
- $E = A \cup B$ since $x \notin E$.

Thus A and B are separated, which is a contradiction. Therefore, we have

$$a < x < b \implies x \in E \quad \text{for all } a, b \in E \text{ and } x \in \mathbb{R}.$$

($\impliedby$) Suppose $E \subseteq \mathbb{R}$ satisfies

$$a < x < b \implies x \in E \quad \text{for all } a, b \in E \text{ and } x \in \mathbb{R}.$$

Assume E is disconnected. Then there exist separated sets $A, B \subseteq \mathbb{R}$ such that

- $\emptyset \neq A \subseteq E$ and $\emptyset \neq B \subseteq E$.
- $\overline{A} \cap B = A \cap \overline{B} = \emptyset$.
- $E = A \cup B$.

Let $a \in A$ and $b \in B$. Without loss of generality, suppose that $a < b$. Let

$$\alpha := \sup(A \cap [a, b]) \in \mathbb{R},$$

where we have

- $a \in A \implies A \cap [a, b] \neq \emptyset$.

- $A \cap [a, b]$ is bounded above by b .

Then the *existence* of α is guaranteed by the *least upper bound property*. By [theorem 2.4.16](#), we have

$$\alpha \in \overline{(A \cap [a, b])} = (A \cap [a, b]) \cup (A \cap [a, b])'.$$

By [definition 2.4.4](#), it is clear that

$$(A \cap [a, b])' \subseteq A'.$$

Thus we have

$$\alpha \in (A \cap [a, b]) \subseteq A \quad \text{or} \quad \alpha \in (A \cap [a, b])' \subseteq A' \implies \alpha \in \bar{A}.$$

Then the following statements hold:

- $\alpha = \sup(A \cap [a, b]) \implies a \leq \alpha \leq b$.
- $\alpha \in \bar{A} \implies \alpha \notin B \implies \alpha \neq b$.

If $\alpha \notin A$, then we have

- $\alpha \neq a \implies a < \alpha < b$
- $\alpha \notin A$ and $\alpha \notin B \implies \alpha \notin E$

which is a contradiction. Otherwise, we have

$$\alpha \in A \implies \alpha \notin \bar{B}.$$

Then there exists $r \in (0, b - \alpha)$ such that

$$\tilde{N}_r(\alpha) \cap B = \emptyset \implies \beta := \alpha + r \notin B.$$

Then we have

- $a \leq \alpha < \beta < b$
- $\beta > \alpha \implies \beta \notin A \implies \beta \notin E$

which is a contradiction. Therefore, we conclude that E is connected. □

Chapter 3

Sequences and Series

3.1 Convergent Sequences

Definition 3.1.1. Suppose (X, d) is a metric space. A **sequence** $(x_n)_{n=1}^{\infty}$ is a function $f : \mathbb{N} \rightarrow X$, where $x_n := f(n)$ for each $n \in \mathbb{N}$.

A sequence $(x_n)_{n=1}^{\infty}$ is said to be **convergent** if there exists $x \in X$ such that

$$\forall \varepsilon \in \mathbb{R}^+, \exists N \in \mathbb{N}, \forall n \geq N, d(x_n, x) < \varepsilon$$

In this case, we say that

- $(x_n)_{n=1}^{\infty}$ **converges** to x .
- x is the **limit** of $(x_n)_{n=1}^{\infty}$.

We denote the limit of $(x_n)_{n=1}^{\infty}$ by

- $\lim_{n \rightarrow \infty} x_n = x$.
- $x_n \rightarrow x$ as $n \rightarrow \infty$.

If $(x_n)_{n=1}^{\infty}$ is not convergent, it is said to be **divergent**.

Example 3.1.2. In $(\mathbb{R}, \|\cdot\|)$, define $(x_n)_{n=1}^{\infty}$ by

$$x_n := \frac{1}{n} \quad \text{for each } n \in \mathbb{N}.$$

Then, for each $\varepsilon \in \mathbb{R}^+$, we have

$$n > \frac{1}{\varepsilon} \implies \|x_n - 0\| = \left| \frac{1}{n} - 0 \right| = \frac{1}{n} < \varepsilon.$$

Therefore, $(x_n)_{n=1}^{\infty}$ converges to 0.

In the subspace $(\mathbb{R}^+, \|\cdot\|)$ of $(\mathbb{R}, \|\cdot\|)$, define $(x_n)_{n=1}^{\infty}$ by

$$x_n = \frac{1}{n} \quad \text{for each } n \in \mathbb{N}.$$

By [theorem 1.6.2](#), for each $y \in \mathbb{R}^+$, there exists $N_y \in \mathbb{N}$ such that $N_y > \frac{1}{y}$. Let

$$\varepsilon_y := y - \frac{1}{N_y} \in \mathbb{R}^+ \quad \text{for each } y \in \mathbb{R}^+.$$

Then we have

$$n > N_y \implies \|x_n - y\| = \left| \frac{1}{n} - y \right| > \left| \frac{1}{N_y} - y \right| = \varepsilon_y.$$

Therefore, $(x_n)_{n=1}^{\infty}$ diverges in $(\mathbb{R}^+, \|\cdot\|)$.

Remark. Suppose (X, d) is a metric space. In this textbook, $(x_n)_{n=1}^{\infty}$ represents a sequence in (X, d) , while $\{x_n\}_{n=1}^{\infty}$ represents a set defined by

$$\{x_n\}_{n=1}^{\infty} := \{x_n \mid n \in \mathbb{N}\} \subseteq X.$$

Definition 3.1.3. Suppose (X, d) is a metric space. A sequence $(x_n)_{n=1}^{\infty}$ is said to be **bounded** if $\{x_n\}_{n=1}^{\infty}$ is bounded.

Theorem 3.1.4. Suppose (X, d) is a metric space. Suppose $(x_n)_{n=1}^{\infty}$ is a convergent sequence. Then we have

(A) $(x_n)_{n=1}^{\infty}$ is bounded.

(B) The limit of $(x_n)_{n=1}^{\infty}$ is unique.

Proof. Let x be a limit of $(x_n)_{n=1}^{\infty}$.

(A) Let $\varepsilon = 1$. Then there exists $N \in \mathbb{N}$ such that

$$n \geq N \implies d(x_n, x) < 1.$$

Define

$$S := \{d(x_n, x) \mid 1 \leq n \leq N\} \subseteq [0, \infty).$$

Let $f : [1, N]_{\mathbb{N}} \rightarrow S$ defined by

$$f(k) = d(x_k, x) \quad \text{for all } k \in [1, N]_{\mathbb{N}}.$$

Then f is surjective. By lemma 2.2.7, S is finite. Then, by Theorem 2.2.9, S has a greatest element, say $M := \max S$. Then we have

$$d(x_n, x) < M + 1 \quad \text{for all } n \in \mathbb{N}.$$

Therefore, $(x_n)_{n=1}^{\infty}$ is bounded.

(B) Suppose y be a limit of $(x_n)_{n=1}^{\infty}$. Assume $x \neq y$. Then we have $r := d(x, y) > 0$. Let $\varepsilon = \frac{r}{2}$. Then we have

- There exists $N_x \in \mathbb{N}$ such that $n \geq N_x \implies d(x_n, x) < \frac{r}{2}$.
- There exists $N_y \in \mathbb{N}$ such that $n \geq N_y \implies d(x_n, y) < \frac{r}{2}$.

Thus we have

$$n \geq \max\{N_x, N_y\} \implies d(x, y) \leq d(x, x_n) + d(x_n, y) < r,$$

which is a contradiction. Therefore, we have $x = y$.

□

Theorem 3.1.5. Suppose (X, d) is a metric space. Then the following statements hold:

(A) A sequence $(x_n)_{n=1}^{\infty}$ converges to $x \in X$ if and only if

For every $\varepsilon \in \mathbb{R}^+$, $N_{\varepsilon}(x)$ contains all but finitely many terms of $(x_n)_{n=1}^{\infty}$.

(B) Suppose $E \subseteq X$. If x is a limit point of E , then there exists $(x_n)_{n=1}^{\infty}$ in (X, d) such that

$$\lim_{n \rightarrow \infty} x_n = x.$$

Proof.

(A) ($\implies$) Suppose $(x_n)_{n=1}^{\infty}$ converge to $x \in X$. Then, for each $\varepsilon \in \mathbb{R}^+$, there exists $M_{\varepsilon} \in \mathbb{N}$ such that

$$n \geq M_{\varepsilon} \implies d(x_n, x) < \varepsilon,$$

which implies that

$$\{x_n\}_{n=M_{\varepsilon}}^{\infty} \subseteq N_{\varepsilon}(x).$$

($\Leftarrow$) Suppose $(x_n)_{n=1}^{\infty}$ in (X, d) such that

For every $\varepsilon \in \mathbb{R}^+$, $N_{\varepsilon}(x)$ contains all but finitely many terms of $(x_n)_{n=1}^{\infty}$.

(B) Suppose $E \subseteq X$ and x is a limit point of E . Then for every $\varepsilon \in \mathbb{R}^+$, we have $(N_{\varepsilon}(x) - \{x\}) \cap E \neq \emptyset$. Thus we can choose $(y_n)_{n=1}^{\infty}$ in (X, d) such that

$$y_n \in (N_{\frac{1}{n}}(x) \setminus \{x\}) \cap E.$$

Then we have

$$d(y_n, x) < \frac{1}{n} \quad \text{for each } n \in \mathbb{N},$$

which implies that $\lim_{n \rightarrow \infty} y_n = x$.

□

3.2 Subsequences**3.3 Cauchy Sequences****3.4 Upper and Lower Limits****3.5 Some Special Sequences****3.6 Series****3.7 Series of Nonnegative Terms****3.8 The Number e** **3.9 The Root and Ratio Tests****3.10 Power Series****3.11 Summation by Parts****3.12 Absolute Convergence****3.13 Addition and Multiplication of Series****3.14 Rearrangements**

Chapter 4

Continuity

- 4.1 Limits of Functions**
- 4.2 Continuous Functions**
- 4.3 Continuity and Compactness**
- 4.4 Continuity and Connectedness**
- 4.5 Discontinuities**
- 4.6 Monotonic Functions**
- 4.7 Infinite Limits and Limits at Infinity**

Chapter 5

Differentiation

5.1 The Derivative of a Real Function

5.2 Mean Value Theorems

5.3 The Continuity of Derivatives

5.4 L'Hospital's Rule

5.5 Derivatives of Higher Order

5.6 Taylor's Theorem

5.7 Differentiation of Vector-valued Functions

Chapter 6

The Riemann-Stieltjes Integral

6.1 Definition and Existence of the Integral

6.2 Properties of the Integral

6.3 Integration and Differentiation

6.4 Integration of Vector-valued Functions

6.5 Rectifiable Curves

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